

Cross-Domain State Preservation Functor

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Abstract

We mechanize, in Isabelle/HOL, the preservation of state transitions synchronized across heterogeneous domains (blockchains and off-chain ledgers). A hierarchy of locales captures cross-domain state preservation as a functor: we prove the three category axioms on preservation morphisms (identity, composition, associativity), lift them to a tower of synchronization-degree functors connected by natural transformations that are closed under composition, and establish degree monotonicity (over-provisioning is safe, under-provisioning is not). A compositional safety/liveness layer yields *guarded bounded convergence*: from an arbitrary finite-domain, lock-free state, with no assumption that cross-chain consistency holds initially, the synchronization oracle reaches a valid state within a bounded number of steps. We couple the functor to an authenticated data structure by lifting the merge and blinding operations of a Merkle interface to the global-state level, building directly on Lochbihler and Marić’s `ADS_Functor` entry, and instantiate the construction both on its blindable-position functor and on a recursive model of the Canton transaction tree. The liveness locales abstract priority resolution and fair scheduling under an assume-guarantee fairness bound; the Byzantine threshold ($n \geq 3f+1$) is load-bearing, yielding a fair schedule that makes the liveness layer inhabitable, witnessed by a non-degenerate four-node instance with one Byzantine node. Every generic locale is instantiated on the regulatory state model of Oraclizer, a state synchronization oracle for tokenized assets; to exhibit domain independence, the generic safety layer is additionally instantiated on a TCP connection tracker. The synchronization model is atomic; lifting it to a partially synchronous network is left to subsequent entries.

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1 Overview

Methodological positioning and contributions. Regarding state machines as objects and the structure-preserving synchronization maps of `State_Preservation.thy` as morphisms, `Functor_Laws.thy` proves the three category axioms on these morphisms: identity is a preservation morphism (`preservation_id`), preservation morphisms compose (`preservation_compose`), and their composition is associative (`preservation_assoc`). The functorial reading is carried at two further places: each transition system is a functor from its one-object action category into partial maps on states, its two functor laws carried by the identity word and word concatenation (`apply_actions_append`); and `Hierarchy.thy` exhibits a tower of degree-indexed functors with natural transformations between them. This is the cross-domain counterpart of the “closed under composition” structure of Lochbihler and Marić’s `ADS_Functor` entry, whose authenticated-data-structure functor is single-domain; the present entry carries the functorial reading across heterogeneous domains and couples it with that authenticated layer through two glue predicates, `state_join` and `state_refines`, that lift the ADS merge and blinding (partial-order) operations to the global-state level.

We claim no new proof technique. The contribution is a mechanization, in three layers, of structure that, to the best of our knowledge, has not previously been formalized across domains: (i) the cross-domain state-preservation *functor* and its category axioms, above; (ii) *guarded bounded convergence* within the Byzantine D-quencer model ($n \geq 3f + 1$, deterministic fair-leader assumption) — from an *arbitrary* finite-domain, lock-free initial global state, with no assumption that the cross-chain consistency invariant holds initially, the synchronization oracle reaches a valid state within a bounded number of evolution steps (`oraclizer_guarded_bounded_convergence`, `Functor_Laws.thy`); and (iii) a *tower* of synchronization-degree functors connected by natural

transformations (`degree_natural_transformation`) that are closed under composition (`nt_compose`, via `nt_vertical_compose`), a layer which the `ADS_Functor` entry, to our knowledge, does not develop (`Hierarchy.thy`).

The entry consists of ten theory files; the theory text carries the detailed exposition alongside the formal material.

State_Preservation.thy domain-independent locales for state machines, pairwise state preservation (with naturality in both the enabled and the disabled direction), symmetric bidirectional preservation, and multi-domain preservation — instantiable by any domain with finite states and actions, deterministic transitions, and optional terminal states.

Regulatory_Instance.thy the regulatory safety instance: five states, seven actions, transition exclusions with legal justification, a synchronization protocol with preemptive locking, and the main safety theorems — *regulatory homomorphism*, *valid state preservation*, and *multi-domain parametric instantiation* — together with heterogeneous-action and layer-crossing bridge instances.

Priority_Resolution.thy domain-independent liveness locales: deterministic selection from a finite message set with injective priorities (`priority_system`), and starvation freedom under periodic honest leader scheduling (`fair_leader_system`).

DQuencer_Instance.thy the regulatory liveness instance: a four-component priority key with proven injectivity under well-formedness bounds, a Byzantine fault model with the BFT threshold $n \geq 3f + 1$, the liveness consequences (*select highest deterministic*, *starvation freedom*, *eventual completion*), and the safety-side corollary *conditional safety preservation*.

Proof_Automation.thy reusable Eisbach discharge methods with entry points `discharge_state_machine` and `discharge_preservation`, driven by three named theorem collections; see the *Proof Automation* subsection below.

Composition.thy domain-independent locales composing a guarded safety invariant with an eventual-progress scheduler, extended by a natural-number progress measure to *guarded bounded convergence* from an arbitrary carrier state (`bounded_convergence_from_arbitrary`).

Functor_Laws.thy the category axioms, the ADS glue predicates (`state_join`, `state_refines`), the `authenticated_state` locale over the `merkle_interface` with soundness theorems whose proofs invoke the Merkle interface explicitly, and the oracle’s convergence instantiation (`oraclizer_guarded_bounded_convergence`).

Hierarchy.thy the synchronization-degree tower: a forgetful natural transformation closed under composition (`degree_forget`, `degree_natural_transformation`, `nt_compose`), reduction containment, the causal-consistency boundary, and a genuinely one-directional degree monotonicity (`over_provisioning_guarantees`, `no_downward_safety`).

Canton_Bridge.thy the categorical laws of the authenticated layer (`cdsp_ads_compose`, `cdsp_ads_merge_assoc`), sequence-level authenticity, and instantiations on the blindable-position functor of `ADS_Functor.ADS_Construction` and on a recursive model of the Canton transaction tree, with a declared consensus-scope limit.

External_Instance.thy an instance outside the regulatory domain — the TCP connection state machine (an RFC 793 subset) observed by a stateful connection tracker — importing only the generic theories and discharged by the generic methods: the evidence that the generic layer carries no hidden regulatory assumptions.

1.1 Theory Dependency Structure

The four base theories form two generic-instance pairs that meet at `DQuencer_Instance.thy`: `State_Preservation.thy` is imported by `Regulatory_Instance.thy`, and `Priority_Resolution.thy`, which depends only on `Main`, is imported by `DQuencer_Instance.thy` together with the regulatory instance. The compositional theories sit on top: `Composition.thy` and `Proof_Automation.thy` import `State_Preservation.thy`; `Functor_Laws.thy` imports both instances, `ADS_Functor.Merkle_Interface`, `Composition.thy`, and `Proof_Automation.thy`; `Hierarchy.thy` and `Canton_Bridge.thy` import `Functor_Laws.thy`; and `External_Instance.thy` imports only `Proof_Automation.thy`, deliberately independent of the regulatory instance. The dependency on `ADS_Functor` is the entry’s only external session dependency beyond the Isabelle distribution (`HOL-Library`, `HOL-Eisbach`); `Canton_Bridge.thy` additionally imports `ADS_Functor.ADS_Construction` and `ADS_Functor.Canton_Transaction_Tree`.

1.2 Design Decisions

The regulatory state model incorporates several deliberate design decisions based on legal precedence and operational considerations:

- The state machine models seven regulatory `reg_actions`: the four escalation actions `FREEZE`, `SEIZE`, `CONFISCATE`, `RESTRICT` and their de-escalation counterparts `UNFREEZE`, `UNRESTRICT`, `RELEASE`. `RECOVER` and `LIQUIDATE`, two of the product’s six enforcement actions,

are excluded as they involve force transfers or external DEX interactions rather than regulatory state transitions.

- The direct transition `SEIZED` $\rightarrow$ `FROZEN` is excluded because seizure is a strictly stronger legal constraint than freezing; relaxation requires explicit release first.
- The direct transition `FROZEN` $\rightarrow$ `RESTRICTED` is excluded to enforce passage through `ACTIVE`.
- Preemptive locking is modelled as a guard on the synchronization function: `sync` acquires the lock, updates, and releases in a single atomic step. Under this atomic abstraction the guard expresses the *intended* exclusion of competing regulatory actions on the same asset rather than serialising them dynamically (there is no interleaving in the model to serialise); the dynamic correctness of contention-time locking is deferred to the preemptive-lock property. The guard does not address double-spending, which is handled at a different layer.
- Deadlock (a concurrency phenomenon) does not arise within the model's scope: the atomic `sync` model has no concurrent lock contention. Forced lock release under contention is out of scope, deferred to the preemptive-lock property; the entry states no proved deadlock-freedom theorem.

1.3 Proof Automation

The entry ships a reusable proof-automation layer, `Proof_Automation.thy`: two Eisbach methods discharge the obligations of `state_machine` and `state_preservation` instances on finite, enumerated domains — `discharge_state_machine` and `discharge_preservation`. An instance theory declares the defining equations of its state/action sets and transition functions into three named theorem collections (`discharge_simps` for equational and conditional-rewrite content, `discharge_intros` for closure rules, `discharge_dels` for default rewrites that must be suppressed to keep structured state maps opaque), after which whole instances are discharged by a single method invocation.

The methods are exercised against the entry's own instances, with the statements kept identical to their manual counterparts. Four instances are discharged by pure search, with no registered interpretation to fall back on — the escalation-scoped DAML state machine (6 obligations), the escalation-scoped layer-crossing bridge (11 obligations), the composed three-domain morphism re-derived from atomic obligations without invoking `preservation_compose` (11 obligations), and a forward escalation-scoped bridge with no manual counterpart, which the method constructs outright

(11 obligations) — closing 39 of 39 atomic obligations, with no obligation weakened or left to a manual fallback. Three further one-line restatements of the registered interpretations are discharged from the registrations. The methods and their collections are part of the entry’s reusable surface: an importer instantiating the locales on its own domain populates the collections and reuses the same two entry points, as `External_Instance.thy` does.

The synchronization model is atomic; lifting it to a partially synchronous network with message delays, redelivery, and out-of-order arrival is left to subsequent entries.

```
theory State-Preservation
imports Main
begin
```

2 Generic State Machine Locale

A state machine with finite state set, deterministic partial transition function, and optional terminal states. Terminal states accept no transitions.

```
locale state-machine =
  fixes states :: 's set
    and actions :: 'a set
    and transition :: 's  $\Rightarrow$  'a  $\Rightarrow$  's option
    and terminal :: 's set
  assumes finite-states: finite states
    and finite-actions: finite actions
    and terminal-subset: terminal  $\subseteq$  states
    and terminal-absorbing:  $\llbracket s \in \text{terminal}; a \in \text{actions} \rrbracket \Longrightarrow \text{transition } s \ a = \text{None}$ 
    and transition-closed:  $\llbracket s \in \text{states}; a \in \text{actions}; \text{transition } s \ a = \text{Some } s' \rrbracket \Longrightarrow s' \in \text{states}$ 
    and transition-domain:  $\llbracket s \notin \text{states} \rrbracket \Longrightarrow \text{transition } s \ a = \text{None}$ 
begin
```

Determinism is inherent: transition is a function, not a relation.

```
lemma transition-deterministic:
  transition s a = Some s1  $\Longrightarrow$  transition s a = Some s2  $\Longrightarrow$  s1 = s2
by simp
```

Sequential composition of actions (partial, returns None on first failure).

```
fun apply-actions :: 's  $\Rightarrow$  'a list  $\Rightarrow$  's option where
  apply-actions s [] = Some s
| apply-actions s (a # as) = (case transition s a of
    None  $\Rightarrow$  None
  | Some s'  $\Rightarrow$  apply-actions s' as)
```

```
lemma apply-actions-closed:
```

$\llbracket s \in \text{states}; \forall a \in \text{set } as. a \in \text{actions}; \text{apply-actions } s \text{ as} = \text{Some } s' \rrbracket$
 $\implies s' \in \text{states}$
by (induction as arbitrary: s) (auto split: option.splits dest: transition-closed)

Note: $\text{apply-actions } ?s \llbracket = \text{Some } ?s$ already provides the simp rule $\text{apply-actions } s \llbracket = \text{Some } s$, so no separate [simp] lemma is needed.

lemma *apply-actions-terminal*:

$\llbracket s \in \text{terminal}; a \in \text{actions} \rrbracket \implies \text{apply-actions } s (a \# as) = \text{None}$
by (simp add: terminal-absorbing)

Concatenation of action words is the Kleisli composition of the two segment maps: together with $\text{apply-actions } ?s \llbracket = \text{Some } ?s$ (the identity word), this is the functor reading of a transition system on the free monoid of action words.

lemma *apply-actions-append*:

$\text{apply-actions } s (as @ bs) =$
 $(\text{case } \text{apply-actions } s \text{ as of } \text{None} \Rightarrow \text{None} \mid \text{Some } s' \Rightarrow \text{apply-actions } s' bs)$
by (induction as arbitrary: s) (auto split: option.splits)

Outside the state set no action sequence makes progress: the domain discipline **transition_domain** lifts from single transitions to words.

lemma *apply-actions-outside-domain*:

$s \notin \text{states} \implies \text{apply-actions } s (a \# as) = \text{None}$
by (simp add: transition-domain)

end

3 Cross-Domain State Preservation Locale

Two state machines connected by a synchronization map. The key property (naturality / homomorphism) states that synchronizing after a local transition yields the same result as transitioning after synchronization.

More precisely: for state machines (**S_src**, **T_src**) and (**S_tgt**, **T_tgt**) connected by **sync** (mapping source states to target states), the following diagram commutes:

$\text{s_src} \xrightarrow{\text{T_src}(a)} \text{s_src}' \mid \mid \text{sync } \text{sync} \mid \mid \text{v v } \text{s_tgt} \xrightarrow{\text{T_tgt}(a)} \text{s_tgt}'$

That is: $\text{sync } (\text{T_src } s \text{ a}) = \text{T_tgt } (\text{sync } s) \text{ a}$

This is the structural preservation guarantee: a transition performed at the source domain, when synchronized, produces the same result as performing the corresponding transition at the target domain.

locale *state-preservation* =

source: state-machine $\text{states}_s \text{ actions}_s \text{ transition}_s \text{ terminal}_s +$
target: state-machine $\text{states}_t \text{ actions}_t \text{ transition}_t \text{ terminal}_t$
for $\text{states}_s :: 's \text{ set}$ **and** $\text{actions}_s :: 'a \text{ set}$
and $\text{transition}_s :: 's \Rightarrow 'a \Rightarrow 's \text{ option}$

```

and  $terminal_s :: 's \text{ set}$ 
and  $states_t :: 't \text{ set}$  and  $actions_t :: 'b \text{ set}$ 
and  $transition_t :: 't \Rightarrow 'b \Rightarrow 't \text{ option}$ 
and  $terminal_t :: 't \text{ set} +$ 
fixes  $state\text{-}map :: 's \Rightarrow 't$ 
and  $action\text{-}map :: 'a \Rightarrow 'b$ 
assumes  $state\text{-}map\text{-}well\text{-}defined: s \in states_s \implies state\text{-}map\ s \in states_t$ 
and  $action\text{-}map\text{-}well\text{-}defined: a \in actions_s \implies action\text{-}map\ a \in actions_t$ 
and  $terminal\text{-}preservation: s \in terminal_s \implies state\text{-}map\ s \in terminal_t$ 
— The naturality / homomorphism condition:
and  $naturality:$ 
   $\llbracket s \in states_s; a \in actions_s; transition_s\ s\ a = Some\ s' \rrbracket$ 
   $\implies transition_t\ (state\text{-}map\ s)\ (action\text{-}map\ a) = Some\ (state\text{-}map\ s')$ 
and  $naturality\text{-}none:$ 
   $\llbracket s \in states_s; a \in actions_s; transition_s\ s\ a = None \rrbracket$ 
   $\implies transition_t\ (state\text{-}map\ s)\ (action\text{-}map\ a) = None$ 
begin

```

The fundamental theorem: state preservation extends to sequences of actions. If a sequence of transitions is valid at the source, the mapped sequence is valid at the target with the mapped final state.

```

theorem sequential-preservation:
  assumes  $s \in states_s$ 
  and  $\forall a \in set\ as. a \in actions_s$ 
  and  $source.apply\text{-}actions\ s\ as = Some\ s'$ 
  shows  $target.apply\text{-}actions\ (state\text{-}map\ s)\ (map\ action\text{-}map\ as) = Some\ (state\text{-}map\ s')$ 
  using assms
proof (induction as arbitrary: s)
  case Nil
  then show ?case by simp
next
  case (Cons a as)
  then obtain s-mid where
    step: transition_s s a = Some s-mid and
    rest: source.apply-actions s-mid as = Some s'
  by (auto split: option.splits)
  from  $Cons.prems(1)\ Cons.prems(2)\ step$  have  $s\text{-}mid \in states_s$ 
  using source.transition-closed by auto
  from  $Cons.prems(1)\ Cons.prems(2)\ step$ 
  have  $mapped\text{-}step: transition_t\ (state\text{-}map\ s)\ (action\text{-}map\ a) = Some\ (state\text{-}map\ s\text{-}mid)$ 
  using naturality by auto
  from  $\langle s\text{-}mid \in states_s \rangle\ Cons.prems(2)\ rest$ 
  have  $target.apply\text{-}actions\ (state\text{-}map\ s\text{-}mid)\ (map\ action\text{-}map\ as) = Some\ (state\text{-}map\ s')$ 
  using Cons.IH by auto
  with mapped-step show ?case by simp
qed

```

```

theorem sequential-preservation-none:
  assumes  $s \in \text{states}_s$ 
    and  $\forall a \in \text{set as. } a \in \text{actions}_s$ 
    and  $\text{source.apply-actions } s \text{ as} = \text{None}$ 
  shows  $\text{target.apply-actions } (\text{state-map } s) (\text{map action-map as}) = \text{None}$ 
  using assms
proof (induction as arbitrary: s)
  case Nil
    then show ?case by simp
next
  case (Cons a as)
  show ?case
  proof (cases transitions s a)
    case None
      then have  $\text{transition}_t (\text{state-map } s) (\text{action-map } a) = \text{None}$ 
      using naturality-none Cons.prem(1) Cons.prem(2) by auto
      then show ?thesis by simp
    next
    case (Some s-mid)
    then have mid-in: s-mid ∈ statess
      using source.transition-closed Cons.prem(1) Cons.prem(2) by auto
    from Some Cons.prem(3) have  $\text{source.apply-actions } s\text{-mid as} = \text{None}$ 
    by simp
    then have ih: target.apply-actions (state-map s-mid) (map action-map as) =
None
      using Cons.IH mid-in Cons.prem(2) by auto
    from Some have  $\text{transition}_t (\text{state-map } s) (\text{action-map } a) = \text{Some } (\text{state-map } s\text{-mid})$ 
      using naturality Cons.prem(1) Cons.prem(2) by auto
    with ih show ?thesis by simp
  qed
qed

  Terminal states are preserved: if a source state is terminal, its image is
  terminal.

corollary terminal-image-absorbing:
  assumes  $s \in \text{terminal}_s$  and  $a \in \text{actions}_s$ 
  shows  $\text{transition}_t (\text{state-map } s) (\text{action-map } a) = \text{None}$ 
proof –
  from assms(1) have  $s \in \text{states}_s$  using source.terminal-subset by auto
  from assms have  $\text{transition}_s s a = \text{None}$ 
  using source.terminal-absorbing by auto
  then show ?thesis
    using naturality-none ⟨s ∈ statess⟩ assms(2) by auto
qed

end

```

4 Symmetric State Preservation

For bidirectional synchronization, both directions must preserve structure. This locale composes two `state_preservation` instances with inverse maps, establishing that synchronization in either direction is structure-preserving.

```

locale symmetric-state-preservation =
  forward: state-preservation
    statess actionss transitions terminals
    statest actionst transitiont terminalt
    state-map action-map +
  backward: state-preservation
    statest actionst transitiont terminalt
    statess actionss transitions terminals
    state-map-inv action-map-inv
  for statess :: 's set and actionss :: 'a set
    and transitions :: 's  $\Rightarrow$  'a  $\Rightarrow$  's option
    and terminals :: 's set
    and statest :: 't set and actionst :: 'b set
    and transitiont :: 't  $\Rightarrow$  'b  $\Rightarrow$  't option
    and terminalt :: 't set
    and state-map :: 's  $\Rightarrow$  't and action-map :: 'a  $\Rightarrow$  'b
    and state-map-inv :: 't  $\Rightarrow$  's and action-map-inv :: 'b  $\Rightarrow$  'a +
  assumes roundtrip-state-src:  $s \in \text{states}_s \implies \text{state-map-inv } (\text{state-map } s) = s$ 
    and roundtrip-state-tgt:  $t \in \text{states}_t \implies \text{state-map } (\text{state-map-inv } t) = t$ 
    and roundtrip-action-src:  $a \in \text{actions}_s \implies \text{action-map-inv } (\text{action-map } a) = a$ 
    and roundtrip-action-tgt:  $b \in \text{actions}_t \implies \text{action-map } (\text{action-map-inv } b) = b$ 
begin

```

Under symmetric preservation with roundtrip maps, `state_map` is injective on the state set — distinct source states map to distinct target states. This guarantees no information loss during synchronization.

```

lemma state-map-injective:
   $\llbracket s1 \in \text{states}_s; s2 \in \text{states}_s; \text{state-map } s1 = \text{state-map } s2 \rrbracket \implies s1 = s2$ 
  using roundtrip-state-src by metis

```

The roundtrip pair upgrades injectivity to a bijection between the two state sets and between the two action sets — the formal content of “no information loss” for bidirectional synchronization.

```

lemma state-map-bij: bij-betw state-map statess statest
proof (rule bij-betw-byWitness[where  $f' = \text{state-map-inv}$ ])
  show  $\forall s \in \text{states}_s. \text{state-map-inv } (\text{state-map } s) = s$ 
    using roundtrip-state-src by blast
  show  $\forall t \in \text{states}_t. \text{state-map } (\text{state-map-inv } t) = t$ 
    using roundtrip-state-tgt by blast
  show  $\text{state-map} \text{ ' } \text{states}_s \subseteq \text{states}_t$ 
    using forward.state-map-well-defined by auto
  show  $\text{state-map-inv} \text{ ' } \text{states}_t \subseteq \text{states}_s$ 
    using backward.state-map-well-defined by auto

```

qed

lemma *action-map-bij*: *bij-betw action-map actions_s actions_t*
proof (*rule bij-betw-byWitness*[**where** $f' = \text{action-map-inv}$])
show $\forall a \in \text{actions}_s. \text{action-map-inv} (\text{action-map } a) = a$
 using *roundtrip-action-src* **by** *blast*
show $\forall b \in \text{actions}_t. \text{action-map} (\text{action-map-inv } b) = b$
 using *roundtrip-action-tgt* **by** *blast*
show $\text{action-map} \text{ 'actions}_s \subseteq \text{actions}_t$
 using *forward.action-map-well-defined* **by** *auto*
show $\text{action-map-inv} \text{ 'actions}_t \subseteq \text{actions}_s$
 using *backward.action-map-well-defined* **by** *auto*
 qed

end

5 Multi-Domain State Preservation

Generalization to N domains sharing the same abstract state machine. This models scenarios where a single asset exists on multiple domains (e.g., blockchain networks), all of which must reflect the same state.

The key property: if one domain transitions, the synchronization brings ALL other domains to the same resulting state.

We model this as: all domains are instances of the same abstract state machine, connected by identity-like state/action maps. The "same abstract state machine" assumption holds when the protocol enforces a uniform state model across all domains.

locale *multi-domain-preservation* =
fixes *domains* :: 'd set
 and *states* :: 's set
 and *actions* :: 'a set
 and *transition* :: 's $\Rightarrow$ 'a $\Rightarrow$'s option
 and *terminal* :: 's set
 and *domain-state* :: 'd $\Rightarrow$ 'id $\Rightarrow$'s option
 — Current state of asset 'id in domain 'd
assumes *fin-domains*: *finite domains*
 and *sm*: *state-machine states actions transition terminal*
 and *consistent-init*:
 — Precondition: all domains holding an asset agree on its state
 $\llbracket d1 \in \text{domains}; d2 \in \text{domains};$
 $\text{domain-state } d1 \text{ aid} = \text{Some } s1; \text{domain-state } d2 \text{ aid} = \text{Some } s2 \rrbracket$
 $\Rightarrow s1 = s2$
begin

Which domains hold a given asset.

definition *connected-domains* :: 'id $\Rightarrow$ 'd set **where**
connected-domains aid = $\{d \in \text{domains}. \text{domain-state } d \text{ aid} \neq \text{None}\}$

The global consensus state of an asset (well-defined by `consistent_init`).

definition *consensus-state* :: 'id $\Rightarrow$'s option **where**
consensus-state aid =
 (if *connected-domains aid* = {} then None
 else *domain-state* (SOME d. d $\in$ *connected-domains aid*) aid)

The finiteness of the domain roster bounds every connected set.

lemma *connected-domains-finite*: finite (*connected-domains aid*)

proof –

have *connected-domains aid* $\subseteq$ *domains*
unfolding *connected-domains-def* **by** auto
then show ?thesis **using** *fin-domains* **by** (rule *finite-subset*)

qed

The consensus state is well-defined: by `consistent_init` the choice of witnessing domain is immaterial, so the `SOME` in the definition is sound — every connected domain reports the consensus value.

lemma *consensus-state-well-defined*:

assumes d $\in$ *connected-domains aid*
shows *consensus-state aid* = *domain-state d aid*

proof –

have *some-in*: (SOME d'. d' $\in$ *connected-domains aid*) $\in$ *connected-domains aid*
using *assms* **by** (rule *someI*)
from *assms* **have** *ne*: *connected-domains aid* $\neq$ {} **by** auto
obtain s **where** s: *domain-state d aid* = Some s
using *assms* **unfolding** *connected-domains-def* **by** auto
obtain s' **where** s': *domain-state* (SOME d'. d' $\in$ *connected-domains aid*) aid
 = Some s'
using *some-in* **unfolding** *connected-domains-def* **by** auto
have d $\in$ *domains* (SOME d'. d' $\in$ *connected-domains aid*) $\in$ *domains*
using *assms* *some-in* **unfolding** *connected-domains-def* **by** auto
then have s' = s **using** *consistent-init s s'* **by** blast
then show ?thesis
unfolding *consensus-state-def* **using** *ne s s'* **by** simp

qed

After synchronizing an action on an asset, all connected domains reflect the new state. This is the cross-domain consistency theorem.

definition *sync-all* ::

'd $\Rightarrow$ 'a $\Rightarrow$ 'id $\Rightarrow$ ('d $\Rightarrow$ 'id $\Rightarrow$'s option) $\Rightarrow$ ('d $\Rightarrow$ 'id $\Rightarrow$'s option) option

where

sync-all source action aid ds =
 (case ds source aid of
 None $\Rightarrow$ None
 | Some s $\Rightarrow$
 (case transition s action of
 None $\Rightarrow$ None
 | Some s' $\Rightarrow$

Some ($\lambda d \text{ aid}'.$
if $\text{aid}' = \text{aid} \wedge d \in \text{connected-domains aid}$
then *Some* s'
else $ds \ d \ \text{aid}'$))

Terminal states admit no synchronization: the state-machine assumption `sm` transports terminal absorption to `sync_all`.

lemma *sync-all-terminal-none*:

assumes *domain-state source aid = Some s*
and $s \in \text{terminal}$ **and** $\text{action} \in \text{actions}$
shows *sync-all source action aid domain-state = None*
proof –
have *transition s action = None*
using *state-machine.terminal-absorbing[OF sm assms(2) assms(3)]* .
then show *?thesis*
unfolding *sync-all-def* **using** *assms(1)* **by** *simp*
qed

The cross-domain consistency theorem: after `sync_all`, every connected domain agrees on the new state.

theorem *cross-domain-consistency*:

assumes $\text{source} \in \text{domains}$
and *domain-state source aid = Some s*
and $s \in \text{states}$
and $\text{action} \in \text{actions}$
and *transition s action = Some s'*
and *sync-all source action aid domain-state = Some ds'*
and $d \in \text{connected-domains aid}$
shows $ds' \ d \ \text{aid} = \text{Some } s'$
proof –
from *assms(2,5,6)* **have**
 $ds' = (\lambda d \ \text{aid}'.$
if $\text{aid}' = \text{aid} \wedge d \in \text{connected-domains aid}$
then *Some* s'
else *domain-state d aid')*
unfolding *sync-all-def* **by** *auto*
with *assms(7)* **show** *?thesis* **by** *simp*
qed

`sync_all` does not affect other assets.

theorem *sync-isolation*:

assumes *sync-all source action aid domain-state = Some ds'*
and $\text{aid}' \neq \text{aid}$
shows $ds' \ d \ \text{aid}' = \text{domain-state d aid}'$
proof –
from *assms(1)* **obtain** $s \ s'$ **where**
domain-state source aid = Some s
transition s action = Some s'
unfolding *sync-all-def* **by** (*auto split: option.splits*)

```

then have  $ds' = (\lambda d \text{ aid}')$ 
  if  $\text{aid}' = \text{aid} \wedge d \in \text{connected-domains aid}$ 
  then Some s'
  else domain-state d aid'
  using assms(1) unfolding sync-all-def by auto
  with assms(2) show ?thesis by auto
qed

end

end

```

```

theory Regulatory-Instance
  imports State-Preservation
begin

```

6 Regulatory State and Action Definitions

```

datatype reg-state = ACTIVE | FROZEN | SEIZED | CONFISCATED | RE-
RESTRICTED

```

```

datatype reg-action = FREEZE | SEIZE | CONFISCATE | RESTRICT
  | UNFREEZE | UNRESTRICT | RELEASE

```

The seven constructors are the four escalation actions (*FREEZE*, *SEIZE*, *CONFISCATE*, *RESTRICT*) and their de-escalation counterparts (*UNFREEZE*, *UNRESTRICT*, *RELEASE*). RECOVER and LIQUIDATE, two of the product's six enforcement actions, are excluded from *reg-action*: they involve force transfers or external DEX interactions rather than regulatory state transitions, and are modelled at a different layer.

The regulatory transition function. Partial: returns None for invalid (state, action) combinations. The transition table encodes legal precedence: seizure is stronger than freezing, confiscation is terminal and universally reachable, and intermediate states must return to Active before lateral transitions.

```

fun reg-transition :: reg-state  $\Rightarrow$  reg-action  $\Rightarrow$  reg-state option where
  — From ACTIVE: all forward actions valid
  reg-transition ACTIVE FREEZE      = Some FROZEN
| reg-transition ACTIVE SEIZE       = Some SEIZED
| reg-transition ACTIVE CONFISCATE = Some CONFISCATED
| reg-transition ACTIVE RESTRICT  = Some RESTRICTED
| reg-transition ACTIVE UNFREEZE   = None
| reg-transition ACTIVE UNRESTRICT = None
| reg-transition ACTIVE RELEASE    = None
  — From FROZEN: unfreeze, escalate to SEIZED or CONFISCATED
| reg-transition FROZEN UNFREEZE   = Some ACTIVE

```

$\text{reg-transition FROZEN SEIZE} = \text{Some SEIZED}$
 $\text{reg-transition FROZEN CONFISCATE} = \text{Some CONFISCATED}$
 $\text{reg-transition FROZEN FREEZE} = \text{None}$
 $\text{reg-transition FROZEN RESTRICT} = \text{None}$
 $\text{reg-transition FROZEN UNRESTRICT} = \text{None}$
 $\text{reg-transition FROZEN RELEASE} = \text{None}$
 — From SEIZED: release or final confiscation only
 $\text{reg-transition SEIZED RELEASE} = \text{Some ACTIVE}$
 $\text{reg-transition SEIZED CONFISCATE} = \text{Some CONFISCATED}$
 $\text{reg-transition SEIZED FREEZE} = \text{None}$
 $\text{reg-transition SEIZED SEIZE} = \text{None}$
 $\text{reg-transition SEIZED RESTRICT} = \text{None}$
 $\text{reg-transition SEIZED UNFREEZE} = \text{None}$
 $\text{reg-transition SEIZED UNRESTRICT} = \text{None}$
 — From CONFISCATED: terminal no transitions out
 $\text{reg-transition CONFISCATED FREEZE} = \text{None}$
 $\text{reg-transition CONFISCATED SEIZE} = \text{None}$
 $\text{reg-transition CONFISCATED CONFISCATE} = \text{None}$
 $\text{reg-transition CONFISCATED RESTRICT} = \text{None}$
 $\text{reg-transition CONFISCATED UNFREEZE} = \text{None}$
 $\text{reg-transition CONFISCATED UNRESTRICT} = \text{None}$
 $\text{reg-transition CONFISCATED RELEASE} = \text{None}$
 — From RESTRICTED: unrestrict, escalate to FROZEN or CONFISCATED
 $\text{reg-transition RESTRICTED UNRESTRICT} = \text{Some ACTIVE}$
 $\text{reg-transition RESTRICTED FREEZE} = \text{Some FROZEN}$
 $\text{reg-transition RESTRICTED CONFISCATE} = \text{Some CONFISCATED}$
 $\text{reg-transition RESTRICTED RESTRICT} = \text{None}$
 $\text{reg-transition RESTRICTED SEIZE} = \text{None}$
 $\text{reg-transition RESTRICTED UNFREEZE} = \text{None}$
 $\text{reg-transition RESTRICTED RELEASE} = \text{None}$

7 Fundamental Properties of the Transition Function

7.1 Invariant I1: CONFISCATED is terminal

lemma *confiscated-terminal:*

$\text{reg-transition CONFISCATED } a = \text{None}$
 by (cases a) auto

7.2 Invariant I2: CONFISCATE reaches CONFISCATED from any non-terminal state

lemma *confiscate-universal:*

$s \neq \text{CONFISCATED} \implies \text{reg-transition } s \text{ CONFISCATE} = \text{Some CONFISCATED}$
 by (cases s) auto

7.3 Invariant I3: Determinism (inherent from function definition)

Determinism follows directly from the **fun** definition.

7.4 SEIZED to FROZEN exclusion

SEIZED is a strictly stronger constraint than FROZEN. Direct transition from SEIZED to FROZEN is legally nonsensical (cannot "weaken" a court-ordered seizure to a mere freeze). The path is: RELEASE then ACTIVE then FREEZE.

lemma *seized-no-freeze*:
 reg-transition SEIZED FREEZE = None
 by *simp*

lemma *seized-no-unfreeze*:
 reg-transition SEIZED UNFREEZE = None
 by *simp*

lemma *seized-no-restrict*:
 reg-transition SEIZED RESTRICT = None
 by *simp*

7.5 FROZEN to RESTRICTED exclusion

Must go through ACTIVE: UNFREEZE then ACTIVE then RESTRICT.

lemma *frozen-no-restrict*:
 reg-transition FROZEN RESTRICT = None
 by *simp*

7.6 Transition completeness: every non-terminal state has at least one valid action

lemma *non-terminal-has-action*:
 assumes $s \neq \text{CONFISCATED}$
 shows $\exists a. \text{reg-transition } s \ a \neq \text{None}$
proof (*cases s*)
 case *ACTIVE*
 show *?thesis*
 apply (*rule exI[where x=FREEZE]*)
 using *ACTIVE* **by** *simp*
 next
 case *FROZEN*
 show *?thesis*
 apply (*rule exI[where x=UNFREEZE]*)
 using *FROZEN* **by** *simp*
 next
 case *SEIZED*

```

  show ?thesis
  apply (rule exI[where x=RELEASE])
  using SEIZED by simp
next
  case CONFISCATED
  then show ?thesis using assms by contradiction
next
  case RESTRICTED
  show ?thesis
  apply (rule exI[where x=UNRESTRICT])
  using RESTRICTED by simp
qed

```

7.7 Reachability from ACTIVE

Every state is reachable from ACTIVE via some sequence of actions.

lemma *active-reaches-frozen: reg-transition ACTIVE FREEZE = Some FROZEN*
by simp

lemma *active-reaches-seized: reg-transition ACTIVE SEIZE = Some SEIZED*
by simp

lemma *active-reaches-confiscated: reg-transition ACTIVE CONFISCATE = Some CONFISCATED*
by simp

lemma *active-reaches-restricted: reg-transition ACTIVE RESTRICT = Some RESTRICTED*
by simp

7.8 Return paths to ACTIVE

lemma *frozen-returns: reg-transition FROZEN UNFREEZE = Some ACTIVE*
by simp

lemma *seized-returns: reg-transition SEIZED RELEASE = Some ACTIVE*
by simp

lemma *restricted-returns: reg-transition RESTRICTED UNRESTRICT = Some ACTIVE*
by simp

8 State Machine Instance

We show that `reg_state` / `reg_action` / `reg_transition` satisfy the `state_machine` locale assumptions.

definition *reg-states :: reg-state set where*
reg-states = {ACTIVE, FROZEN, SEIZED, CONFISCATED, RESTRICTED}

```

definition reg-actions :: reg-action set where
  reg-actions = {FREEZE, SEIZE, CONFISCATE, RESTRICT, UNFREEZE,
UNRESTRICT, RELEASE}

definition reg-terminal :: reg-state set where
  reg-terminal = {CONFISCATED}

lemma reg-states-UNIV: reg-states = UNIV
  unfolding reg-states-def by (auto, case-tac x, auto)

lemma reg-actions-UNIV: reg-actions = UNIV
  unfolding reg-actions-def by (auto, case-tac x, auto)

lemma reg-transition-closed:
   $\llbracket \text{reg-transition } s \ a = \text{Some } s' \rrbracket \implies s' \in \text{reg-states}$ 
  unfolding reg-states-def by (cases s; cases a; auto)

interpretation reg-sm: state-machine reg-states reg-actions reg-transition
reg-terminal
proof unfold-locales
  show finite reg-states unfolding reg-states-def by auto
next
  show finite reg-actions unfolding reg-actions-def by auto
next
  show reg-terminal  $\subseteq$  reg-states unfolding reg-terminal-def reg-states-def by auto
next
  fix s a
  assume s  $\in$  reg-terminal a  $\in$  reg-actions
  then show reg-transition s a = None
    unfolding reg-terminal-def by (auto simp: confiscated-terminal)
next
  fix s a s'
  assume s  $\in$  reg-states a  $\in$  reg-actions reg-transition s a = Some s'
  then show s'  $\in$  reg-states
    using reg-transition-closed by auto
next
  fix s :: reg-state and a :: reg-action
  assume s  $\notin$  reg-states
  then show reg-transition s a = None
    using reg-states-UNIV by auto
qed

```

9 Asset and Global State Modeling

```

type-synonym asset-id = nat
type-synonym chain-id = nat
type-synonym timestamp = nat

```

```
record asset-state =
  as-reg-state :: reg-state
```

```
type-synonym chain-state = asset-id  $\Rightarrow$  asset-state option
```

```
record global-state =
  gs-chains    :: chain-id  $\Rightarrow$  chain-state
  gs-locks    :: asset-id  $\Rightarrow$  bool
```

The synchronization message, reduced to the fields this model consumes: the regulatory action and its timestamp. The product's full message also carries routing and attribution data (asset, authority, source, targets); in this model routing lives at the global-state level (`connected_chains` and the target set of `update_all_chains`), and the D-quencer layer adds exactly the attribution fields its priority order consumes (see `dq_message`).

```
record oss-message =
  msg-action    :: reg-action
  msg-timestamp :: timestamp
```

10 State Accessors and Predicates

```
definition get-asset-state :: global-state  $\Rightarrow$  chain-id  $\Rightarrow$  asset-id  $\Rightarrow$  asset-state option where
  get-asset-state gs cid aid = gs-chains gs cid aid
```

```
definition get-reg-state :: global-state  $\Rightarrow$  chain-id  $\Rightarrow$  asset-id  $\Rightarrow$  reg-state option where
  get-reg-state gs cid aid =
    (case get-asset-state gs cid aid of
      None  $\Rightarrow$  None
    | Some ast  $\Rightarrow$  Some (as-reg-state ast))
```

```
definition asset-exists :: global-state  $\Rightarrow$  chain-id  $\Rightarrow$  asset-id  $\Rightarrow$  bool where
  asset-exists gs cid aid = (get-asset-state gs cid aid  $\neq$  None)
```

```
definition is-locked :: global-state  $\Rightarrow$  asset-id  $\Rightarrow$  bool where
  is-locked gs aid = gs-locks gs aid
```

Connected chains: all chains holding a given asset.

```
definition connected-chains :: global-state  $\Rightarrow$  asset-id  $\Rightarrow$  chain-id set where
  connected-chains gs aid = {cid. asset-exists gs cid aid}
```

11 Global State Validity

A global state is valid if all chains holding the same asset agree on its regulatory state. This is the cross-chain consistency invariant that synchronization must preserve.

definition *consistent-state* :: *global-state* $\Rightarrow$ *bool* **where**

consistent-state *gs* $\equiv$
 $\forall c1\ c2\ aid\ s1\ s2.$
 $get_reg_state\ gs\ c1\ aid = Some\ s1 \longrightarrow$
 $get_reg_state\ gs\ c2\ aid = Some\ s2 \longrightarrow$
 $s1 = s2$

definition *no-locked-without-reason* :: *global-state* $\Rightarrow$ *bool* **where**

no-locked-without-reason *gs* $\equiv$
 $\forall aid. \neg is_locked\ gs\ aid$
— Total absence of outstanding locks: in a quiescent valid state no asset is locked at all (locks are transient, held only inside a synchronization), not merely no unjustified lock.

definition *valid-state* :: *global-state* $\Rightarrow$ *bool* **where**

valid-state *gs* $\equiv consistent_state\ gs \wedge no_locked_without_reason\ gs$

12 Synchronization Protocol

12.1 Lock acquisition

definition *acquire-lock* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state option* **where**

acquire-lock *gs* *aid* =
(if *is-locked* *gs* *aid* then *None*
else *Some* (*gs* | *gs*-locks := (*gs*-locks *gs*)(*aid* := *True*) |))

definition *release-lock* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state* **where**

release-lock *gs* *aid* = *gs* | *gs*-locks := (*gs*-locks *gs*)(*aid* := *False*) |

12.2 State update across chains

Update all connected chains to a new regulatory state.

We define this directly rather than using `Finite_Set.fold`, because the update for a given asset on each chain is independent we simply construct a new global state where every chain's view of the asset is updated. This avoids the need for `comp_fun_commute` proofs.

definition *update-all-chains* ::

global-state $\Rightarrow$ *asset-id* $\Rightarrow$ *reg-state* $\Rightarrow$ *chain-id set* $\Rightarrow$ *global-state*

where

update-all-chains *gs* *aid* *new-st* *targets* =
gs | *gs*-chains := ($\lambda cid.$
if *cid* $\in$ *targets* then
($\lambda aid'.$ if *aid'* = *aid* then
(case *gs*-chains *gs* *cid* *aid* of
None $\Rightarrow$ *None*
| *Some* *ast* $\Rightarrow$ *Some* (*ast* | *as*-reg-state := *new-st* |))
else *gs*-chains *gs* *cid* *aid'*)
else *gs*-chains *gs* *cid*) |

12.3 The synchronization function

`sync`: the core synchronization operation.

Given a source chain, action, and `asset_id`: 1. Verify the asset exists on the source chain 2. Check the transition is valid 3. Acquire global lock (fails if already locked) 4. Update all connected chains to the new state 5. Release lock

Returns `None` if any step fails (invalid transition, asset already locked, etc.)

definition `sync` ::

chain-id $\Rightarrow$ *reg-action* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state* $\Rightarrow$ *global-state option*

where

sync source action aid gs =
 (case *get-reg-state gs source aid* of
 None $\Rightarrow$ *None*
 | *Some current-st* $\Rightarrow$
 (case *reg-transition current-st action* of
 None $\Rightarrow$ *None*
 | *Some new-st* $\Rightarrow$
 (case *acquire-lock gs aid* of
 None $\Rightarrow$ *None*
 | *Some gs-locked* $\Rightarrow$
 let *targets* = *connected-chains gs aid* in
 let *gs-updated* = *update-all-chains gs-locked aid new-st targets* in
 Some (release-lock gs-updated aid)))))

13 Synchronization Properties

13.1 No self-loops

The transition table has no self-loops: a successful transition always changes the regulatory state, so a successful synchronization never rewrites an asset to the state it already carries.

lemma *no-self-loops*:

reg-transition s a = *Some s* $\implies$ *False*
by (*cases s*; *cases a*; *auto*)

lemma *transition-changes-state*:

reg-transition s a = *Some s'* $\implies$ *s* $\neq$ *s'*
using *no-self-loops* **by** *auto*

13.2 Properties of `update_all_chains`

`update_all_chains` correctly updates any chain in the target set.

lemma *update-all-chains-in-targets*:

assumes *cid* $\in$ *targets*
and *gs-chains gs cid aid* = *Some ast*

shows $gs-chains$ ($update-all-chains$ gs aid $new-st$ $targets$) cid $aid =$
 $Some$ ($ast \mid as-reg-state := new-st \mid$)
unfolding $update-all-chains-def$ **using** $assms$ **by** $auto$

lemma $update-all-chains-outside-targets$:
assumes $cid \notin targets$
shows $gs-chains$ ($update-all-chains$ gs aid $new-st$ $targets$) $cid = gs-chains$ gs cid
unfolding $update-all-chains-def$ **using** $assms$ **by** $auto$

lemma $update-all-chains-other-asset$:
assumes $aid' \neq aid$
shows $gs-chains$ ($update-all-chains$ gs aid $new-st$ $targets$) cid $aid' =$
 $gs-chains$ gs cid aid'
unfolding $update-all-chains-def$ **using** $assms$ **by** $auto$

lemma $update-all-chains-locks$:
 $gs-locks$ ($update-all-chains$ gs aid $new-st$ $targets$) = $gs-locks$ gs
unfolding $update-all-chains-def$ **by** $auto$

Key lemma: get_reg_state after $update_all_chains$ for a chain in targets.

lemma $update-all-chains-reg-state$:
assumes $cid \in targets$
and $get-asset-state$ gs cid $aid = Some$ ast
shows $get-reg-state$ ($update-all-chains$ gs aid $new-st$ $targets$) cid $aid = Some$ $new-st$
proof –
from $assms(2)$ **have** $chain$: $gs-chains$ gs cid $aid = Some$ ast
unfolding $get-asset-state-def$ **by** $simp$
then have $gs-chains$ ($update-all-chains$ gs aid $new-st$ $targets$) cid $aid =$
 $Some$ ($ast \mid as-reg-state := new-st \mid$)
using $update-all-chains-in-targets[OF assms(1)]$ **by** $simp$
then show $?thesis$
unfolding $get-reg-state-def$ $get-asset-state-def$ **by** $simp$
qed

13.3 Lock mutual exclusion

If a lock is held, a further lock acquisition on the same asset fails ($acquire_lock$ returns $None$). In the atomic model this guard expresses the intended mutual exclusion of regulatory actions on an asset; there is no concurrent execution in the model to serialise.

lemma $lock-acquire-success$:
assumes $\neg is-locked$ gs aid
shows $\exists gs'. acquire-lock$ gs $aid = Some$ $gs' \wedge is-locked$ gs' aid
proof –
from $assms$ **have** $\neg gs-locks$ gs aid
unfolding $is-locked-def$ **by** $simp$

```

then have acquire-lock gs aid = Some (gs | gs-locks := (gs-locks gs)(aid := True)
|)
  unfolding acquire-lock-def is-locked-def by simp
  moreover have is-locked (gs | gs-locks := (gs-locks gs)(aid := True) |) aid
  unfolding is-locked-def by simp
  ultimately show ?thesis by auto
qed

```

14 Main Theorem: Cross-Chain Regulatory Consistency

The central theorem of this theory. If a regulatory action is applied via *sync* — from an arbitrary global state — every connected chain reflects the new regulatory state; under a valid input state, validity is preserved as well (*valid_state_preservation* below).

This is the regulatory instantiation of the cross-domain consistency theorem from *State_Preservation.thy*.

The proof proceeds by: 1. Unfolding *sync* into *lock*, then *update_all_chains*, then *unlock* 2. Showing *acquire_lock* and *release_lock* preserve chain state 3. Showing *update_all_chains* updates every chain in the target set 4. Combining these facts to establish the conclusion

Auxiliary: *release_lock* does not change chains.

```

lemma release-lock-chains:
  gs-chains (release-lock gs aid) = gs-chains gs
  unfolding release-lock-def by simp

lemma release-lock-reg-state:
  get-reg-state (release-lock gs aid) cid aid' = get-reg-state gs cid aid'
  unfolding get-reg-state-def get-asset-state-def release-lock-chains by simp

```

Auxiliary: *acquire_lock* does not change chains.

```

lemma acquire-lock-chains:
  assumes acquire-lock gs aid = Some gs'
  shows gs-chains gs' = gs-chains gs
  using assms unfolding acquire-lock-def is-locked-def
  by (auto split: if-splits)

```

Auxiliary: *connected_chains* uses original *gs*, not locked *gs*.

The cross-chain regulatory homomorphism theorem.

```

theorem regulatory-homomorphism:
  assumes current: get-reg-state gs source aid = Some s
  and trans: reg-transition s action = Some s'
  and synced: sync source action aid gs = Some gs'
  and connected: c ∈ connected-chains gs aid
  shows get-reg-state gs' c aid = Some s'

```

proof –

— Step 1: Unfold sync to extract intermediate states
from *synced current trans*
obtain *gs-locked* **where**
 lock: *acquire-lock gs aid = Some gs-locked*
and *gs'-def*: *gs' = release-lock*
 (*update-all-chains gs-locked aid s' (connected-chains gs aid)*) *aid*
unfolding *sync-def*
by (*auto split: option.splits simp: Let-def*)

— Step 2: **acquire_lock** preserves chains
have *locked-chains*: *gs-chains gs-locked = gs-chains gs*
using *acquire-lock-chains[OF lock]* **by** *simp*

— Step 3: *c* is in **connected_chains**, so the asset exists on *c*
from *connected* **have** *asset-exists gs c aid*
unfolding *connected-chains-def* **by** *simp*
then obtain *ast-c* **where** *ast-c*: *gs-chains gs c aid = Some ast-c*
unfolding *asset-exists-def get-asset-state-def* **by** *auto*

— Step 4: Therefore asset also exists in **gs_locked** on chain *c*
have *ast-locked*: *gs-chains gs-locked c aid = Some ast-c*
using *ast-c locked-chains* **by** *simp*

— Step 5: **update_all_chains** updates chain *c*
have *updated*: *gs-chains (update-all-chains gs-locked aid s' (connected-chains gs aid)) c aid =*
 Some (ast-c | as-reg-state := s' |)
using *update-all-chains-in-targets[OF connected ast-locked]* **by** *simp*

— Step 6: **release_lock** does not change chains
have *gs-chains gs' c aid = Some (ast-c | as-reg-state := s' |)*
using *updated unfolding gs'-def release-lock-chains* **by** *simp*

— Step 7: Extract **reg_state**
then show *?thesis*
unfolding *get-reg-state-def get-asset-state-def* **by** *simp*
qed

15 Valid State Preservation

The central safety property: synchronization preserves global state validity.

If the global state is valid (consistent and unlocked) before sync, and sync succeeds, then the resulting global state is also valid.

This closes the inductive invariant: valid initial state + **valid_state** preservation under sync = **valid_state** holds at all reachable states.

The proof decomposes into two parts: (1) **consistent_state** preservation: sync updates all connected chains to the same new state and leaves

other assets untouched. (2) `no_locked_without_reason` preservation: `sync` acquires and then releases the lock within the same operation, so the output has no outstanding locks (assuming no other locks were held).

15.1 Part 1: `consistent_state` preservation

After `sync`, any two chains holding asset `aid` agree on its new regulatory state (it is `s'`), and any two chains holding a different asset `aid'` still agree (their states are unchanged from the consistent input).

lemma *sync-preserves-consistent-state*:

assumes *valid*: *valid-state* *gs*

and *current*: *get-reg-state* *gs* *source* *aid* = *Some* *s*

and *trans*: *reg-transition* *s* *action* = *Some* *s'*

and *synced*: *sync* *source* *action* *aid* *gs* = *Some* *gs'*

shows *consistent-state* *gs'*

proof —

— Unfold `sync` to extract the structure

from *synced* *current* *trans*

obtain *gs-locked* **where**

lock: *acquire-lock* *gs* *aid* = *Some* *gs-locked*

and *gs'-def*: *gs'* = *release-lock*

(*update-all-chains* *gs-locked* *aid* *s'* (*connected-chains* *gs* *aid*)) *aid*

unfolding *sync-def*

by (*auto* *split*: *option.splits* *simp*: *Let-def*)

have *locked-chains*: *gs-chains* *gs-locked* = *gs-chains* *gs*

using *acquire-lock-chains*[*OF* *lock*] **by** *simp*

show *?thesis*

unfolding *consistent-state-def*

proof (*intro* *allI* *impI*)

fix *c1* *c2* *aid'* *s1-final* *s2-final*

assume *s1-eq*: *get-reg-state* *gs'* *c1* *aid'* = *Some* *s1-final*

and *s2-eq*: *get-reg-state* *gs'* *c2* *aid'* = *Some* *s2-final*

show *s1-final* = *s2-final*

proof (*cases* *aid'* = *aid*)

case *True*

— Case: the synchronized asset. Two sub-cases per chain.

show *?thesis*

proof (*cases* *c1* ∈ *connected-chains* *gs* *aid*)

case *c1-in*: *True*

then have *ae1*: *asset-exists* *gs* *c1* *aid*

unfolding *connected-chains-def* **by** *simp*

then obtain *ast1* **where** *ast1*: *gs-chains* *gs* *c1* *aid* = *Some* *ast1*

unfolding *asset-exists-def* *get-asset-state-def* **by** *auto*

have *ast1-locked*: *gs-chains* *gs-locked* *c1* *aid* = *Some* *ast1*

using *ast1* *locked-chains* **by** *simp*

```

    have upd1: gs-chains (update-all-chains gs-locked aid s' (connected-chains
gs aid)) c1 aid =
      Some (ast1 (as-reg-state := s' ))
    using update-all-chains-in-targets[OF c1-in ast1-locked] by simp
  have reg1: get-reg-state gs' c1 aid = Some s'
    unfolding gs'-def release-lock-chains get-reg-state-def get-asset-state-def
    using upd1 by simp
  show ?thesis
  proof (cases c2 ∈ connected-chains gs aid)
    case c2-in: True
    then have ae2: asset-exists gs c2 aid
      unfolding connected-chains-def by simp
    then obtain ast2 where ast2: gs-chains gs c2 aid = Some ast2
      unfolding asset-exists-def get-asset-state-def by auto
    have ast2-locked: gs-chains gs-locked c2 aid = Some ast2
      using ast2 locked-chains by simp
    have reg2: get-reg-state gs' c2 aid = Some s'
      unfolding gs'-def release-lock-chains get-reg-state-def get-asset-state-def
      using update-all-chains-in-targets[OF c2-in ast2-locked] by simp
    from reg1 reg2 s1-eq s2-eq True show ?thesis by simp
  next
    case c2-out: False
    — c2 is not connected, so asset doesn't exist on c2 for aid
    have gs-chains (update-all-chains gs-locked aid s' (connected-chains gs
aid)) c2 =
      gs-chains gs-locked c2
      using update-all-chains-outside-targets[OF c2-out] by simp
    then have gs-chains gs' c2 aid = gs-chains gs c2 aid
      unfolding gs'-def release-lock-chains using locked-chains by simp
    moreover have gs-chains gs c2 aid = None
      using c2-out unfolding connected-chains-def asset-exists-def
get-asset-state-def
      by auto
    ultimately have get-reg-state gs' c2 aid = None
      unfolding get-reg-state-def get-asset-state-def by simp
    with s2-eq True show ?thesis by simp
  qed
next
  case c1-out: False
  — c1 not connected: asset doesn't exist there
  have gs-chains (update-all-chains gs-locked aid s' (connected-chains gs aid))
c1 =
    gs-chains gs-locked c1
    using update-all-chains-outside-targets[OF c1-out] by simp
  then have gs-chains gs' c1 aid = gs-chains gs c1 aid
    unfolding gs'-def release-lock-chains using locked-chains by simp
  moreover have gs-chains gs c1 aid = None
    using c1-out unfolding connected-chains-def asset-exists-def
get-asset-state-def

```

```

      by auto
    ultimately have get-reg-state gs' c1 aid = None
      unfolding get-reg-state-def get-asset-state-def by simp
    with s1-eq True show ?thesis by simp
  qed
next
case diff: False
  — Case: a different asset. Sync does not touch it.
  have other1: gs-chains (update-all-chains gs-locked aid s' (connected-chains
    gs aid)) c1 aid' =
    gs-chains gs-locked c1 aid'
    using update-all-chains-other-asset[OF diff] by simp
  have other2: gs-chains (update-all-chains gs-locked aid s' (connected-chains
    gs aid)) c2 aid' =
    gs-chains gs-locked c2 aid'
    using update-all-chains-other-asset[OF diff] by simp
  have gs-chains gs' c1 aid' = gs-chains gs c1 aid'
    unfolding gs'-def release-lock-chains using other1 locked-chains by simp
  moreover have gs-chains gs' c2 aid' = gs-chains gs c2 aid'
    unfolding gs'-def release-lock-chains using other2 locked-chains by simp
  ultimately have get-reg-state gs' c1 aid' = get-reg-state gs c1 aid'
    and get-reg-state gs' c2 aid' = get-reg-state gs c2 aid'
    unfolding get-reg-state-def get-asset-state-def by simp-all
  with s1-eq s2-eq have
    get-reg-state gs c1 aid' = Some s1-final
    get-reg-state gs c2 aid' = Some s2-final
    by simp-all
  with valid show ?thesis
    unfolding valid-state-def consistent-state-def by blast
  qed
qed
qed

```

15.2 Part 2: no_locked_without_reason preservation

After sync, no assets are locked. The sync operation acquires a lock on the target asset and releases it before returning. For all other assets, the lock state is unchanged (and was False by `valid_state`).

lemma *sync-preserves-no-locks*:

```

assumes valid: valid-state gs
  and current: get-reg-state gs source aid = Some s
  and trans: reg-transition s action = Some s'
  and synced: sync source action aid gs = Some gs'
shows no-locked-without-reason gs'

```

proof —

```

from synced current trans
obtain gs-locked where
  lock: acquire-lock gs aid = Some gs-locked
  and gs'-def: gs' = release-lock

```

```

    (update-all-chains gs-locked aid s' (connected-chains gs aid)) aid
  unfolding sync-def
  by (auto split: option.splits simp: Let-def)

— Locks after acquire: only aid is locked
from lock have locked-locks: gs-locks gs-locked = (gs-locks gs)(aid := True)
  unfolding acquire-lock-def is-locked-def
  by (auto split: if-splits)

— update_all_chains does not change locks
have upd-locks: gs-locks (update-all-chains gs-locked aid s' (connected-chains gs
aid)) =
  gs-locks gs-locked
  using update-all-chains-locks by simp

— release_lock clears the lock on aid
have final-locks: gs-locks gs' = (gs-locks gs-locked)(aid := False)
  unfolding gs'-def release-lock-def using upd-locks by simp

show ?thesis
unfolding no-locked-without-reason-def is-locked-def
proof
  fix aid'
  show ¬ gs-locks gs' aid'
  proof (cases aid' = aid)
    case True
    then show ?thesis using final-locks by simp
  next
    case False
    then have gs-locks gs' aid' = gs-locks gs-locked aid'
      using final-locks by simp
    also have ... = gs-locks gs aid'
      using locked-locks False by simp
    also have ... = False
      using valid unfolding valid-state-def no-locked-without-reason-def
is-locked-def
      by simp
    finally show ?thesis by simp
  qed
qed
qed

```

15.3 The valid_state preservation theorem

Combining the two parts: sync preserves `valid_state`. This is the inductive invariant that guarantees `valid_state` holds at every reachable global state.

theorem *valid-state-preservation:*

assumes *valid:* `valid-state gs`
and *current:* `get-reg-state gs source aid = Some s`

```

and trans: reg-transition s action = Some s'
and synced: sync source action aid gs = Some gs'
shows valid-state gs'
unfolding valid-state-def
using sync-preserves-consistent-state[OF assms]
       sync-preserves-no-locks[OF assms]
by auto

```

16 Heterogeneous-Action State Preservation Instance

This section instantiates the `state_preservation` locale with a concrete heterogeneous-action scenario between two chains.

The scenario models the following operational situation. Two chains A and B share the same regulatory state space (ACTIVE, FROZEN, SEIZED, CONFISCATED, RESTRICTED), but their on-chain action vocabularies differ. Chain A supports the full seven-action set used elsewhere in this theory. Chain B is a chain whose jurisdiction handles de-escalation (UN-FREEZE, UNRESTRICT, RELEASE) exclusively through separate judicial procedures rather than as on-chain actions; correspondingly, Chain B's on-chain action vocabulary is the four-action escalation subset only. The synchronisation map between the two chains is therefore non-trivial in two ways:

- The source action type is `reg_action` and the target action type is a separate datatype `chain_b_action` with four constructors. This exercises the locale's heterogeneous source / target action types ('a vs. 'b).
- The locale parameter `actions\<^sub>s` is instantiated with a strict subset of the full `reg_action` set, namely the four escalation actions. De-escalation actions exist in `reg_action` as a datatype but are out of scope for this synchronisation instance.

The naturality assumption then has to hold only on the escalation subset, which is exactly the design intent of the locale's `actions\<^sub>s` parameter: the user can scope structural preservation to a subset of source actions rather than to the full source action type.

```

datatype chain-b-action = B-FREEZE | B-SEIZE | B-CONFISCATE |
B-RESTRICT

```

The escalation subset of `reg_action` that this instance synchronises across to Chain B.

definition *escalation-actions* :: *reg-action set* **where**
escalation-actions = {*FREEZE*, *SEIZE*, *CONFISCATE*, *RESTRICT*}

definition *chain-b-actions* :: *chain-b-action set* **where**
chain-b-actions = {*B-FREEZE*, *B-SEIZE*, *B-CONFISCATE*, *B-RESTRICT*}

Chain B's transition function. Its values mirror Chain A's transition function on the four escalation actions: when Chain A maps (*s*, *escalation action*) to *Some s'*, Chain B does the same; when Chain A rejects, Chain B rejects. Both chains share the regulatory state space, so no state translation is required (the state map is the identity).

fun *chain-b-transition* :: *reg-state* $\Rightarrow$ *chain-b-action* $\Rightarrow$ *reg-state option* **where**
chain-b-transition *s* *B-FREEZE* = *reg-transition* *s* *FREEZE*
| *chain-b-transition* *s* *B-SEIZE* = *reg-transition* *s* *SEIZE*
| *chain-b-transition* *s* *B-CONFISCATE* = *reg-transition* *s* *CONFISCATE*
| *chain-b-transition* *s* *B-RESTRICT* = *reg-transition* *s* *RESTRICT*

The escalation action map: a 1-to-1 correspondence between Chain A's four escalation actions and Chain B's four constructors.

fun *escalation-action-map* :: *reg-action* $\Rightarrow$ *chain-b-action* **where**
escalation-action-map *FREEZE* = *B-FREEZE*
| *escalation-action-map* *SEIZE* = *B-SEIZE*
| *escalation-action-map* *CONFISCATE* = *B-CONFISCATE*
| *escalation-action-map* *RESTRICT* = *B-RESTRICT*
| *escalation-action-map* *UNFREEZE* = *B-FREEZE*
| *escalation-action-map* *UNRESTRICT* = *B-FREEZE*
| *escalation-action-map* *RELEASE* = *B-FREEZE*

— The last three clauses are unused by the locale instance, since *actions\<sub>s* = *escalation_actions* excludes the de-escalation actions. They are present only to make the function total on *reg_action*.

Chain B is also a state machine over the same regulatory state space.

lemma *chain-b-transition-closed*:

chain-b-transition *s* *a* = *Some s'* $\implies$ *s'* $\in$ *reg-states*

proof —

assume *chain-b-transition* *s* *a* = *Some s'*

then have $\exists a'. \text{reg-transition } s \ a' = \text{Some } s'$

by (*cases a*) *auto*

then show *s'* $\in$ *reg-states* **using** *reg-transition-closed* **by** *auto*

qed

lemma *chain-b-terminal*:

s $\in$ *reg-terminal* $\implies$ *chain-b-transition* *s* *a* = *None*

unfolding *reg-terminal-def*

by (*cases a*) (*auto simp: confiscated-terminal*)

lemma *chain-b-state-machine*:

state-machine *reg-states* *chain-b-actions* *chain-b-transition* *reg-terminal*

proof —

have *f1*: *finite* *reg-states* **unfolding** *reg-states-def* **by** *auto*

have *f2*: *finite* *chain-b-actions* **unfolding** *chain-b-actions-def* **by** *auto*

```

have f3: reg-terminal  $\subseteq$  reg-states
  unfolding reg-terminal-def reg-states-def by auto
have f4:  $\bigwedge s\ a. s \in \text{reg-terminal} \implies a \in \text{chain-b-actions} \implies \text{chain-b-transition}$ 
s a = None
  using chain-b-terminal by auto
have f5:  $\bigwedge s\ a\ s'. s \in \text{reg-states} \implies a \in \text{chain-b-actions} \implies$ 
chain-b-transition s a = Some s'  $\implies$  s'  $\in$  reg-states
  using chain-b-transition-closed by auto
have f6:  $\bigwedge s\ a. (s :: \text{reg-state}) \notin \text{reg-states} \implies \text{chain-b-transition s a} = \text{None}$ 
  using reg-states-UNIV by auto
show ?thesis
  by (intro state-machine.intro) (use f1 f2 f3 f4 f5 f6 in auto)
qed

```

The naturality conditions of `state_preservation` hold for the escalation subset. Both directions (`Some` and `None` branches) follow by case analysis on the escalation action: by construction, Chain B's transition function on each `B_X` constructor mirrors Chain A's transition function on the corresponding `X` action.

```

lemma escalation-naturality-some:
  assumes a  $\in$  escalation-actions
  and reg-transition s a = Some s'
  shows chain-b-transition s (escalation-action-map a) = Some s'
proof -
  from assms(1) have a  $\in$  {FREEZE, SEIZE, CONFISCATE, RESTRICT}
  unfolding escalation-actions-def by simp
  then consider a = FREEZE | a = SEIZE | a = CONFISCATE | a = RE-
STRICT
  by auto
  then show ?thesis using assms(2) by (cases) auto
qed

```

```

lemma escalation-naturality-none:
  assumes a  $\in$  escalation-actions
  and reg-transition s a = None
  shows chain-b-transition s (escalation-action-map a) = None
proof -
  from assms(1) have a  $\in$  {FREEZE, SEIZE, CONFISCATE, RESTRICT}
  unfolding escalation-actions-def by simp
  then consider a = FREEZE | a = SEIZE | a = CONFISCATE | a = RE-
STRICT
  by auto
  then show ?thesis using assms(2) by (cases) auto
qed

```

The escalation instance: `state_preservation` with `actions\<sub>s` the escalation subset of `reg_action`, target action type `chain_b_action`, and identity state map (both chains share the regulatory state space).

Source and target state-machine bundles for the heterogeneous-action in-

stance. The target side reuses the existing `chain_b_state_machine` lemma; the source side is on the escalation subset of `reg_actions` rather than the full action set, so we discharge it inline.

```

lemma escalation-source-state-machine:
  state-machine reg-states escalation-actions reg-transition reg-terminal
proof unfold-locales
  show finite reg-states unfolding reg-states-def by auto
next
  show finite escalation-actions unfolding escalation-actions-def by auto
next
  show reg-terminal  $\subseteq$  reg-states unfolding reg-terminal-def reg-states-def by auto
next
  fix s a
  assume s  $\in$  reg-terminal a  $\in$  escalation-actions
  then show reg-transition s a = None
    unfolding reg-terminal-def by (auto simp: confiscated-terminal)
next
  fix s a s'
  assume s  $\in$  reg-states a  $\in$  escalation-actions reg-transition s a = Some s'
  then show s'  $\in$  reg-states using reg-transition-closed by auto
next
  fix s :: reg-state and a :: reg-action
  assume s  $\notin$  reg-states
  then show reg-transition s a = None using reg-states-UNIV by auto
qed

```

Heterogeneous-action instance: the source action set is the `escalation_actions` subset of `reg_action`, the target action set is `chain_b_actions`, and the state map is the identity. Both source and target state machines share the regulatory state space, so two of the state-machine obligations (`finite reg_states` and `reg_terminal \<subsepeq> reg_states`) collapse between source and target; the proof structure below reflects this by issuing each `show` exactly once for the merged obligation.

```

interpretation escalation-preservation:
  state-preservation
  reg-states escalation-actions reg-transition reg-terminal
  reg-states chain-b-actions chain-b-transition reg-terminal
  id :: reg-state  $\Rightarrow$  reg-state escalation-action-map
proof unfold-locales
  show finite reg-states unfolding reg-states-def by auto
next
  show finite escalation-actions unfolding escalation-actions-def by auto
next
  show reg-terminal  $\subseteq$  reg-states unfolding reg-terminal-def reg-states-def by auto
next
  fix s a
  assume s  $\in$  reg-terminal a  $\in$  escalation-actions
  then show reg-transition s a = None

```

```

    unfolding reg-terminal-def by (auto simp: confiscated-terminal)
next
  fix s a s'
  assume s ∈ reg-states a ∈ escalation-actions reg-transition s a = Some s'
  then show s' ∈ reg-states using reg-transition-closed by auto
next
  fix s :: reg-state and a :: reg-action
  assume s ∉ reg-states
  then show reg-transition s a = None using reg-states-UNIV by auto
next
  show finite chain-b-actions unfolding chain-b-actions-def by auto
next
  fix s a
  assume s ∈ reg-terminal a ∈ chain-b-actions
  then show chain-b-transition s a = None using chain-b-terminal by auto
next
  fix s a s'
  assume s ∈ reg-states a ∈ chain-b-actions chain-b-transition s a = Some s'
  then show s' ∈ reg-states using chain-b-transition-closed by auto
next
  fix s :: reg-state and a :: chain-b-action
  assume s ∉ reg-states
  then show chain-b-transition s a = None using reg-states-UNIV by auto
next
  fix s
  assume s ∈ reg-states
  then show id s ∈ reg-states by simp
next
  fix a
  assume a ∈ escalation-actions
  then show escalation-action-map a ∈ chain-b-actions
    unfolding escalation-actions-def chain-b-actions-def by auto
next
  fix s
  assume s ∈ reg-terminal
  then show id s ∈ reg-terminal by simp
next
  fix s a s'
  assume s ∈ reg-states a ∈ escalation-actions reg-transition s a = Some s'
  then show chain-b-transition (id s) (escalation-action-map a) = Some (id s')
    using escalation-naturality-some by simp
next
  fix s a
  assume s ∈ reg-states a ∈ escalation-actions reg-transition s a = None
  then show chain-b-transition (id s) (escalation-action-map a) = None
    using escalation-naturality-none by simp
qed

```

As a corollary of the locale interpretation, sequential preservation (the locale's main theorem) applies to escalation action sequences: any valid se-

quence of escalation actions on Chain A produces, when mapped through `escalation_action_map`, a valid sequence on Chain B ending in the same regulatory state.

corollary *escalation-sequential-preservation*:

```

assumes  $s \in \text{reg-states}$ 
and  $\forall a \in \text{set } as. a \in \text{escalation-actions}$ 
and  $\text{escalation-preservation.source.apply-actions } s \text{ } as = \text{Some } s'$ 
shows  $\text{escalation-preservation.target.apply-actions } s \text{ (map escalation-action-map } as)$ 
     $= \text{Some } s'$ 
using  $\text{assms escalation-preservation.sequential-preservation [where } as = as]$ 
by simp

```

17 Layer-Crossing Symmetric State Preservation Instance

This section instantiates the `symmetric_state_preservation` locale with a concrete bidirectional binding between two representations of the same regulatory state.

The on-chain representation is the five-element `reg_state` enum used throughout this theory. The off-chain representation is a structured permission record (`daml_perm`) modelling a DAML party-permission ledger entry: a status tag plus auxiliary fields that carry the party / scope metadata associated with a non-trivial regulatory status (the seizing party for `SEIZED`, the restriction scope for `RESTRICTED`). The identifier values themselves are representative placeholders; what the invariant tracks is their presence, and the layer-crossing content lives in the status tag. A type-level invariant (`valid_daml_perm`) ties the auxiliary fields to the status tag; the bijection domain itself is the image of the representation map (`daml_states`), every element of which satisfies the invariant.

The action vocabularies on the two layers coincide: both layers process the seven actions of `reg_action`, and the action map is the identity. All of the non-trivial content of the symmetric instance therefore lives in the layer-crossing state mapping. The roundtrip assumptions of `symmetric_state_preservation` become a pair of bijection conditions between `reg_state` and the image `daml_states` of the representation map.

```

datatype daml-status-tag =
  D-Active | D-Frozen | D-Seized | D-Confiscated | D-Restricted

```

```

record daml-perm =
  status-tag      :: daml-status-tag
  seized-by      :: nat option
  restriction-scope :: nat option

```

```

definition default-party-id :: nat where

```

$default-party-id = 0$

definition $default-scope-id :: nat$ **where**
 $default-scope-id = 0$

The well-formedness invariant: the auxiliary fields are non-None exactly on the status tags that semantically require them.

definition $valid-daml-perm :: daml-perm \Rightarrow bool$ **where**
 $valid-daml-perm\ p \longleftrightarrow$
 $(status-tag\ p = D-Seized \longleftrightarrow seized-by\ p \neq None) \wedge$
 $(status-tag\ p = D-Restricted \longleftrightarrow restriction-scope\ p \neq None)$

The two state maps between the two representations.

fun $reg-to-daml :: reg-state \Rightarrow daml-perm$ **where**
 $reg-to-daml\ ACTIVE = (\mid status-tag = D-Active,$
 $seized-by = None,$
 $restriction-scope = None \mid)$
 $\mid reg-to-daml\ FROZEN = (\mid status-tag = D-Frozen,$
 $seized-by = None,$
 $restriction-scope = None \mid)$
 $\mid reg-to-daml\ SEIZED = (\mid status-tag = D-Seized,$
 $seized-by = Some\ default-party-id,$
 $restriction-scope = None \mid)$
 $\mid reg-to-daml\ CONFISCATED = (\mid status-tag = D-Confiscated,$
 $seized-by = None,$
 $restriction-scope = None \mid)$
 $\mid reg-to-daml\ RESTRICTED = (\mid status-tag = D-Restricted,$
 $seized-by = None,$
 $restriction-scope = Some\ default-scope-id \mid)$

fun $daml-to-reg :: daml-perm \Rightarrow reg-state$ **where**
 $daml-to-reg\ p = (case\ status-tag\ p\ of$
 $D-Active \Rightarrow ACTIVE$
 $\mid D-Frozen \Rightarrow FROZEN$
 $\mid D-Seized \Rightarrow SEIZED$
 $\mid D-Confiscated \Rightarrow CONFISCATED$
 $\mid D-Restricted \Rightarrow RESTRICTED)$

The DAML target state space: the image of reg_to_daml .

definition $daml-states :: daml-perm\ set$ **where**
 $daml-states = range\ reg-to-daml$

definition $daml-terminal :: daml-perm\ set$ **where**
 $daml-terminal = \{reg-to-daml\ CONFISCATED\}$

The DAML transition function: lifts $reg_transition$ through the representation map. By construction, $daml_transition$ respects the reg_to_daml map.

definition $daml-transition :: daml-perm \Rightarrow reg-action \Rightarrow daml-perm\ option$ **where**

```

daml-transition p a =
  (if p ∈ daml-states then
    (case reg-transition (daml-to-reg p) a of
      None    ⇒ None
    | Some s' ⇒ Some (reg-to-daml s'))
  else None)

```

Roundtrip lemmas establishing the layer-crossing bijection.

```

lemma daml-to-reg-to-daml-id:
  daml-to-reg (reg-to-daml s) = s
  by (cases s) auto

```

```

lemma reg-to-daml-to-reg-on-image:
  assumes p ∈ daml-states
  shows reg-to-daml (daml-to-reg p) = p
proof -
  from assms obtain s where p = reg-to-daml s
  unfolding daml-states-def by auto
  then show ?thesis using daml-to-reg-to-daml-id by simp
qed

```

```

lemma reg-to-daml-valid:
  valid-daml-perm (reg-to-daml s)
  unfolding valid-daml-perm-def by (cases s) auto

```

The DAML side is a state machine on the image of `reg_to_daml`.

```

lemma daml-states-finite: finite daml-states
proof -
  have fin-reg: finite reg-states unfolding reg-states-def by auto
  hence fin-img: finite (reg-to-daml ' reg-states) by (rule finite-imageI)
  have eq: daml-states = reg-to-daml ' reg-states
  unfolding daml-states-def using reg-states-UNIV by simp
  from fin-img eq show ?thesis by simp
qed

```

```

lemma daml-terminal-subset: daml-terminal ⊆ daml-states
  unfolding daml-terminal-def daml-states-def by blast

```

```

lemma daml-terminal-absorbing:
  assumes p ∈ daml-terminal and a ∈ reg-actions
  shows daml-transition p a = None
proof -
  from assms(1) have p = reg-to-daml CONFISCATED
  unfolding daml-terminal-def by simp
  then have daml-to-reg p = CONFISCATED using daml-to-reg-to-daml-id by
simp
  moreover have p ∈ daml-states
  using assms(1) daml-terminal-subset by auto
  ultimately show ?thesis

```

unfolding *daml-transition-def* using *confiscated-terminal* by *simp*
qed

lemma *daml-transition-closed*:
 assumes $p \in \text{daml-states}$ and $a \in \text{reg-actions}$
 and $\text{daml-transition } p \ a = \text{Some } p'$
 shows $p' \in \text{daml-states}$
proof –
 from *assms*(1,3) obtain s' where
 $s'\text{-eq}$: $\text{reg-transition } (\text{daml-to-reg } p) \ a = \text{Some } s'$
 and $p'\text{-def}$: $p' = \text{reg-to-daml } s'$
 unfolding *daml-transition-def*
 by (*auto split: option.splits if-splits*)
 then show *?thesis*
 unfolding *daml-states-def* by *auto*
qed

lemma *daml-transition-outside-states*:
 $p \notin \text{daml-states} \implies \text{daml-transition } p \ a = \text{None}$
 unfolding *daml-transition-def* by *simp*

lemma *daml-state-machine*:
state-machine daml-states reg-actions daml-transition daml-terminal
proof
 show *finite daml-states* by (*rule daml-states-finite*)
next
 show *finite reg-actions* unfolding *reg-actions-def* by *auto*
next
 show $\text{daml-terminal} \subseteq \text{daml-states}$ by (*rule daml-terminal-subset*)
next
 fix $s \ a$
 assume $s \in \text{daml-terminal}$ $a \in \text{reg-actions}$
 then show $\text{daml-transition } s \ a = \text{None}$ by (*rule daml-terminal-absorbing*)
next
 fix $s \ a \ s'$
 assume $s \in \text{daml-states}$ $a \in \text{reg-actions}$ $\text{daml-transition } s \ a = \text{Some } s'$
 then show $s' \in \text{daml-states}$ by (*rule daml-transition-closed*)
next
 fix $s :: \text{daml-perm}$ and $a :: \text{reg-action}$
 assume $s \notin \text{daml-states}$
 then show $\text{daml-transition } s \ a = \text{None}$ by (*rule daml-transition-outside-states*)
qed

Forward naturality: *reg_to_daml* commutes with transitions.

lemma *reg-to-daml-naturality-some*:
 assumes $s \in \text{reg-states}$ and $a \in \text{reg-actions}$
 and $\text{reg-transition } s \ a = \text{Some } s'$
 shows $\text{daml-transition } (\text{reg-to-daml } s) \ a = \text{Some } (\text{reg-to-daml } s')$
proof –

```

have in-states: reg-to-daml s ∈ daml-states
  unfolding daml-states-def by auto
have daml-to-reg (reg-to-daml s) = s using daml-to-reg-to-daml-id by simp
with assms(3) have
  reg-transition (daml-to-reg (reg-to-daml s)) a = Some s' by simp
with in-states show ?thesis
  unfolding daml-transition-def by simp
qed

```

```

lemma reg-to-daml-naturality-none:
  assumes s ∈ reg-states and a ∈ reg-actions
  and reg-transition s a = None
  shows daml-transition (reg-to-daml s) a = None
proof -
  have in-states: reg-to-daml s ∈ daml-states
    unfolding daml-states-def by auto
  have daml-to-reg (reg-to-daml s) = s using daml-to-reg-to-daml-id by simp
  with assms(3) have
    reg-transition (daml-to-reg (reg-to-daml s)) a = None by simp
  with in-states show ?thesis
    unfolding daml-transition-def by simp
qed

```

Backward naturality: `daml_to_reg` on the image of `reg_to_daml` commutes with transitions in the opposite direction.

```

lemma daml-to-reg-naturality-some:
  assumes p ∈ daml-states and a ∈ reg-actions
  and daml-transition p a = Some p'
  shows reg-transition (daml-to-reg p) a = Some (daml-to-reg p')
proof -
  from assms(1,3) obtain s' where
    s'-eq: reg-transition (daml-to-reg p) a = Some s'
    and p'-def: p' = reg-to-daml s'
  unfolding daml-transition-def
  by (auto split: option.splits if-splits)
  from p'-def have daml-to-reg p' = s' using daml-to-reg-to-daml-id by simp
  with s'-eq show ?thesis by simp
qed

```

```

lemma daml-to-reg-naturality-none:
  assumes p ∈ daml-states and a ∈ reg-actions
  and daml-transition p a = None
  shows reg-transition (daml-to-reg p) a = None
proof (rule ccontr)
  assume reg-transition (daml-to-reg p) a ≠ None
  then obtain s' where reg-transition (daml-to-reg p) a = Some s' by auto
  with assms(1) have daml-transition p a = Some (reg-to-daml s')
    unfolding daml-transition-def by simp
  with assms(3) show False by simp

```

qed

Forward state preservation: `reg_to_daml` as a structure-preserving map.

Forward state preservation. The source state machine is on `(reg_states, reg_actions, reg_transition, reg_terminal)`, which exactly matches the `reg_sm` interpretation; consequently `unfold_locales` auto-discharges all six source state-machine obligations. The target side is the DAML structured-permission record, for which only individual lemmas (not a registered interpretation) are available, so all six target state-machine obligations remain pending alongside the five preservation-specific axioms eleven obligations in total. The asymmetry with `backward_layer_preservation` below (which has only the five preservation axioms remaining) reflects the order in which the locale extension presents the two state-machine sub-locales.

interpretation *forward-layer-preservation*:

state-preservation

reg-states reg-actions reg-transition reg-terminal

daml-states reg-actions daml-transition daml-terminal

reg-to-daml id

proof *unfold-locales*

show *finite daml-states* **by** (*rule daml-states-finite*)

next

show *finite reg-actions* **unfolding** *reg-actions-def* **by** *auto*

next

show *daml-terminal* $\subseteq$ *daml-states* **by** (*rule daml-terminal-subset*)

next

fix *s a*

assume *s* $\in$ *daml-terminal* *a* $\in$ *reg-actions*

then show *daml-transition s a = None* **by** (*rule daml-terminal-absorbing*)

next

fix *s a s'*

assume *s* $\in$ *daml-states* *a* $\in$ *reg-actions* *daml-transition s a = Some s'*

then show *s'* $\in$ *daml-states* **by** (*rule daml-transition-closed*)

next

fix *s :: daml-perm* **and** *a :: reg-action*

assume *s* $\notin$ *daml-states*

then show *daml-transition s a = None* **by** (*rule daml-transition-outside-states*)

next

fix *s*

assume *s* $\in$ *reg-states*

then show *reg-to-daml s* $\in$ *daml-states* **unfolding** *daml-states-def* **by** *auto*

next

fix *a*

assume *a* $\in$ *reg-actions*

then show *id a* $\in$ *reg-actions* **by** *simp*

next

fix *s*

assume *s* $\in$ *reg-terminal*

```

    then show reg-to-daml  $s \in \text{daml-terminal}$ 
      unfolding reg-terminal-def daml-terminal-def by simp
next
  fix  $s \ a \ s'$ 
  assume  $s \in \text{reg-states} \ a \in \text{reg-actions} \ \text{reg-transition } s \ a = \text{Some } s'$ 
  then show daml-transition (reg-to-daml  $s$ ) (id  $a$ ) = Some (reg-to-daml  $s'$ )
    using reg-to-daml-naturality-some by simp
next
  fix  $s \ a$ 
  assume  $s \in \text{reg-states} \ a \in \text{reg-actions} \ \text{reg-transition } s \ a = \text{None}$ 
  then show daml-transition (reg-to-daml  $s$ ) (id  $a$ ) = None
    using reg-to-daml-naturality-none by simp
qed

```

Backward state preservation: *daml_to_reg* as a structure-preserving map on the image of *reg_to_daml*.

Backward state preservation. The source side is DAML and the target side matches *reg_sm*. *unfold_locales* auto-discharges all twelve state-machine obligations from the previously registered interpretations, so only the five preservation axioms remain pending.

interpretation *backward-layer-preservation*:

state-preservation

daml-states *reg-actions* *daml-transition* *daml-terminal*

reg-states *reg-actions* *reg-transition* *reg-terminal*

daml-to-reg id

proof *unfold-locales*

```

  fix  $p$ 
  assume  $p \in \text{daml-states}$ 
  then show daml-to-reg  $p \in \text{reg-states}$  using reg-states-UNIV by auto
next
  fix  $a$ 
  assume  $a \in \text{reg-actions}$ 
  then show id  $a \in \text{reg-actions}$  by simp
next
  fix  $p$ 
  assume  $p \in \text{daml-terminal}$ 
  then have  $p = \text{reg-to-daml } \text{CONFISCATED}$  unfolding daml-terminal-def by simp
  then have daml-to-reg  $p = \text{CONFISCATED}$  using daml-to-reg-to-daml-id by simp
  then show daml-to-reg  $p \in \text{reg-terminal}$  unfolding reg-terminal-def by simp
next
  fix  $p \ a \ p'$ 
  assume  $p \in \text{daml-states} \ a \in \text{reg-actions} \ \text{daml-transition } p \ a = \text{Some } p'$ 
  then show reg-transition (daml-to-reg  $p$ ) (id  $a$ ) = Some (daml-to-reg  $p'$ )
    using daml-to-reg-naturality-some by simp
next
  fix  $p \ a$ 

```

```

assume  $p \in \text{daml-states}$   $a \in \text{reg-actions}$   $\text{daml-transition } p \ a = \text{None}$ 
then show  $\text{reg-transition } (\text{daml-to-reg } p) \ (\text{id } a) = \text{None}$ 
using  $\text{daml-to-reg-naturality-none}$  by  $\text{simp}$ 
qed

```

The symmetric instance: forward and backward state preservation composed with roundtrip guarantees. The non-trivial content is the bijection between `reg_state` and the image of `reg_to_daml` in `daml_perm`; the action map is the identity because both layers share the same regulatory action vocabulary.

interpretation *onchain-daml-bridge:*

```

symmetric-state-preservation
   $\text{reg-states}$   $\text{reg-actions}$   $\text{reg-transition}$   $\text{reg-terminal}$ 
   $\text{daml-states}$   $\text{reg-actions}$   $\text{daml-transition}$   $\text{daml-terminal}$ 
   $\text{reg-to-daml}$   $\text{id}$ 
   $\text{daml-to-reg}$   $\text{id}$ 

```

proof *unfold-locales*

```

fix  $s$ 
assume  $s \in \text{reg-states}$ 
then show  $\text{daml-to-reg } (\text{reg-to-daml } s) = s$  using  $\text{daml-to-reg-to-daml-id}$  by  $\text{simp}$ 
next
fix  $p$ 
assume  $p \in \text{daml-states}$ 
then show  $\text{reg-to-daml } (\text{daml-to-reg } p) = p$  using  $\text{reg-to-daml-to-reg-on-image}$ 
by  $\text{simp}$ 
next
fix  $a$ 
assume  $a \in \text{reg-actions}$ 
then show  $\text{id } (\text{id } a) = a$  by  $\text{simp}$ 
qed

```

As a corollary of the symmetric interpretation, `reg_to_daml` is injective on `reg_states`: distinct on-chain states map to distinct off-chain DAML records, so the layer-crossing representation incurs no information loss.

corollary *reg-to-daml-injective-on-states:*

```

 $\llbracket s1 \in \text{reg-states}; s2 \in \text{reg-states}; \text{reg-to-daml } s1 = \text{reg-to-daml } s2 \rrbracket$ 
 $\implies s1 = s2$ 
using  $\text{onchain-daml-bridge.state-map-injective}$  .

```

18 Multi-Domain Locale Instantiation

We now instantiate the `multi_domain_preservation` locale from `State_Preservation.thy` with the regulatory state machine.

The instantiation is parametric: given ANY finite set of domains and ANY initial `domain_state` function satisfying the consistency precondition, the locale is satisfied. This proves that the abstract framework applies to the concrete regulatory model without fixing a specific topology.

The key bridge: we construct a `domain_state` function from our `global_state`, and show that `valid_state` implies the consistency precondition required by `multi_domain_preservation`.

Bridge from our `global_state` model to the `multi_domain_preservation` locale's `domain_state` signature.

```
multi_domain_preservation expects: domain_state :: 'd => 'id
=> 's option
```

We instantiate with: `'d = chain_id`, `'id = asset_id`, `'s = reg_state`

The bridge function extracts `reg_state` from our richer `asset_state`.

definition `reg-domain-state :: global-state => chain-id => asset-id => reg-state option` **where**

```
reg-domain-state gs cid aid = get-reg-state gs cid aid
```

The parametric instantiation theorem: for any finite set of `chain_ids` and any valid `global_state`, we can interpret the `multi_domain_preservation` locale with the regulatory state machine.

The `state_machine` predicate as a standalone fact. This is needed for instantiating `multi_domain_preservation`, which takes `state_machine` as an opaque predicate assumption.

lemma `reg-state-machine-pred`:

```
state-machine reg-states reg-actions reg-transition reg-terminal
```

proof –

```
have f1: finite reg-states unfolding reg-states-def by auto
```

```
have f2: finite reg-actions unfolding reg-actions-def by auto
```

```
have f3: reg-terminal ⊆ reg-states
```

```
unfolding reg-terminal-def reg-states-def by auto
```

```
have f4: ⋀ s a. s ∈ reg-terminal ⟹ a ∈ reg-actions ⟹ reg-transition s a =
None
```

```
unfolding reg-terminal-def by (auto simp: confiscated-terminal)
```

```
have f5: ⋀ s a s'. s ∈ reg-states ⟹ a ∈ reg-actions ⟹
reg-transition s a = Some s' ⟹ s' ∈ reg-states
```

```
using reg-transition-closed by auto
```

```
have f6: ⋀ s a. (s :: reg-state) ∉ reg-states ⟹ reg-transition s a = None
```

```
using reg-states-UNIV by auto
```

```
show ?thesis
```

```
by (intro state-machine.intro) (use f1 f2 f3 f4 f5 f6 in auto)
```

qed

theorem `reg-multi-domain-instantiation`:

```
assumes fin-doms: finite (doms :: chain-id set)
```

```
and valid: valid-state gs
```

```
shows multi-domain-preservation doms reg-states reg-actions reg-transition
reg-terminal (reg-domain-state gs)
```

proof `(intro multi-domain-preservation.intro)`

```
show finite doms using fin-doms by simp
```

next

```

show state-machine reg-states reg-actions reg-transition reg-terminal
  by (rule reg-state-machine-pred)
next
  fix d1 d2 :: chain-id and aid :: asset-id and s1 s2 :: reg-state
  assume d1 ∈ doms d2 ∈ doms
    and reg-domain-state gs d1 aid = Some s1
    and reg-domain-state gs d2 aid = Some s2
  then have get-reg-state gs d1 aid = Some s1
    and get-reg-state gs d2 aid = Some s2
  unfolding reg-domain-state-def by simp-all
  with valid show s1 = s2
  unfolding valid-state-def consistent-state-def by blast
qed

```

Corollary: the generic `cross_domain_consistency` theorem from `State_Preservation.thy` applies to regulatory synchronization.

This connects the abstract theory to the concrete model: the generic theorem guarantees that after `sync_all` (the abstract synchronization), all connected domains agree. Combined with our concrete `regulatory_homomorphism` theorem, we have both the abstract and concrete guarantees.

The instantiation above (`reg_multi_domain_instantiation`) enables using all theorems proven in `multi_domain_preservation` in particular `cross_domain_consistency` and `sync_isolation` for the regulatory model without reproving them. Any consumer of this theory can invoke:

```

multi_domain_preservation.cross_domain_consistency      [OF
reg_multi_domain_instantiation[OF fin_doms valid_gs]]
to obtain the concrete guarantee for their domain set and global state.

```

Summary of what this theory establishes:

1. The regulatory state machine (`reg_state`, `reg_action`, `reg_transition`) satisfies the `state_machine` locale from `State_Preservation.thy`.
2. `CONFISCATED` is terminal (I1), `CONFISCATE` is universally reachable (I2), and the transition function is deterministic (I3).
3. `SEIZED` to `FROZEN` and `FROZEN` to `RESTRICTED` direct transitions are excluded by design, with legal and complexity-reduction justifications.
4. The synchronization protocol (lock, then update, then unlock) preserves cross-chain consistency for the same asset while isolating other assets.
5. Preemptive locking guards the `sync` function against a second action while the lock is held; under the atomic model this expresses the intended exclusion of competing regulatory actions on the same asset (NOT double-spend prevention that is handled at a different layer).
6. The `regulatory_homomorphism` theorem establishes that after `sync`, all connected chains agree on the new regulatory state.
7. The `valid_state_preservation` theorem establishes that `sync` pre-

serves the global validity invariant: `consistent_state` AND no outstanding locks. This closes the inductive invariant.

8. The heterogeneous-action instance (`escalation_preservation`) instantiates `state_preservation` with the four-action escalation subset of `reg_action` on the source side, the dedicated four-constructor datatype `chain_b_action` on the target side, and the identity state map. This exercises both the locale's `actions\<^sub>s` subset parameter and the heterogeneous source / target action types.

9. The layer-crossing instance (`onchain_daml_bridge`) instantiates `symmetric_state_preservation` with the on-chain enum representation and a structured DAML permission record (`daml_perm`) carrying auxiliary fields. The action vocabularies coincide so the action map is the identity; the non-trivial content is the bijection between the enum and the image `daml_states` of the representation map, with a type-level invariant (`valid_daml_perm`) tying the auxiliary fields to the status tag.

10. The `multi_domain_preservation` locale is instantiated parametrically: for any finite domain set and valid global state, the abstract framework from `State_Preservation.thy` applies to the regulatory model.

Open work for subsequent entries: - Eventual consistency under the partially synchronous network model (with message delays, redelivery, and out-of-order arrival).

end

theory *Priority-Resolution*

imports *Main*

begin

19 Priority-Based Deterministic Selection

A system where messages carry priorities from a linear order. Given a finite non-empty set of messages with injective priorities, there exists a unique highest-priority message, and selection is deterministic.

The injectivity assumption models tiebreaking: in practice, a composite key (e.g., authority level, timestamp, severity, node ID) ensures no two distinct messages share the same priority.

locale *priority-system* =

fixes *priority* :: '*m* $\Rightarrow$ '*k*::*linorder*

assumes *priority-injective*:

$\llbracket \text{priority } m1 = \text{priority } m2 \rrbracket \implies m1 = m2$

begin

In a finite non-empty set with injective priorities, there exists a unique element with the maximum priority.

lemma *highest-priority-exists*:

assumes *finite S* and $S \neq \{\}$
 shows $\exists!m. m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$
proof –
 have *fin-ing*: *finite (priority ‘ S)* **using** *assms(1)* **by** *simp*
 have *ne-ing*: *priority ‘ S $\neq \{\}$* **using** *assms(2)* **by** *simp*
 obtain *m* **where** *m-in*: $m \in S$ **and** *m-pri*: $\text{priority } m = \text{Max } (\text{priority ‘ } S)$
 using *Max-in[OF fin-ing ne-ing]* **by** *auto*
 have *m-max*: $\forall m' \in S. \text{priority } m' \leq \text{priority } m$
proof
 fix *m'* **assume** $m' \in S$
 hence *priority m' $\in$ priority ‘ S* **by** *simp*
 hence *priority m' $\leq$ Max (priority ‘ S)* **using** *Max-ge[OF fin-ing]* **by** *simp*
 thus *priority m' $\leq$ priority m* **using** *m-pri* **by** *simp*
qed
 show *?thesis*
proof (*rule ex1I[of - m]*)
 show $m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$
 using *m-in m-max* **by** *auto*
 next
 fix *m2* **assume** $m2 \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m2)$
 then have *m2-in*: $m2 \in S$ **and** *m2-max*: $\forall m' \in S. \text{priority } m' \leq \text{priority } m2$
by *auto*
 from *m-max m2-in* **have** *priority m2 $\leq$ priority m* **by** *auto*
 from *m2-max m-in* **have** *priority m $\leq$ priority m2* **by** *auto*
 then have *priority m = priority m2* **using** $\langle \text{priority } m2 \leq \text{priority } m \rangle$ **by** *auto*
 then show $m2 = m$ **using** *priority-injective* **by** *auto*
qed
qed

definition *select-highest* :: $'m \text{ set} \Rightarrow 'm \text{ option}$ **where**
 select-highest S =
 (*if S = $\{\}$ then None*
 else *Some (THE m. m $\in$ S $\wedge$ ($\forall m' \in S. \text{priority } m' \leq \text{priority } m$)))*)

theorem *select-highest-deterministic*:
 assumes *ne*: $S \neq \{\}$
 shows $\exists!m. \text{select-highest } S = \text{Some } m$
proof –
 from *ne* **obtain** *v* **where** *hv*: $\text{select-highest } S = \text{Some } v$
 unfolding *select-highest-def* **by** *simp*
 show *?thesis*
proof (*rule ex1I[of - v]*)
 show $\text{select-highest } S = \text{Some } v$ **by** (*rule hv*)
 next
 fix *w* **assume** $\text{select-highest } S = \text{Some } w$
 with *hv* **show** $w = v$ **by** *simp*
qed
qed

```

theorem select-highest-in-set:
  assumes fin: finite S and sel: select-highest S = Some m
  shows  $m \in S$ 
proof –
  from sel have ne:  $S \neq \{\}$ 
    unfolding select-highest-def by (simp split: if-splits)
  from highest-priority-exists[OF fin ne] have uniq:
     $\exists!x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x)$  .
  from theI'[OF uniq] have the-in:
     $(\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x)) \in S$  by auto
  from sel ne have m =  $(\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x))$ 
    unfolding select-highest-def by simp
  with the-in show ?thesis by simp
qed

```

```

theorem select-highest-is-max:
  assumes fin: finite S and ne:  $S \neq \{\}$  and sel: select-highest S = Some m
  shows  $\forall m' \in S. \text{priority } m' \leq \text{priority } m$ 
proof –
  from highest-priority-exists[OF fin ne] have uniq:
     $\exists!x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x)$  .
  from theI'[OF uniq] have the-max:
     $\forall m' \in S. \text{priority } m' \leq$ 
     $\text{priority } (\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x))$ 
    by auto
  from sel ne have m =  $(\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x))$ 
    unfolding select-highest-def by simp
  with the-max show ?thesis by simp
qed

```

end

20 Fair Leader Starvation Freedom

A leader-based system where an honest leader always processes at least one pending request, and a fairness assumption guarantees that honest leaders appear periodically.

The fairness assumption is a sufficient condition that abstracts over the probabilistic guarantees of VRF-based leader election. Under $f < n / (3::'a)$ Byzantine faults, the probability that a Byzantine leader is elected k consecutive times is $(f / n)^k$, which decreases exponentially. The fairness bound k captures this as a deterministic assumption.

The key theorem: under the fairness assumption, any pending request is processed within a bounded number of epochs.

```

locale fair-leader-system =
  fixes leader-at :: nat  $\Rightarrow$  'n
  and is-honest :: 'n  $\Rightarrow$  bool

```

```

and pending :: nat  $\Rightarrow$  nat
and fairness-bound :: nat
assumes fair-leader:
   $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge \text{is-honest} (\text{leader-at } e)$ 
and honest-progress:
   $\llbracket \text{is-honest} (\text{leader-at } e); \text{pending } e > 0 \rrbracket \implies \text{pending} (\text{Suc } e) < \text{pending } e$ 
and non-honest-bounded:
   $\text{pending} (\text{Suc } e) \leq \text{pending } e$ 
begin

```

Positivity of the fairness bound is implied by fairness itself: the window starting at any epoch must be non-empty to contain an honest slot. It is therefore derived rather than assumed.

```

lemma fairness-bound-positive: fairness-bound > 0
proof –
  obtain e where  $(0::\text{nat}) \leq e$  and  $e < 0 + \text{fairness-bound}$ 
  using fair-leader by blast
  then show ?thesis by simp
qed

```

Pending count is monotonically non-increasing.

```

lemma pending-monotone:
  assumes  $e1 \leq e2$ 
  shows  $\text{pending } e2 \leq \text{pending } e1$ 
  using assms
proof (induction rule: inc-induct)
  case base
  then show ?case by simp
next
  case (step e)
  have  $\text{pending } e2 \leq \text{pending} (\text{Suc } e)$  by (rule step.IH)
  also have  $\text{pending} (\text{Suc } e) \leq \text{pending } e$  using non-honest-bounded by auto
  finally show ?case .
qed

```

If there are pending requests, within *fairness-bound* epochs at least one will be processed (the pending count strictly decreases).

```

theorem starvation-bound:
  assumes pos:  $\text{pending epoch} > 0$ 
  shows  $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge \text{pending} (\text{Suc } e) < \text{pending } e$ 
proof –
  from fair-leader obtain h where
    h-range:  $\text{epoch} \leq h \wedge h < \text{epoch} + \text{fairness-bound}$  and
    h-honest:  $\text{is-honest} (\text{leader-at } h)$ 
  by auto
  show ?thesis
  proof (cases pending h > 0)

```

```

    case True
      with h-honest have pending (Suc h) < pending h using honest-progress by
auto
      with h-range show ?thesis by auto
    next
      case False
      then have ph0: pending h = 0 by simp
      — pending dropped from >0 (at epoch) to 0 (at h): find a strict decrease step
      have key:  $\forall n. 0 < n \longrightarrow \text{pending } (\text{epoch} + n) = 0 \longrightarrow$ 
         $(\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + n \wedge \text{pending } (\text{Suc } e) < \text{pending } e)$ 
      proof (rule allI)
        fix n
        show  $0 < n \longrightarrow \text{pending } (\text{epoch} + n) = 0 \longrightarrow$ 
           $(\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + n \wedge \text{pending } (\text{Suc } e) < \text{pending } e)$ 
        proof (induction n)
          case 0 show ?case by simp
        next
          case (Suc k)
          show ?case
          proof (intro impI)
            assume hpk: pending (epoch + Suc k) = 0
            show  $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{Suc } k \wedge \text{pending } (\text{Suc } e) < \text{pending } e$ 
            proof (cases k = 0)
              case True
              from hpk True have pending (Suc epoch) < pending epoch using pos by
simp
              thus ?thesis by (intro exI[of - epoch]) auto
            next
              case False
              then have kpos:  $0 < k$  by simp
              show ?thesis
              proof (cases pending (epoch + k) = 0)
                case True
                then have pk0: pending (epoch + k) = 0 .
                from Suc.IH[THEN mp, OF kpos, THEN mp, OF pk0]
                obtain e' where epoch  $\leq e'$   $e' < \text{epoch} + k$  pending (Suc e') < pending
e'
                by auto
                thus ?thesis by auto
              next
                case False
                from False hpk have pending (epoch + Suc k) < pending (epoch + k)
by simp
                thus ?thesis by (intro exI[of - epoch + k]) auto
              qed
            qed
          qed
        qed
      qed
    qed
  qed

```

```

have dpos: 0 < h - epoch
proof -
  have h ≠ epoch using ph0 pos by auto
  with h-range(1) show ?thesis by arith
qed
have pd: pending (epoch + (h - epoch)) = 0
proof -
  have epoch + (h - epoch) = h using h-range(1) by arith
  with ph0 show ?thesis by simp
qed
from key[rule-format, OF dpos, OF pd]
obtain e' where he': epoch ≤ e' e' < epoch + (h - epoch) pending (Suc e') <
pending e'
  by auto
have epoch + (h - epoch) ≤ h using h-range(1) by arith
with he' h-range(2) show ?thesis by auto
qed
qed

```

Corollary: all pending requests are eventually processed. By induction on the pending count (which is a natural number and thus a well-founded decreasing measure), repeated application of `starvation_bound` drives the count to zero.

theorem *eventual-completion*:

shows $\exists e\text{-final}. \text{pending } e\text{-final} = 0$

proof -

define *P* **where**

$P\ n \equiv \forall \text{epoch}. \text{pending epoch} = n \longrightarrow (\exists e\text{-final}. \text{pending } e\text{-final} = 0)$

for $n :: \text{nat}$

have *all-P*: $\bigwedge n. P\ n$

proof -

fix *n* **show** *P n*

unfolding *P-def*

proof (*induction n rule: less-induct*)

case (*less n*)

show $\forall \text{epoch}. \text{pending epoch} = n \longrightarrow (\exists e\text{-final}. \text{pending } e\text{-final} = 0)$

proof (*intro allI impI*)

fix *epoch* **assume** *eq*: $\text{pending epoch} = n$

show $\exists e\text{-final}. \text{pending } e\text{-final} = 0$

proof (*cases n = 0*)

case *True* **thus** ?thesis **using** *eq* **by** *auto*

next

case *False*

have *pos*: $\text{pending epoch} > 0$ **using** *eq False* **by** *auto*

from *starvation-bound*[*OF pos*] **obtain** *e* **where**

e-ge: $\text{epoch} \leq e$ **and** *e-dec*: $\text{pending (Suc } e) < \text{pending } e$

by *auto*

have *lt*: $\text{pending (Suc } e) < n$

using *pending-monotone*[*OF e-ge*] *eq e-dec* **by** *simp*

```

      from less.IH[OF lt] show ?thesis
      unfolding P-def by blast
    qed
  qed
  qed
  from all-P[of pending 0] show ?thesis
  unfolding P-def by blast
qed

end

end

theory DQuencer-Instance
  imports Priority-Resolution Regulatory-Instance HOL-Library.Product-Lexorder
begin

```

21 Authority Level and Action Severity

Regulatory authority hierarchy. International authorities (e.g., FATF) take precedence over national (e.g., SEC), which take precedence over regional authorities. This reflects the RCP framework’s jurisdictional priority model.

datatype *authority-level* = *Regional* | *National* | *International*

```

fun authority-rank :: authority-level  $\Rightarrow$  nat where
  authority-rank Regional = 1
| authority-rank National = 2
| authority-rank International = 3

```

```

lemma authority-rank-injective:
  authority-rank a1 = authority-rank a2  $\implies$  a1 = a2
by (cases a1; cases a2; auto)

```

Action severity determines priority among actions from the same authority level and timestamp (it is the third component of the lexicographic key). Stronger enforcement actions (CONFISCATE, SEIZE) take precedence over weaker ones (RESTRICT, UNFREEZE). This reflects the legal principle that more conservative (asset-protective) actions prevail when conflicting orders must be ranked against one another.

```

fun action-severity :: reg-action  $\Rightarrow$  nat where
  action-severity UNRESTRICT = 1
| action-severity UNFREEZE = 2
| action-severity RELEASE = 3
| action-severity RESTRICT = 4
| action-severity FREEZE = 5

```

| *action-severity SEIZE* = 6
| *action-severity CONFISCATE* = 7

lemma *action-severity-injective*:

action-severity a1 = action-severity a2 $\implies$ a1 = a2

by (*cases a1; cases a2; auto*)

22 Priority Key

The priority key is a 4-tuple of natural numbers, using Isabelle's built-in lexicographic order on products. No manual `linorder` instance registration is needed.

Components (highest to lowest significance):

1. **authority_rank**: higher authority = higher priority
2. **inverted_timestamp**: earlier timestamp = higher priority (inverted)
3. **action_severity**: stronger action = higher priority
4. **node_id**: deterministic tiebreaker (lower node ID = higher priority, so we invert to make larger = higher in the nat order)

For timestamp and **node_id** inversion, we use *max-val* – *actual-val* to convert "smaller is better" to "larger is better" for compatibility with the nat product order where larger = higher.

type-synonym *priority-key* = *nat* $\times$ *nat* $\times$ *nat* $\times$ *nat*

23 Extended Message Type

Extends `oss_message` from `Regulatory_Instance` with authority level and source node ID. Uses Isabelle record inheritance so that existing functions on `oss_message` remain applicable.

record *dq-message* = *oss-message* +
dqm-authority-level :: *authority-level*
dqm-source-node :: *nat*

24 Priority Construction

Construct a priority key from message fields. Uses two upper bounds for inversion: **max_time** for timestamps and **max_node** for node IDs.

definition *make-priority-key* ::

nat $\Rightarrow$ *nat* $\Rightarrow$ *dq-message* $\Rightarrow$ *priority-key*

where

```

make-priority-key max-time max-node msg =
  (authority-rank (dqm-authority-level msg),
   max-time - msg-timestamp msg,
   action-severity (msg-action msg),
   max-node - dqm-source-node msg)

```

25 Node and System Model

```

datatype node-behavior = Honest | Byzantine

```

```

record node-info =
  ni-id      :: nat
  ni-behavior :: node-behavior

```

```

definition honest-nodes :: node-info set  $\Rightarrow$  node-info set where
  honest-nodes ns = {n  $\in$  ns. ni-behavior n = Honest}

```

```

definition byzantine-nodes :: node-info set  $\Rightarrow$  node-info set where
  byzantine-nodes ns = {n  $\in$  ns. ni-behavior n = Byzantine}

```

```

lemma honest-byzantine-partition:
  honest-nodes ns  $\cup$  byzantine-nodes ns = ns

```

```

proof (rule set-eqI)
  fix x
  show x  $\in$  honest-nodes ns  $\cup$  byzantine-nodes ns  $\longleftrightarrow$  x  $\in$  ns
    unfolding honest-nodes-def byzantine-nodes-def
    by (cases ni-behavior x; auto)
qed

```

```

lemma honest-byzantine-disjoint:
  honest-nodes ns  $\cap$  byzantine-nodes ns = {}
  unfolding honest-nodes-def byzantine-nodes-def by auto

```

26 D-quencer System Locale

The D-quencer system locale encapsulates the BFT assumption $((3::'a) * f + 1 \leq n)$ and system parameters.

```

locale dquencer-system =
  fixes nodes :: node-info set
  and f-max :: nat
  and fairness-bound :: nat
  and max-time :: nat
  and max-node :: nat
  assumes finite-nodes: finite nodes
  and bft-threshold: card nodes  $\geq$  3 * f-max + 1
  and byzantine-bound: card (byzantine-nodes nodes)  $\leq$  f-max
  and fairness-positive: fairness-bound > 0
begin

```

```

lemma honest-majority:
  card (honest-nodes nodes) > 2 * f-max
proof –
  have card nodes = card (honest-nodes nodes) + card (byzantine-nodes nodes)
  using honest-byzantine-partition honest-byzantine-disjoint finite-nodes
  by (metis card-Un-disjoint finite-Un)
  with bft-threshold byzantine-bound show ?thesis by linarith
qed

```

```

lemma honest-nonempty:
  honest-nodes nodes ≠ {}
proof –
  have fin: finite (honest-nodes nodes)
  unfolding honest-nodes-def using finite-nodes by simp
  have pos: 0 < card (honest-nodes nodes)
  using honest-majority by linarith
  with fin pos show ?thesis by (auto simp: card-eq-0-iff)
qed

```

end

27 Priority System Instantiation

We instantiate the `priority_system` locale from `Priority_Resolution.thy` with D-quencer message priorities.

The priority function maps messages to `priority_key` (which is `nat \<times> nat \<times> nat \<times> nat` with the built-in lexicographic order).

Injectivity follows from: distinct messages have either different authority levels, different timestamps, different actions, or different source nodes. The source node ID serves as the final tiebreaker ensuring no two distinct messages share the same key.

For a concrete message set with the injectivity precondition, the `priority_system` locale provides deterministic selection.

```

lemma priority-key-injectivity:
  assumes make-priority-key mt mn m1 = make-priority-key mt mn m2
  and msg-timestamp m1 ≤ mt and msg-timestamp m2 ≤ mt
  and dqm-source-node m1 ≤ mn and dqm-source-node m2 ≤ mn
  shows dqm-authority-level m1 = dqm-authority-level m2
   $\wedge$  msg-timestamp m1 = msg-timestamp m2
   $\wedge$  msg-action m1 = msg-action m2
   $\wedge$  dqm-source-node m1 = dqm-source-node m2
proof –
  from assms(1) have
    eq1: authority-rank (dqm-authority-level m1) = authority-rank (dqm-authority-level m2) and

```

```

eq2:  $mt - \text{msg-timestamp } m1 = mt - \text{msg-timestamp } m2$  and
eq3:  $\text{action-severity } (\text{msg-action } m1) = \text{action-severity } (\text{msg-action } m2)$  and
eq4:  $mn - \text{dqm-source-node } m1 = mn - \text{dqm-source-node } m2$ 
unfolding make-priority-key-def by auto
from eq1 have al:  $\text{dqm-authority-level } m1 = \text{dqm-authority-level } m2$ 
using authority-rank-injective by auto
from eq2 assms(2,3) have ts:  $\text{msg-timestamp } m1 = \text{msg-timestamp } m2$ 
by linarith
from eq3 have act:  $\text{msg-action } m1 = \text{msg-action } m2$ 
using action-severity-injective by auto
from eq4 assms(4,5) have nd:  $\text{dqm-source-node } m1 = \text{dqm-source-node } m2$ 
by linarith
from al ts act nd show ?thesis by auto
qed

```

Auxiliary corollary: under the well-formedness bounds, equal priority keys imply equal messages on every priority-relevant field.

The generic `priority_system` locale assumes priority injectivity unconditionally on its carrier type. We instantiate it with the identity function on `priority_key` as the carrier; injectivity holds by reflexivity on the carrier. The recovery of the unique D-quencer message corresponding to a chosen priority key is provided as a corollary inside the `dquencer_priority_concrete` locale, using the well-formedness bounds and the additional priority-key distinctness assumption.

```

interpretation dq-priority:
  priority-system id :: priority-key  $\Rightarrow$  priority-key
by unfold-locales auto

```

The `dquencer_priority_context` locale extends `dquencer_system` with a fixed message set whose elements satisfy the well-formedness bounds. Together with the bound-preservation, this set induces the priority key set `msg_priority_keys` on which priority-based selection operates.

```

locale dquencer-priority-context = dquencer-system +
  fixes msg-set :: dq-message set
  assumes msg-set-nonempty:  $\text{msg-set} \neq \{\}$ 
  and msg-set-well-formed:
     $\bigwedge m. m \in \text{msg-set} \implies \text{msg-timestamp } m \leq \text{max-time}$ 
     $\wedge \text{dqm-source-node } m \leq \text{max-node}$ 

```

begin

```

definition msg-priority-keys :: priority-key set where
  msg-priority-keys = make-priority-key max-time max-node ' msg-set

```

```

lemma msg-priority-keys-nonempty:  $\text{msg-priority-keys} \neq \{\}$ 
using msg-set-nonempty unfolding msg-priority-keys-def by simp

```

```

lemma msg-priority-keys-finite:
   $\text{finite } \text{msg-set} \implies \text{finite } \text{msg-priority-keys}$ 

```

unfolding *msg-priority-keys-def* **by** *simp*

The well-formedness bounds are what make the priority key faithful to the message fields: under them, equal keys force equal priority-relevant fields. This lifts *priority-key-injectivity* into the context, consuming the bounds.

lemma *msg-priority-key-field-injective*:

assumes $m1 \in \text{msg-set}$ **and** $m2 \in \text{msg-set}$
and *make-priority-key max-time max-node* $m1 = \text{make-priority-key max-time max-node } m2$
shows *dqm-authority-level* $m1 = \text{dqm-authority-level } m2$
 $\wedge$ *msg-timestamp* $m1 = \text{msg-timestamp } m2$
 $\wedge$ *msg-action* $m1 = \text{msg-action } m2$
 $\wedge$ *dqm-source-node* $m1 = \text{dqm-source-node } m2$
using *priority-key-injectivity*[*OF assms*(3)]
 $\text{msg-set-well-formed}$ [*OF assms*(1)] $\text{msg-set-well-formed}$ [*OF assms*(2)]
by *blast*

end

The *dquencer_priority_concrete* locale strengthens the context with the priority-key distinctness assumption on *msg_set*. This is the natural BFT-consensus precondition: distinct authority / timestamp / severity / source-node tuples ensure determinism. Under this assumption the priority key uniquely identifies a message in *msg_set*, so selecting the highest priority key in *msg_priority_keys* is equivalent to selecting the highest-priority message in *msg_set*.

locale *dquencer-priority-concrete* = *dquencer-priority-context* +

assumes *msg-set-priority-distinct*:

$\bigwedge m1\ m2. m1 \in \text{msg-set} \implies m2 \in \text{msg-set}$
 $\implies \text{make-priority-key max-time max-node } m1$
 $= \text{make-priority-key max-time max-node } m2$
 $\implies m1 = m2$

begin

Recovery: given a priority key from *msg_priority_keys*, return the unique *msg_set* message with that key.

definition *recover-msg* :: *priority-key* $\Rightarrow$ *dq-message* **where**

recover-msg $k = (\text{THE } m. m \in \text{msg-set} \wedge \text{make-priority-key max-time max-node } m = k)$

lemma *recover-msg-unique-existence*:

assumes $k \in \text{msg-priority-keys}$

shows $\exists! m. m \in \text{msg-set} \wedge \text{make-priority-key max-time max-node } m = k$

proof –

from *assms* **obtain** m

where *m-in*: $m \in \text{msg-set}$

and *m-key*: $\text{make-priority-key max-time max-node } m = k$

unfolding *msg-priority-keys-def* **by** *auto*

```

show ?thesis
proof (rule ex1I[of - m])
  show  $m \in \text{msg-set} \wedge \text{make-priority-key max-time max-node } m = k$ 
  using m-in m-key by auto
next
  fix m'
  assume  $m' \in \text{msg-set} \wedge \text{make-priority-key max-time max-node } m' = k$ 
  then have m'-in:  $m' \in \text{msg-set}$ 
  and m'-key:  $\text{make-priority-key max-time max-node } m' = k$ 
  by auto
  from m-key m'-key
  have  $\text{make-priority-key max-time max-node } m' = \text{make-priority-key max-time max-node } m$ 
  by simp
  then show  $m' = m$ 
  using msg-set-priority-distinct[OF m'-in m-in] by simp
qed
qed

lemma recover-msg-correct:
  assumes  $k \in \text{msg-priority-keys}$ 
  shows  $\text{recover-msg } k \in \text{msg-set}$ 
   $\wedge \text{make-priority-key max-time max-node } (\text{recover-msg } k) = k$ 
proof -
  from recover-msg-unique-existence[OF assms]
  have ex1:  $\exists! m. m \in \text{msg-set} \wedge \text{make-priority-key max-time max-node } m = k$  .
  show ?thesis
  using theI'[OF ex1] unfolding recover-msg-def .
qed

```

Concrete consequence: deterministic selection of the highest-priority key from a non-empty finite `msg_priority_keys` set, and recovery of the unique well-formed D-quencer message that produced it.

corollary dq-select-highest-deterministic:

```

 $\exists! k. \text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$ 
using dq-priority.select-highest-deterministic[OF msg-priority-keys-nonempty] .

```

corollary dq-select-highest-in-set:

```

assumes finite msg-set
  and  $\text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$ 
shows  $k \in \text{msg-priority-keys}$ 
using dq-priority.select-highest-in-set
  [OF msg-priority-keys-finite[OF assms(1)] assms(2)] .

```

corollary dq-select-highest-message:

```

assumes finite msg-set
  and  $\text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$ 
shows  $\text{recover-msg } k \in \text{msg-set}$ 
   $\wedge \text{make-priority-key max-time max-node } (\text{recover-msg } k) = k$ 

```

using *recover-msg-correct*[*OF dq-select-highest-in-set*[*OF assms*]] .

The recovered message is maximal: no well-formed message in the set carries a strictly higher priority key. This projects the key-level maximality of **select_highest** back onto messages, closing the loop of the “highest-priority message” reading.

corollary *dq-select-highest-message-maximal*:

assumes *finite msg-set*

and *dq-priority.select-highest msg-priority-keys = Some k*

shows $\forall m \in \text{msg-set}. \text{make-priority-key max-time max-node } m$
 $\leq \text{make-priority-key max-time max-node } (\text{recover-msg } k)$

proof –

have *max*: $\forall k' \in \text{msg-priority-keys}. k' \leq k$

using *dq-priority.select-highest-is-max*

[*OF msg-priority-keys-finite*[*OF assms*(1)] *msg-priority-keys-nonempty*
assms(2)]

by *simp*

have *rk*: *make-priority-key max-time max-node (recover-msg k) = k*

using *recover-msg-correct*[*OF dq-select-highest-in-set*[*OF assms*]] **by** *simp*

show *?thesis*

proof

fix *m* **assume** *m* $\in \text{msg-set}$

then have *make-priority-key max-time max-node m* $\in \text{msg-priority-keys}$

unfolding *msg-priority-keys-def* **by** *simp*

with *max rk* **show** *make-priority-key max-time max-node m*

$\leq \text{make-priority-key max-time max-node } (\text{recover-msg } k)$ **by** *simp*

qed

qed

end

28 Fair Leader Starvation Freedom Instantiation

We instantiate the **fair_leader_system** locale with the D-quencer’s leader election and pending request processing.

The fairness assumption abstracts VRF randomness: within any **fairness_bound** consecutive epochs, at least one epoch has an honest leader. Under $f < n / (3::'a)$, the probability of *fairness-bound* consecutive Byzantine leaders is $(f / n)^{\text{fairness-bound}} < (1 / (3::'a))^{\text{fairness-bound}}$.

For concrete instantiation, we need a leader schedule and pending count function that satisfy the locale assumptions. These are provided as parameters in the instantiation context.

locale *dquencer-liveness* = *dquencer-system* +

fixes *leader-schedule* :: *nat* $\Rightarrow$ *node-info*

and *pending-count* :: *nat* $\Rightarrow$ *nat*

assumes *fair-leader*:

```

 $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$ 
 $\text{ni-behavior}(\text{leader-schedule } e) = \text{Honest}$ 
and honest-processes:
 $\llbracket \text{ni-behavior}(\text{leader-schedule } e) = \text{Honest}; \text{pending-count } e > 0 \rrbracket$ 
 $\implies \text{pending-count}(\text{Suc } e) < \text{pending-count } e$ 
and pending-non-increasing:
 $\text{pending-count}(\text{Suc } e) \leq \text{pending-count } e$ 
begin

interpretation dq-fair: fair-leader-system
 $\text{leader-schedule } \lambda n. \text{ni-behavior } n = \text{Honest pending-count fairness-bound}$ 
proof unfold-locales
show  $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$ 
 $\text{ni-behavior}(\text{leader-schedule } e) = \text{Honest}$ 
using fair-leader .
next
fix  $e$ 
assume  $\text{ni-behavior}(\text{leader-schedule } e) = \text{Honest } 0 < \text{pending-count } e$ 
then show  $\text{pending-count}(\text{Suc } e) < \text{pending-count } e$ 
using honest-processes by auto
next
fix  $e$ 
show  $\text{pending-count}(\text{Suc } e) \leq \text{pending-count } e$ 
using pending-non-increasing .
qed

```

Concrete starvation freedom for the D-quencer: if there are pending regulatory requests, at least one will be processed within `fairness_bound` epochs.

```

corollary dq-starvation-bound:
assumes  $\text{pending-count } \text{epoch} > 0$ 
shows  $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$ 
 $\text{pending-count}(\text{Suc } e) < \text{pending-count } e$ 
using dq-fair.starvation-bound[OF assms] .

```

end

29 Conditional Safety alongside the Liveness Results

This section records how the safety result of Property 1 (cross-domain state preservation) sits alongside the liveness results of Property 2 (deterministic selection and fair scheduling).

The theorem below, `conditional_safety_preservation`, is a *conditional safety* statement: from a valid global state in which the target asset is unlocked and a transition is enabled, the synchronization function succeeds and preserves the global validity invariant. Its proof uses only

the safety side (`lock_acquire_success` and `valid_state_preservation`); it does not invoke the liveness interpretations (`dq_priority`, `dq_fair`) and makes no Byzantine, determinism, deadlock, or starvation claim of its own. Those liveness properties are the separate theorems of this theory (`dq_select_highest_deterministic` and `dq_starvation_bound`), each stated in its own right.

The genuine fusion of the two sides — safety freed of its initial-validity hypothesis by a well-founded progress measure on cross-chain inconsistency, under the fair-leader assumption — is `oracclizer_guarded_bounded_convergence` in `Functor_Laws.thy`, which drops the initial-validity hypothesis assumed here.

The statement below is a leaf corollary: it restates `valid_state_preservation` under the explicit precondition that the asset is unlocked — the precondition the lock discipline is designed to establish. It is deliberately not the place where liveness and safety are fused (see `Functor_Laws.thy`).

theorem *conditional-safety-preservation:*

```

assumes valid: valid-state gs
and exists: asset-exists gs source aid
and current: get-reg-state gs source aid = Some s
and trans: reg-transition s action = Some s'
and not-locked: ¬ is-locked gs aid
shows  $\exists gs'. \text{sync source action aid gs} = \text{Some } gs' \wedge \text{valid-state } gs'$ 
proof –
from not-locked obtain gs-locked where
  lock: acquire-lock gs aid = Some gs-locked
using lock-acquire-success by auto
from valid current trans lock have
   $\exists gs'. \text{sync source action aid gs} = \text{Some } gs'$ 
unfolding sync-def
using exists current trans lock
by (auto simp: get-reg-state-def get-asset-state-def
      split: option.splits
      intro!: exI)
then obtain gs' where synced: sync source action aid gs = Some gs'
by auto
have valid-state gs'
using valid-state-preservation[OF valid current trans synced] .
with synced show ?thesis by auto
qed

```

Summary of what Properties 1 + 2 together establish:

1. **Safety** (Property 1): After synchronization, all connected chains agree on the regulatory state. The global validity invariant is preserved.
2. **Determinism** (Property 2): Conflicting regulatory actions are resolved by a total order on priority keys. The consensus output

— abstracted here as deterministic priority selection — is unique. The `priority_system` locale is instantiated on the `priority_key` type as carrier (with the corresponding messages recovered inside `dquencer_priority_concrete` via `recover_msg`), yielding deterministic selection from any finite non-empty set of valid candidate messages. Priority here resolves conflicts *among regulatory messages*; precedence of regulatory actions over ordinary (non-regulatory) transactions is not modelled — the message universe of this theory contains regulatory actions only.

3. **Deadlock** (Property 2, scope note): deadlock is a concurrency phenomenon. The atomic sync model has no concurrent lock contention (lock holding is a boolean without contended holders), so deadlock does not arise within the model’s scope; forced lock release under contention is out of scope, deferred to the preemptive-lock property. The theory states no proved deadlock-freedom theorem.
4. **Starvation freedom** (Property 2): Under the fair leader assumption, every pending regulatory request is processed within a bounded number of epochs. The pending-count assumptions encode a closed system — no new requests arrive within the analysis horizon; liveness under continuous arrivals belongs to the partially synchronous lift left to subsequent entries.

Open work for subsequent entries: - Compositional assurance across heterogeneous verification regimes (Canton + OSS + EVM). - Refinement: formal correspondence between these models and the Rust / Solidity implementation.

end

```
theory Composition
  imports State-Preservation
begin
```

30 Guarded Invariants

A guarded transition system: a carrier set closed under stepping, together with an invariant that is preserved by every *guarded* step. The guard isolates exactly the operations that may fire while maintaining the invariant.

```
locale guarded-invariant =
  fixes carrier :: 's set
  and step :: 's  $\Rightarrow$  'op  $\Rightarrow$  's option
  and ops :: 'op set
  and inv :: 's  $\Rightarrow$  bool
```

```

and guard :: 's ⇒ 'op ⇒ bool
assumes step-closed:
  ⌊ s ∈ carrier; opn ∈ ops; step s opn = Some s' ⌋ ⇒ s' ∈ carrier
and guarded-preservation:
  ⌊ s ∈ carrier; opn ∈ ops; inv s; guard s opn; step s opn = Some s' ⌋
  ⇒ inv s'

```

31 Eventual Dischargers

A scheduler that, within every window of length *window*, produces at least one discharging event. This is a bounded-fairness abstraction: progress-bearing events are never starved beyond *window* time units.

```

locale eventual-discharger =
  fixes schedule :: nat ⇒ 'event
  and discharges :: 'event ⇒ bool
  and window :: nat
  assumes bounded-occurrence:
    ∀ t. ∃ t'. t ≤ t' ∧ t' < t + window ∧ discharges (schedule t')
begin

```

Window positivity is implied by bounded occurrence: the window starting at any time index must be non-empty to contain a discharging slot. It is therefore derived rather than assumed.

```

lemma window-positive: window > 0
proof –
  obtain t' where (0::nat) ≤ t' and t' < 0 + window
  using bounded-occurrence by blast
  then show ?thesis by simp
qed

end

```

32 Compositional Safety and Liveness

The composition couples a guarded invariant with an eventual discharger through a realization map *realize* that turns a discharging event, in the current state, into an operation. The coupling assumption states that any realized operation is admissible and guarded; invariant preservation along a trajectory then follows from the guard, and progress is supplied by the measure discharge of the converging extension below.

```

locale safety-liveness-composition =
  safe: guarded-invariant carrier step ops inv guard +
  live: eventual-discharger schedule discharges window
  for carrier :: 's set
  and step :: 's ⇒ 'op ⇒ 's option
  and ops :: 'op set

```

```

and inv :: 's  $\Rightarrow$  bool
and guard :: 's  $\Rightarrow$  'op  $\Rightarrow$  bool
and schedule :: nat  $\Rightarrow$  'event
and discharges :: 'event  $\Rightarrow$  bool
and window :: nat +
fixes realize :: 'event  $\Rightarrow$  's  $\Rightarrow$  'op option
assumes realize-discharges:
   $\llbracket s \in \text{carrier}; \text{discharges } \text{ev}; \text{realize } \text{ev } s = \text{Some } \text{opn} \rrbracket$ 
   $\implies \text{opn} \in \text{ops} \wedge \text{guard } s \text{ opn}$ 
begin

```

One evolution step at absolute time index t : a scheduled discharging event is realized and applied; any non-discharging event (or a realization that does not enable a step) stutters, leaving the state unchanged.

```

definition evolve-step :: nat  $\Rightarrow$  's  $\Rightarrow$  's where
  evolve-step  $t$   $s$  =
    (if discharges (schedule  $t$ ) then
      (case realize (schedule  $t$ )  $s$  of
        None  $\Rightarrow$   $s$ 
      | Some opn  $\Rightarrow$  (case step  $s$  opn of None  $\Rightarrow$   $s$  | Some  $s'$   $\Rightarrow$   $s'$ ))
    else  $s$ )

```

The trajectory of length n started at time index k .

```

fun run-from :: nat  $\Rightarrow$  's  $\Rightarrow$  nat  $\Rightarrow$  's where
  run-from  $k$   $s$  0 =  $s$ 
| run-from  $k$   $s$  (Suc  $n$ ) = evolve-step ( $k + n$ ) (run-from  $k$   $s$   $n$ )

```

The trajectory started at time 0, and the induced evolution relation.

```

definition run :: 's  $\Rightarrow$  nat  $\Rightarrow$  's where
  run  $s$   $n$  = run-from 0  $s$   $n$ 

```

```

definition evolves-to :: 's  $\Rightarrow$  nat  $\Rightarrow$  's  $\Rightarrow$  bool where
  evolves-to  $s$   $t$   $s'$   $\longleftrightarrow$  run  $s$   $t$  =  $s'$ 

```

Trajectories compose: continuing for b more steps re-roots the schedule index by a .

lemma *run-from-add*:

```

  run-from  $k$   $s$  ( $a + b$ ) = run-from ( $k + a$ ) (run-from  $k$   $s$   $a$ )  $b$ 

```

proof (*induction* b)

case 0

show ?case **by** *simp*

next

case (Suc b)

```

  have run-from  $k$   $s$  ( $a + \text{Suc } b$ ) = evolve-step ( $k + (a + b)$ ) (run-from  $k$   $s$  ( $a + b$ ))

```

by *simp*

```

  also have ... = evolve-step ( $k + (a + b)$ ) (run-from ( $k + a$ ) (run-from  $k$   $s$   $a$ )  $b$ )

```

using *Suc.IH* **by** *simp*

also have $\dots = \text{run-from } (k + a) (\text{run-from } k \ s \ a) (\text{Suc } b)$
 by (simp add: add.assoc)
 finally show ?case .
 qed

lemma *evolve-step-stutter*:
 $\neg \text{discharges } (\text{schedule } t) \implies \text{evolve-step } t \ s = s$
 by (simp add: evolve-step-def)

A single evolution step keeps the carrier.

lemma *evolve-step-carrier*:
 assumes $s \in \text{carrier}$
 shows $\text{evolve-step } t \ s \in \text{carrier}$
proof (cases discharges (schedule t))
 case False
 then show ?thesis using assms by (simp add: evolve-step-def)
 next
 case True
 show ?thesis
proof (cases realize (schedule t) s)
 case None
 with assms True show ?thesis by (simp add: evolve-step-def)
 next
 case (Some opn)
 note $r = \text{this}$
 show ?thesis
proof (cases step s opn)
 case None
 with assms True r show ?thesis by (simp add: evolve-step-def)
 next
 case (Some s')
 note $st = \text{this}$
 have $op\text{-}in: \text{opn} \in \text{ops}$ using realize-discharges[OF assms True r] by simp
 have $s' \in \text{carrier}$ using safe.step-closed[OF assms op-in st] .
 with True r st show ?thesis by (simp add: evolve-step-def)
 qed
 qed
 qed

A single evolution step preserves the invariant on the carrier.

lemma *evolve-step-inv*:
 assumes $s \in \text{carrier}$ and $\text{inv } s$
 shows $\text{inv } (\text{evolve-step } t \ s)$
proof (cases discharges (schedule t))
 case False
 then show ?thesis using assms by (simp add: evolve-step-def)
 next
 case True
 show ?thesis

```

proof (cases realize (schedule t) s)
  case None
  with assms True show ?thesis by (simp add: evolve-step-def)
next
  case (Some opn)
  note r = this
  show ?thesis
  proof (cases step s opn)
    case None
    with assms True r show ?thesis by (simp add: evolve-step-def)
  next
    case (Some s')
    note st = this
    have c: opn ∈ ops ∧ guard s opn using realize-discharges[OF assms(1) True
r] .
    have inv s'
      using safe.guarded-preservation[OF assms(1) conjunct1[OF c] assms(2)
conjunct2[OF c] st] .
    with True r st show ?thesis by (simp add: evolve-step-def)
  qed
qed
qed

```

```

lemma run-from-carrier:
  assumes s ∈ carrier
  shows run-from k s n ∈ carrier
proof (induction n)
  case 0
  show ?case using assms by simp
next
  case (Suc n)
  show ?case using evolve-step-carrier[OF Suc.IH] by simp
qed

```

```

lemma run-from-inv:
  assumes s ∈ carrier and inv s
  shows inv (run-from k s n)
proof (induction n)
  case 0
  show ?case using assms(2) by simp
next
  case (Suc n)
  have car: run-from k s n ∈ carrier using run-from-carrier[OF assms(1)] .
  have inv (evolve-step (k + n) (run-from k s n)) using evolve-step-inv[OF car
Suc.IH] .
  then show ?case by simp
qed

```

If no discharging event occurs in the first n slots from k , the state is unchanged.

```

lemma run-from-stutter:
  assumes  $\forall j < n. \neg \text{discharges } (\text{schedule } (k + j))$ 
  shows  $\text{run-from } k \ s \ n = s$ 
  using assms
proof (induction n)
  case 0
  show ?case by simp
next
  case (Suc n)
  have  $\forall j < n. \neg \text{discharges } (\text{schedule } (k + j))$  using Suc.prems by simp
  then have ih:  $\text{run-from } k \ s \ n = s$  using Suc.IH by simp
  have  $\neg \text{discharges } (\text{schedule } (k + n))$  using Suc.prems by simp
  then have  $\text{evolve-step } (k + n) (\text{run-from } k \ s \ n) = \text{run-from } k \ s \ n$ 
    by (simp add: evolve-step-stutter)
  then show ?case using ih by simp
qed

```

Invariant preservation along the whole trajectory: from an invariant carrier state, every reachable state still satisfies the invariant and stays in the carrier.

```

lemma invariant-preserved:
  assumes  $s \in \text{carrier}$  and  $\text{inv } s$ 
  shows  $\text{inv } (\text{run } s \ t) \wedge \text{run } s \ t \in \text{carrier}$ 
proof –
  have  $\text{inv } (\text{run-from } 0 \ s \ t)$  using run-from-inv[OF assms] .
  moreover have  $\text{run-from } 0 \ s \ t \in \text{carrier}$  using run-from-carrier[OF assms(1)]
  .
  ultimately show ?thesis by (simp add: run-def)
qed
end

```

33 Guarded Bounded Convergence

safety-liveness-composition is extended with a natural-number progress measure whose zero set is contained in the invariant (*measure-zero-inv*) and which every discharging step strictly decreases while the invariant fails (*discharge-progresses*). This is exactly the “well-founded measure connecting non-invariant states to invariant states via discharging events” that drives convergence; well-foundedness is supplied by the natural-number order (*measure progress-measure*).

Together with bounded fairness this yields convergence to the invariant within *progress-measure s * window* steps from an arbitrary carrier state, with *no* assumption that the invariant holds initially.

```

locale converging-composition =
  safety-liveness-composition carrier step ops inv guard schedule discharges window
  realize

```

```

for carrier :: 's set
  and step :: 's  $\Rightarrow$  'op  $\Rightarrow$  's option
  and ops :: 'op set
  and inv :: 's  $\Rightarrow$  bool
  and guard :: 's  $\Rightarrow$  'op  $\Rightarrow$  bool
  and schedule :: nat  $\Rightarrow$  'event
  and discharges :: 'event  $\Rightarrow$  bool
  and window :: nat
  and realize :: 'event  $\Rightarrow$  's  $\Rightarrow$  'op option +
fixes progress-measure :: 's  $\Rightarrow$  nat
assumes discharge-progresses:
   $\llbracket s \in \text{carrier}; \neg \text{inv } s; \text{discharges } \text{ev} \rrbracket$ 
   $\implies \exists \text{opn } s'. \text{realize } \text{ev } s = \text{Some } \text{opn} \wedge \text{step } s \text{ opn} = \text{Some } s'$ 
   $\wedge \text{progress-measure } s' < \text{progress-measure } s$ 
begin

```

The zero set of the measure lies inside the invariant — derived rather than assumed: at measure zero a failing invariant would admit a discharging step whose target measure lies strictly below zero.

```

lemma measure-zero-inv:
  assumes  $s \in \text{carrier}$  and  $\text{progress-measure } s = 0$ 
  shows  $\text{inv } s$ 
proof (rule ccontr)
  assume  $\neg \text{inv } s$ 
  obtain  $t$  where  $\text{discharges } (\text{schedule } t)$ 
  using live.bounded-occurrence by blast
  then obtain  $\text{opn } s'$  where  $\text{progress-measure } s' < \text{progress-measure } s$ 
  using discharge-progresses[OF assms(1) ninv] by blast
  with assms(2) show False by simp
qed

```

```

definition convergence-bound :: 's  $\Rightarrow$  nat where
  convergence-bound  $s = \text{progress-measure } s * \text{window}$ 

```

```

lemma progress-measure-wf: wf (measure progress-measure)
  by (rule wf-measure)

```

The core convergence lemma, by strong induction on the measure. From any carrier state with measure m , an invariant state is reached within $m * \text{window}$ steps, regardless of the starting time index k .

```

lemma convergence-bound-reachable:
   $\text{progress-measure } s = m \implies s \in \text{carrier}$ 
   $\implies \exists t \leq m * \text{window}. \text{inv } (\text{run-from } k \ s \ t)$ 
proof (induction m arbitrary: s k rule: less-induct)
  case (less m)
  show ?case
  proof (cases inv s)
  case True
  have  $\text{inv } (\text{run-from } k \ s \ 0)$  using True by simp

```

moreover have $(0::nat) \leq m * window$ **by** *simp*
ultimately show *?thesis* **by** *blast*
next
case *False*
have *m-eq*: $progress-measure\ s = m$ **using** *less.premis(1)* .
have *car*: $s \in carrier$ **using** *less.premis(2)* .
— Locate the least discharging offset within the fairness window.
obtain *t'* **where** *t'1*: $k \leq t'$ **and** *t'2*: $t' < k + window$
and *t'3*: *discharges* (*schedule* *t'*)
using *live.bounded-occurrence* **by** *blast*
define *Q* **where** $Q = (\lambda j. j < window \wedge discharges\ (schedule\ (k + j)))$
have *w*: $t' - k < window$ **using** *t'1 t'2* **by** *linarith*
have *kk*: $k + (t' - k) = t'$ **using** *t'1* **by** *simp*
have *Qwit*: $Q\ (t' - k)$ **unfolding** *Q-def* **using** *w t'3 kk* **by** *simp*
define *j0* **where** *j0* = (*LEAST* *j*. *Q j*)
have *Qj0*: $Q\ j0$ **unfolding** *j0-def* **using** *Qwit* **by** (*rule LeastI*[**where** $P = Q$])
have *j0w*: $j0 < window$ **using** *Qj0* **unfolding** *Q-def* **by** *simp*
have *disc0*: *discharges* (*schedule* ($k + j0$))) **using** *Qj0* **unfolding** *Q-def* **by**
simp
have *before*: $\forall j < j0. \neg discharges\ (schedule\ (k + j))$
proof (*intro allI impI*)
fix *j* **assume** *jl*: $j < j0$
have *lt*: $j < Least\ Q$ **using** *jl* **by** (*simp add: j0-def*)
have *nQ*: $\neg Q\ j$ **using** *not-less-Least*[*OF lt*] .
have *jw*: $j < window$ **using** *jl j0w* **by** *simp*
show $\neg discharges\ (schedule\ (k + j))$ **using** *nQ jw* **by** (*simp add: Q-def*)
qed
have *stut*: $run-from\ k\ s\ j0 = s$ **using** *run-from-stutter*[*OF before*] .
— The discharging step at offset *j0* strictly decreases the measure.
obtain *opn s'* **where** *r*: $realize\ (schedule\ (k + j0))\ s = Some\ opn$
and *st*: $step\ s\ opn = Some\ s'$
and *dec*: $progress-measure\ s' < progress-measure\ s$
using *discharge-progresses*[*OF car False disc0*] **by** *blast*
have *op-in*: $opn \in ops$ **using** *realize-discharges*[*OF car disc0 r*] **by** *simp*
have *s'-car*: $s' \in carrier$ **using** *safe.step-closed*[*OF car op-in st*] .
have *ev*: $run-from\ k\ s\ (Suc\ j0) = s'$
proof —
have $run-from\ k\ s\ (Suc\ j0) = evolve-step\ (k + j0)\ (run-from\ k\ s\ j0)$ **by** *simp*
also have $\dots = evolve-step\ (k + j0)\ s$ **using** *stut* **by** *simp*
also have $\dots = s'$ **using** *disc0 r st* **by** (*simp add: evolve-step-def*)
finally show *?thesis* .
qed
have *decm*: $progress-measure\ s' < m$ **using** *dec m-eq* **by** *simp*
— Apply the induction hypothesis to the smaller-measure state *s'*.
have $\exists t \leq progress-measure\ s' * window. inv\ (run-from\ (k + Suc\ j0)\ s'\ t)$
using *less.IH*[*OF decm refl s'-car*] **by** *blast*
then obtain *t2* **where** *t2b*: $t2 \leq progress-measure\ s' * window$
and *inv2*: $inv\ (run-from\ (k + Suc\ j0)\ s'\ t2)$ **by** *blast*
have *comp*: $run-from\ k\ s\ (Suc\ j0 + t2) = run-from\ (k + Suc\ j0)\ s'\ t2$

```

proof –
  have run-from  $k\ s\ (Suc\ j0 + t2)$ 
     $= run-from\ (k + Suc\ j0)\ (run-from\ k\ s\ (Suc\ j0))\ t2$ 
    by (rule run-from-add)
  also have  $\dots = run-from\ (k + Suc\ j0)\ s'\ t2$  by (simp only: ev)
  finally show ?thesis .
qed
have inv-fin:  $inv\ (run-from\ k\ s\ (Suc\ j0 + t2))$  using comp inv2 by simp
have  $Suc\ j0 + t2 \leq window + progress-measure\ s' * window$ 
  using j0w t2b by linarith
also have  $\dots = (1 + progress-measure\ s') * window$ 
  by (simp add: add-mult-distrib)
also have  $\dots \leq m * window$ 
proof –
  have  $1 + progress-measure\ s' \leq m$  using decm by linarith
  then show ?thesis by (rule mult-le-mono1)
qed
finally have  $Suc\ j0 + t2 \leq m * window$  .
then show ?thesis using inv-fin by blast
qed
qed

```

Guarded bounded convergence. From an arbitrary carrier state with no assumption that the invariant holds the system reaches, within *convergence-bound* s evolution steps, a state satisfying the invariant.

```

theorem bounded-convergence-from-arbitrary:
  assumes  $s \in carrier$ 
  shows  $\exists t \leq convergence-bound\ s. \exists s'. evolves-to\ s\ t\ s' \wedge inv\ s'$ 
proof –
  obtain  $t$  where  $tb: t \leq progress-measure\ s * window$ 
    and  $inv-t: inv\ (run-from\ 0\ s\ t)$ 
    using convergence-bound-reachable[OF refl assms] by blast
  have  $evolves-to\ s\ t\ (run\ s\ t)$  by (simp add: evolves-to-def)
  with  $tb\ inv-t$  show ?thesis
    by (auto simp: convergence-bound-def run-def evolves-to-def)
qed
end
end

```

```

theory Proof-Automation
  imports State-Preservation HOL-Eisbach.Eisbach
begin

```

34 Discharge Methods for Instance Obligations

Two named theorem collections feed the methods. *discharge-simps* holds the equational content of a concrete instance: defining equations of the state/action/terminal sets, finiteness facts, terminal-absorption equations, and naturality equations of structured state maps (conditional rewrites whose premises the simplifier discharges from the goal's assumptions). *discharge-intros* holds introduction rules whose extra premises must be instantiated by unification against the goal's assumptions — typically closure lemmas, whose condition variables do not all occur in the conclusion and which are therefore out of reach of conditional rewriting.

named-theorems *discharge-simps*

⟨defining equations and conditional rewrites consumed by the discharge methods⟩

named-theorems *discharge-intros*

⟨introduction rules consumed by the discharge methods⟩

named-theorems *discharge-dels*

⟨default simplification rules suppressed by the discharge methods ---
typically the defining equations of structured state maps introduced with
⟨fun⟩, whose eager unfolding would destroy the redexes that the
conditional naturality rewrites of ⟨discharge-simps⟩ match on⟩

The common skeleton: unfold the locale predicate into its atomic obligations (*unfold-locales* also discharges sub-locale obligations already available from registered interpretations), then close each remaining obligation by simplifier-driven exhaustive case analysis — membership of an action in an enumerated set splits into finitely many cases, after which the defining equations of the transition functions decide the goal, with *option/if* splits for partially defined transitions.

discharge-state-machine is the entry point for *state-machine* obligations (finiteness, terminal containment, terminal absorption, closure, domain); *discharge-preservation* is the entry point for *state-preservation* obligations, whose unfolding contains the two sub-machines' obligations together with the morphism axioms (well-definedness, terminal preservation, and the two naturality directions).

method *discharge-state-machine* **declares** *discharge-simps* *discharge-intros* *discharge-dels* =

(*unfold-locales*;
(*auto simp add: discharge-simps simp del: discharge-dels*
intro: discharge-intros
split: option.splits if-splits))

method *discharge-preservation* **declares** *discharge-simps* *discharge-intros* *discharge-dels* =

(*unfold-locales*;

```

(auto simp add: discharge-simps simp del: discharge-dels
  intro: discharge-intros
  split: option.splits if-splits))

```

The two entry points share their skeleton deliberately: a *state-preservation* goal unfolds into the union of the two obligation families, so the preservation method must subsume the machine method. Keeping both names separates the reusable surface by intent — an importer discharging a bare machine instance is not invited to reach for the morphism-level method — and lets the two evolve independently if the obligation families diverge in a future revision.

end

```

theory Functor-Laws
imports
  Composition
  Regulatory-Instance
  DQuencer-Instance
  Proof-Automation
  ADS-Functor.Merkle-Interface
begin

```

35 Functor Laws on State-Preservation Morphisms

We regard the locale *state-preservation* as a morphism between two objects of a category. An object is a state machine (*states*, *actions*, *transition*, *terminal*); a morphism from one object to another is a pair (*state-map*, *action-map*) satisfying the well-definedness, terminal-preservation and naturality conditions of *state-preservation*. The following three theorems are the category axioms for this notion of morphism.

35.1 Identity morphism

The identity pair (*id*, *id*) is a state-preservation morphism from any object to itself: *F* sends identities to identities.

```

theorem preservation-id:
  assumes state-machine states actions transition terminal
  shows state-preservation states actions transition terminal
      states actions transition terminal id id

  using assms
  unfolding state-preservation-def state-preservation-axioms-def
  by simp

```

35.2 Composition of morphisms

The composite of two state-preservation morphisms is a state-preservation morphism: the naturality square of the second morphism is stacked on that of the first. This is the cross-domain analogue of "closed under composition".

theorem *preservation-compose*:

```

assumes f: state-preservation  $S_a A_a T_a F_a S_b A_b T_b F_b f f'$ 
and g: state-preservation  $S_b A_b T_b F_b S_c A_c T_c F_c g g'$ 
shows state-preservation  $S_a A_a T_a F_a S_c A_c T_c F_c (g \circ f) (g' \circ f')$ 
proof –
interpret f: state-preservation  $S_a A_a T_a F_a S_b A_b T_b F_b f f'$  by (rule f)
interpret g: state-preservation  $S_b A_b T_b F_b S_c A_c T_c F_c g g'$  by (rule g)
show ?thesis
proof (unfold-locales)
  fix s assume  $s \in S_a$ 
  then show  $(g \circ f) s \in S_c$ 
    by (simp add: f.state-map-well-defined g.state-map-well-defined)
  next
  fix a assume  $a \in A_a$ 
  then show  $(g' \circ f') a \in A_c$ 
    by (simp add: f.action-map-well-defined g.action-map-well-defined)
  next
  fix s assume  $s \in F_a$ 
  then show  $(g \circ f) s \in F_c$ 
    by (simp add: f.terminal-preservation g.terminal-preservation)
  next
  fix s a s'
  assume A:  $s \in S_a$  and B:  $a \in A_a$  and C:  $T_a s a = \text{Some } s'$ 
  have fs:  $f s \in S_b$  using A f.state-map-well-defined by simp
  have fa:  $f' a \in A_b$  using B f.action-map-well-defined by simp
  have  $T_b (f s) (f' a) = \text{Some } (f s')$  using f.naturality A B C by simp
  then have  $T_c (g (f s)) (g' (f' a)) = \text{Some } (g (f s'))$ 
    using g.naturality fs fa by simp
  then show  $T_c ((g \circ f) s) ((g' \circ f') a) = \text{Some } ((g \circ f) s')$ 
    by (simp add: comp-def)
  next
  fix s a
  assume A:  $s \in S_a$  and B:  $a \in A_a$  and C:  $T_a s a = \text{None}$ 
  have fs:  $f s \in S_b$  using A f.state-map-well-defined by simp
  have fa:  $f' a \in A_b$  using B f.action-map-well-defined by simp
  have  $T_b (f s) (f' a) = \text{None}$  using f.naturality-none A B C by simp
  then have  $T_c (g (f s)) (g' (f' a)) = \text{None}$ 
    using g.naturality-none fs fa by simp
  then show  $T_c ((g \circ f) s) ((g' \circ f') a) = \text{None}$ 
    by (simp add: comp-def)
qed
qed

```

35.3 Associativity of morphism composition

Composition of state-preservation morphisms is associative on both the state and the action components. Since the components are functions, this is exactly associativity of function composition; the three morphism hypotheses fix the objects across which the equation is read, completing the category axioms (identity, composition, associativity).

theorem *preservation-assoc*:

assumes *state-preservation* $S_a A_a T_a F_a S_b A_b T_b F_b f f'$
and *state-preservation* $S_b A_b T_b F_b S_c A_c T_c F_c g g'$
and *state-preservation* $S_c A_c T_c F_c S_d A_d T_d F_d k k'$
shows $((k \circ g) \circ f = k \circ (g \circ f)) \wedge ((k' \circ g') \circ f' = k' \circ (g' \circ f'))$
by (*simp add: comp-assoc*)

36 Composition Exercised: A Heterogeneous Three-Domain Chain

preservation-compose is exercised on a concrete chain of three distinct domains: the off-chain DAML permission ledger (structured records), the on-chain regulatory enum, and Chain B with its dedicated four-action vocabulary. In HOL the morphisms of a heterogeneous chain cannot be packed into one object — the domains carry genuinely different state and action types — so the correct formal expression of an N -domain chain is exactly the composite of pairwise morphisms, with *preservation-assoc* making longer path composites unambiguous. Both legs are scoped to the escalation subset of the action vocabulary: the DAML machine restricts a fortiori (a machine restricted to a subset of its action set is again a machine), and the layer-crossing naturality of *daml-to-reg* holds on the subset because it holds on the full vocabulary.

lemma *daml-escalation-state-machine*:

state-machine daml-states escalation-actions daml-transition daml-terminal

proof *unfold-locales*

show *finite daml-states* **by** (*rule daml-states-finite*)

next

show *finite escalation-actions* **unfolding** *escalation-actions-def* **by** *simp*

next

show *daml-terminal* $\subseteq$ *daml-states* **by** (*rule daml-terminal-subset*)

next

fix $p a$

assume $p \in \text{daml-terminal}$ **and** $a \in \text{escalation-actions}$

then show *daml-transition* $p a = \text{None}$

using *daml-terminal-absorbing reg-actions-UNIV* **by** *auto*

next

fix $p a p'$

assume $p \in \text{daml-states}$ **and** $a \in \text{escalation-actions}$

and *daml-transition* $p a = \text{Some } p'$

```

    then show  $p' \in \text{daml-states}$ 
      using daml-transition-closed reg-actions-UNIV by auto
  next
    fix  $p :: \text{daml-perm}$  and  $a :: \text{reg-action}$ 
    assume  $p \notin \text{daml-states}$ 
    then show  $\text{daml-transition } p \ a = \text{None}$  by (rule daml-transition-outside-states)
qed

```

The escalation-scoped backward bridge: *daml-to-reg* as a morphism from the DAML machine to the regulatory machine, both restricted to the escalation actions.

lemma *daml-to-reg-escalation-bridge*:

state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states escalation-actions reg-transition reg-terminal
daml-to-reg id

proof *unfold-locales*

— Source state-machine obligations (the DAML machine on the escalation subset is not registered, so its six obligations remain pending); the target machine coincides with the source machine of the registered *escalation-preservation* interpretation and is discharged automatically.

```

    show finite daml-states by (rule daml-states-finite)
  next
    show finite escalation-actions unfolding escalation-actions-def by simp
  next
    show  $\text{daml-terminal} \subseteq \text{daml-states}$  by (rule daml-terminal-subset)
  next
    fix  $p \ a$ 
    assume  $p \in \text{daml-terminal}$  and  $a \in \text{escalation-actions}$ 
    then show  $\text{daml-transition } p \ a = \text{None}$ 
      using daml-terminal-absorbing reg-actions-UNIV by auto
  next
    fix  $p \ a \ p'$ 
    assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
      and  $\text{daml-transition } p \ a = \text{Some } p'$ 
    then show  $p' \in \text{daml-states}$ 
      using daml-transition-closed reg-actions-UNIV by auto
  next
    fix  $p :: \text{daml-perm}$  and  $a :: \text{reg-action}$ 
    assume  $p \notin \text{daml-states}$ 
    then show  $\text{daml-transition } p \ a = \text{None}$  by (rule daml-transition-outside-states)
  next
    fix  $p$ 
    assume  $p \in \text{daml-states}$ 
    then show  $\text{daml-to-reg } p \in \text{reg-states}$  using reg-states-UNIV by auto
  next
    fix  $a$ 
    assume  $a \in \text{escalation-actions}$ 
    then show  $\text{id } a \in \text{escalation-actions}$  by simp
  next

```

```

fix  $p$ 
  assume  $p \in \text{daml-terminal}$ 
  then have  $p = \text{reg-to-daml CONFISCATED unfolding daml-terminal-def by simp}$ 
  then have  $\text{daml-to-reg } p = \text{CONFISCATED using daml-to-reg-to-daml-id by simp}$ 
  then show  $\text{daml-to-reg } p \in \text{reg-terminal unfolding reg-terminal-def by simp}$ 
next
  fix  $p \ a \ p'$ 
  assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
  and  $\text{daml-transition } p \ a = \text{Some } p'$ 
  then show  $\text{reg-transition } (\text{daml-to-reg } p) \ (\text{id } a) = \text{Some } (\text{daml-to-reg } p')$ 
  using  $\text{daml-to-reg-naturality-some reg-actions-UNIV by auto}$ 
next
  fix  $p \ a$ 
  assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
  and  $\text{daml-transition } p \ a = \text{None}$ 
  then show  $\text{reg-transition } (\text{daml-to-reg } p) \ (\text{id } a) = \text{None}$ 
  using  $\text{daml-to-reg-naturality-none reg-actions-UNIV by auto}$ 
qed

```

The escalation morphism of *Cross-Domain-State-Preservation.Regulatory-Instance* in predicate (bundle) form, so that it can be fed to *preservation-compose*.

lemma *escalation-preservation-bundle*:

```

state-preservation reg-states escalation-actions reg-transition reg-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
id escalation-action-map

```

— This very instance is registered as the *escalation-preservation* interpretation of *Cross-Domain-State-Preservation.Regulatory-Instance*, so every obligation is discharged from the registration.

by *unfold-locales*

The composed three-domain morphism: DAML permission records, synchronized through the regulatory enum, arrive on Chain B’s vocabulary. The composite is literally *preservation-compose* applied to the two legs; no new naturality argument is needed, which is the point of the functor laws. Arbitrary finite heterogeneous chains are expressed the same way, as composites of pairwise morphisms.

theorem *daml-to-chain-b-composed*:

```

state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
(id o daml-to-reg) (escalation-action-map o id)
using preservation-compose[OF daml-to-reg-escalation-bridge
escalation-preservation-bundle] .

```

The domain guard of the layer-crossing bridge is load-bearing, not a notational convenience. The record below carries the *D-Seized* tag with no seizing party: it violates *valid-daml-perm* and lies outside *daml-states*. The guarded DAML transition rejects *RELEASE* on it, while the bare enum

projection of the same record would accept it — so dropping the guard would break naturality on exactly such records. Preservation holds on the carved-out domain and only there; the guard is the boundary of the preservation guarantee, recorded here permanently as a machine-checked witness.

definition *invalid-daml-perm* :: *daml-perm* **where**

invalid-daml-perm =
 (| *status-tag* = *D-Seized*, *seized-by* = *None*, *restriction-scope* = *None* |)

lemma *guard-is-load-bearing*:

¬ *valid-daml-perm invalid-daml-perm*
invalid-daml-perm ∉ *daml-states*
daml-transition invalid-daml-perm RELEASE = *None*
reg-transition (daml-to-reg invalid-daml-perm) RELEASE = *Some ACTIVE*

proof –

show ¬ *valid-daml-perm invalid-daml-perm*
by (*simp add: valid-daml-perm-def invalid-daml-perm-def*)
have ∧*s. invalid-daml-perm* ≠ *reg-to-daml s*

proof –

fix *s* **show** *invalid-daml-perm* ≠ *reg-to-daml s*
by (*cases s*) (*auto simp: invalid-daml-perm-def*)

qed

then show *notin: invalid-daml-perm* ∉ *daml-states*
by (*auto simp: daml-states-def*)

from notin show *daml-transition invalid-daml-perm RELEASE* = *None*
by (*rule daml-transition-outside-states*)

show *reg-transition (daml-to-reg invalid-daml-perm) RELEASE* = *Some ACTIVE*

by (*simp add: invalid-daml-perm-def*)

qed

37 Proof Automation Exercised on the Regulatory Instances

The discharge methods of *Cross-Domain-State-Preservation.Proof-Automation* are fed here with the equational content of the regulatory, Chain B and DAML domains, and then exercised: each *-via-method* lemma below restates a preservation or machine instance — proved manually above or registered in *Cross-Domain-State-Preservation.Regulatory-Instance* — and discharges it with a single method invocation, giving a one-to-one comparison between the manual proofs and the automated ones. The instance obligations are not weakened in any way: the statements are identical to their manual counterparts.

Finiteness of the enumerated regulatory types, in the normal form the simplifier reaches after rewriting the set definitions to *UNIV*.

lemma *finite-reg-state-UNIV* [*discharge-simps*]: *finite (UNIV :: reg-state set)*

using *reg-sm.finite-states* **by** (*simp add: reg-states-UNIV*)

lemma *finite-reg-action-UNIV* [*discharge-simps*]: *finite (UNIV :: reg-action set)*
using *reg-sm.finite-actions* **by** (*simp add: reg-actions-UNIV*)

Terminal and well-definedness helpers for the layer-crossing maps, in conditional-rewrite form.

lemma *daml-to-reg-terminal-val* [*discharge-simps*]:
 $p \in \text{daml-terminal} \implies \text{daml-to-reg } p = \text{CONFISCATED}$
by (*auto simp: daml-terminal-def daml-to-reg-to-daml-id*)

lemma *reg-to-daml-terminal* [*discharge-simps*]:
 $s \in \text{reg-terminal} \implies \text{reg-to-daml } s \in \text{daml-terminal}$
by (*auto simp: reg-terminal-def daml-terminal-def*)

lemma *reg-to-daml-in-states* [*discharge-simps*]:
 $\text{reg-to-daml } s \in \text{daml-states}$
by (*simp add: daml-states-def*)

The collections. Equational and conditional-rewrite content goes to *discharge-simps*; closure rules, whose condition variables do not all occur in their conclusions, go to *discharge-intros*.

lemmas [*discharge-simps*] =
reg-states-UNIV reg-actions-UNIV
escalation-actions-def chain-b-actions-def reg-terminal-def
confiscated-terminal chain-b-terminal
daml-states-finite daml-terminal-subset
daml-terminal-absorbing daml-transition-outside-states
reg-to-daml-naturality-some reg-to-daml-naturality-none
daml-to-reg-naturality-some daml-to-reg-naturality-none
escalation-naturality-some escalation-naturality-none

lemmas [*discharge-intros*] =
reg-transition-closed chain-b-transition-closed daml-transition-closed

The layer-crossing maps are introduced with *fun*, so their defining equations are default simplification rules; the methods must keep the maps opaque so that the conditional naturality and terminal rewrites above can match their redexes.

lemmas [*discharge-dels*] =
daml-to-reg.simps reg-to-daml.simps

The comparison lemmas. The first three restate instances proved manually in this theory; none of these instances is itself registered, so the goal is not discharged wholesale by a registration — sub-machine obligations already available from prior registrations are absorbed by *unfold-locales*, and the method closes every remaining obligation from the collections alone.

lemma *daml-escalation-state-machine-via-method*:

state-machine daml-states escalation-actions daml-transition daml-terminal
by *discharge-state-machine*

lemma *daml-to-reg-escalation-bridge-via-method:*
state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states escalation-actions reg-transition reg-terminal
daml-to-reg id
by *discharge-preservation*

lemma *daml-to-chain-b-composed-via-method:*
state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
(id ∘ daml-to-reg) (escalation-action-map ∘ id)
by *discharge-preservation*

The forward escalation-scoped bridge is a new instance with no manual counterpart: the method constructs it outright, which is the reuse claim in its sharpest form.

lemma *reg-to-daml-escalation-bridge-via-method:*
state-preservation reg-states escalation-actions reg-transition reg-terminal
daml-states escalation-actions daml-transition daml-terminal
reg-to-daml id
by *discharge-preservation*

The remaining three restate instances registered as interpretations in *Cross-Domain-State-Preservation.Regulatory-Instance*; for these, *unfold-locales* discharges the obligations from the registrations, so the one-line proofs are registration-backed rather than search-backed. The search-backed counterparts above show that the method does not depend on the registrations.

lemma *escalation-preservation-via-method:*
state-preservation reg-states escalation-actions reg-transition reg-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
id escalation-action-map
by *discharge-preservation*

lemma *forward-layer-preservation-via-method:*
state-preservation reg-states reg-actions reg-transition reg-terminal
daml-states reg-actions daml-transition daml-terminal
reg-to-daml id
by *discharge-preservation*

lemma *backward-layer-preservation-via-method:*
state-preservation daml-states reg-actions daml-transition daml-terminal
reg-states reg-actions reg-transition reg-terminal
daml-to-reg id
by *discharge-preservation*

38 State Glue: Join and Refinement

The two glue predicates lift the authenticated-data-structure operations to the global-state level. *state-join* *sa sb sab* says that *sab* is the consistent combination of the partial views *sa* and *sb*: on every chain connected in either view, *sab* carries the value of whichever view knows the chain (preferring *sa*), the two views agree wherever both are defined, and *sab* reveals no chain beyond the union of the two views (the containment clause). This is the state-level reading of the ADS *join*: on revealed holdings *sab* is the least consistent combination of *sa* and *sb* — the lock components are unconstrained — and the agreement clause is the state-level reading of equal hashes (mergeability). Without the containment clause a candidate join could carry an arbitrary holding on a chain outside both views and still be accepted; the regression witness *rogue-join-excluded* below records that such a candidate is now rejected.

state-refines *sa sb* says that *sa* is a partial view of *sb*: every chain known to *sa* is known to *sb* with the same regulatory state. This is the state-level reading of the ADS blinding order *bo*: a more-blinded value agrees with a less-blinded one on everything it can observe.

definition *state-join* :: *global-state* $\Rightarrow$ *global-state* $\Rightarrow$ *global-state* $\Rightarrow$ *bool* **where**

$$\begin{aligned} & \text{state-join } sa \ sb \ sab \equiv \\ & (\forall \text{ aid } c. \\ & \quad (c \in \text{connected-chains } sa \ \text{aid} \vee c \in \text{connected-chains } sb \ \text{aid}) \\ & \quad \longrightarrow \text{get-reg-state } sab \ c \ \text{aid} = \\ & \quad \quad (\text{if } c \in \text{connected-chains } sa \ \text{aid} \text{ then } \text{get-reg-state } sa \ c \ \text{aid} \\ & \quad \quad \text{else } \text{get-reg-state } sb \ c \ \text{aid})) \\ & \wedge (\forall \text{ aid } c1 \ c2. \\ & \quad c1 \in \text{connected-chains } sa \ \text{aid} \wedge c2 \in \text{connected-chains } sb \ \text{aid} \\ & \quad \longrightarrow \text{get-reg-state } sa \ c1 \ \text{aid} = \text{get-reg-state } sb \ c2 \ \text{aid}) \\ & \wedge (\forall \text{ aid } c. \\ & \quad c \in \text{connected-chains } sab \ \text{aid} \\ & \quad \longrightarrow c \in \text{connected-chains } sa \ \text{aid} \vee c \in \text{connected-chains } sb \ \text{aid}) \end{aligned}$$

definition *state-refines* :: *global-state* $\Rightarrow$ *global-state* $\Rightarrow$ *bool* **where**

$$\begin{aligned} & \text{state-refines } sa \ sb \equiv \\ & \quad \forall \text{ aid } c. \ c \in \text{connected-chains } sa \ \text{aid} \\ & \quad \longrightarrow c \in \text{connected-chains } sb \ \text{aid} \wedge \text{get-reg-state } sa \ c \ \text{aid} = \text{get-reg-state } sb \ c \ \text{aid} \end{aligned}$$

Refinement is reflexive and transitive: a partial-view preorder on states.

lemma *state-refines-refl*: *state-refines* *sa sa*

unfolding *state-refines-def* **by** *simp*

lemma *state-refines-trans*:

$$\text{state-refines } sa \ sb \Longrightarrow \text{state-refines } sb \ sc \Longrightarrow \text{state-refines } sa \ sc$$

unfolding *state-refines-def* **by** *metis*

A consistent state refined by another transports its consistency upward:

if sa refines a consistent sb , then sa is consistent.

lemma *state-refines-preserves-consistency*:

assumes *state-refines* $sa\ sb$ **and** *consistent-state* sb

shows *consistent-state* sa

unfolding *consistent-state-def*

proof (*intro allI impI*)

fix $c1\ c2\ aid\ s1\ s2$

assume *get-reg-state* $sa\ c1\ aid = Some\ s1$ **and** *get-reg-state* $sa\ c2\ aid = Some\ s2$

moreover have $c1 \in \text{connected-chains } sa\ aid$ **and** $c2 \in \text{connected-chains } sa\ aid$

using *calculation* **unfolding** *connected-chains-def* *asset-exists-def*
get-reg-state-def *get-asset-state-def*

by (*auto split: option.splits*)

ultimately have *get-reg-state* $sb\ c1\ aid = Some\ s1$ **and** *get-reg-state* $sb\ c2\ aid = Some\ s2$

using *assms(1)* **unfolding** *state-refines-def* **by** *metis+*

then show $s1 = s2$ **using** *assms(2)* **unfolding** *consistent-state-def* **by** *blast*
qed

39 Authenticated Cross-Domain State

An authenticated state structure equips a Merkle interface (h, bo, m) with an extraction map *extract-map* that reads a global state out of an authenticated value. The three coherence assumptions tie the authenticated layer to the state-glue layer:

* *extract-respects-merging*: merging hash-compatible authenticated values yields a state that is the join of the extracted views; * *extract-under-blinding*: a blinding of an authenticated value extracts to a refinement of the extracted state; * *extract-preserves-validity*: every extracted state is valid.

The merging and blinding assumptions are guarded by the hash-equality condition $h\ a = h\ b$, which is the ADS notion of mergeability. The two soundness theorems below discharge this condition from the ADS lemmas *merge-respects-hashes* and *hash*, and use *join* to exhibit the join as a refinement of each input so the Merkle interface is used in the proofs, not merely imported.

Reuse beyond the regulatory instance: the glue layer commits only to a single authoritative value per object identifier, so any system of that shape is a potential transplant target — for instance Merkle-replicated registries, authenticated cross-shard asset records, or the public/witness layering of zero-knowledge proofs, where *state-refines* would express that a public view is a blinding refinement of a full witness. The present entry instantiates the layer only on the regulatory state model; those other domains are not formalized here.

The extraction map is named *extract-map* rather than *extract*, because

extract is a reserved Isabelle command keyword and cannot be used as a locale parameter name.

```

locale authenticated-state =
  mk: merkle-interface h bo m
  for h :: 'd  $\Rightarrow$  'e
  and bo :: 'd  $\Rightarrow$  'd  $\Rightarrow$  bool
  and m :: 'd  $\Rightarrow$  'd  $\Rightarrow$  'd option
  and extract-map :: 'd  $\Rightarrow$  global-state option +
  assumes extract-respects-merging:
     $\llbracket h\ a = h\ b; m\ a\ b = \text{Some}\ ab; \text{extract-map}\ a = \text{Some}\ sa; \text{extract-map}\ b = \text{Some}\ sb \rrbracket$ 
     $\implies \exists\ sab. \text{extract-map}\ ab = \text{Some}\ sab \wedge \text{state-join}\ sa\ sb\ sab$ 
  and extract-under-blinding:
     $\llbracket h\ a = h\ b; bo\ a\ b; \text{extract-map}\ b = \text{Some}\ sb \rrbracket$ 
     $\implies \exists\ sa. \text{extract-map}\ a = \text{Some}\ sa \wedge \text{state-refines}\ sa\ sb$ 
  and extract-preserves-validity:
     $\text{extract-map}\ a = \text{Some}\ s \implies \text{valid-state}\ s$ 
begin

```

Blinding preserves hashes: the explicit state-level use of the *mk.hash* lemma of the Merkle interface.

```

lemma bo-hash-eq:
  assumes bo x y
  shows h x = h y
proof –
  have vimage2p h h (=) x y using mk.hash assms by (rule predicate2D)
  then show ?thesis by (simp add: vimage2p-def)
qed

```

Soundness of authenticated preservation under merging. If two authenticated values merge, then their extracted states have a valid join, and that join refines each input view. The proof calls *merge-respects-hashes* (to certify mergeability as hash-equality), *join* (to obtain the blinding relation of each input to the merge), and *hash* (via *bo-hash-eq*, to transport that blinding relation to the extracted states).

```

theorem authenticated-preservation-soundness:
  assumes mab: m a b = Some ab
  and ea: extract-map a = Some sa
  and eb: extract-map b = Some sb
  shows  $\exists\ sab. \text{extract-map}\ ab = \text{Some}\ sab \wedge \text{valid-state}\ sab$ 
     $\wedge \text{state-refines}\ sa\ sab \wedge \text{state-refines}\ sb\ sab$ 
proof –
  — ADS lemma 1: a merge exists, so the hashes agree (mergeability).
  have ex-merge:  $\exists\ x. m\ a\ b = \text{Some}\ x$  using mab by blast
  have hab: h a = h b using mk.merge-respects-hashes ex-merge by blast
  — ADS lemma 2: the merge ab is the least upper bound of a and b (join).
  have lub: bo a ab  $\wedge$  bo b ab  $\wedge$  ( $\forall\ u. bo\ a\ u \longrightarrow bo\ b\ u \longrightarrow bo\ ab\ u$ )
    using mk.join mab by blast

```

then have *boa*: *bo a ab* **and** *bob*: *bo b ab* **by** *simp-all*
 — State-level join from the merging assumption, using the hash equality *hab*.
obtain *sab* **where** *esab*: *extract-map ab = Some sab* **and** *sj*: *state-join sa sb sab*
using *extract-respects-merging*[*OF hab mab ea eb*] **by** *blast*
have *vsab*: *valid-state sab* **using** *extract-preserves-validity*[*OF esab*] .
 — ADS lemma 3 (hash): each input blinds to *ab*, hence shares its hash, so the
 extracted join refines each input view.
have *h a = h ab* **using** *bo-hash-eq*[*OF boa*] .
then have *ra*: *state-refines sa sab*
using *extract-under-blinding*[*OF ⟨h a = h ab⟩ boa esab*] *ea* **by** *auto*
have *h b = h ab* **using** *bo-hash-eq*[*OF bob*] .
then have *rb*: *state-refines sb sab*
using *extract-under-blinding*[*OF ⟨h b = h ab⟩ bob esab*] *eb* **by** *auto*
from *esab vsab ra rb* **show** *?thesis* **by** *blast*
qed

Blinded views preserve validity. A blinding *a* of an authenticated value *b* extracts to a valid state that refines the extraction of *b*. This is the need-to-know guarantee: a partial (blinded) view is still a valid, consistency-respecting view. The proof calls *hash* (via *bo-hash-eq*) to certify the hash equality that licenses the blinding coherence assumption.

theorem *blinded-view-preserves-validity*:
assumes *boab*: *bo a b* **and** *eb*: *extract-map b = Some sb*
shows $\exists sa. \text{extract-map } a = \text{Some } sa \wedge \text{valid-state } sa \wedge \text{state-refines } sa sb$
proof —
have *hab*: *h a = h b* **using** *bo-hash-eq*[*OF boab*] .
obtain *sa* **where** *ea*: *extract-map a = Some sa* **and** *sr*: *state-refines sa sb*
using *extract-under-blinding*[*OF hab boab eb*] **by** *blast*
have *valid-state sa* **using** *extract-preserves-validity*[*OF ea*] .
with *ea sr* **show** *?thesis* **by** *blast*
qed

end

40 Authenticated Cross-Domain State: the Oraculizer Instance

We discharge the model obligation of *authenticated-state* by exhibiting a concrete, non-degenerate instance. Without it the two soundness theorems above would range over a possibly empty class of structures; the instance below pins them to an extraction map that reads a genuine cross-domain state out of authenticated data.

An authenticated datum is a pair (r, P) : a consensus regulatory state *r* together with the set *P* of chains that have revealed — are known to hold — the shared asset in that state. The hash *auth-hash* commits to the consensus state *r* alone, so two data are mergeable exactly when they agree on *r*; the merge *auth-merge* then unions the revealed-chain sets, and the

blinding order *auth-bo* forgets some of them. This is the state-level reading the glue layer asks for: *auth-merge* realises the ADS *join*, *auth-bo* the ADS blinding order, and *auth-extract* sends (r, P) to the global state in which every chain of P holds the asset in state r . Because the hash fixes r , merging never forces two chains to disagree, so the extracted join is consistent and the three obligations hold with no weakening of the hypotheses.

definition *auth-hash* :: *reg-state* $\times$ *chain-id set* $\Rightarrow$ *reg-state* **where**
auth-hash $a = \text{fst } a$

definition *auth-bo* :: *reg-state* $\times$ *chain-id set* $\Rightarrow$ *reg-state* $\times$ *chain-id set* $\Rightarrow$ *bool*
where
auth-bo $a \ b \longleftrightarrow \text{fst } a = \text{fst } b \wedge \text{snd } a \subseteq \text{snd } b$

definition *auth-merge* ::
reg-state $\times$ *chain-id set* $\Rightarrow$ *reg-state* $\times$ *chain-id set* $\Rightarrow$ (*reg-state* $\times$ *chain-id set*)
option **where**
auth-merge $a \ b = (\text{if } \text{fst } a = \text{fst } b \text{ then } \text{Some } (\text{fst } a, \text{snd } a \cup \text{snd } b) \text{ else } \text{None})$

The merge is the union of revealed-chain sets, guarded by agreement on the consensus state. Idempotence, commutativity and associativity descend from the corresponding laws of set union, and the blinding order is exactly the subset order on revealed chains; hence the triple is a Merkle interface.

lemma *merkle-interface-auth* [*locale-witness*]:

merkle-interface *auth-hash* *auth-bo* *auth-merge*

proof (*rule merkle-interface.intro*)

show $\bigwedge a \ b. (\text{auth-hash } a = \text{auth-hash } b) = (\exists ab. \text{auth-merge } a \ b = \text{Some } ab)$

by (*auto simp: auth-hash-def auth-merge-def*)

show $\bigwedge a. \text{auth-merge } a \ a = \text{Some } a$

by (*simp add: auth-merge-def*)

show $\bigwedge a \ b. \text{auth-merge } a \ b = \text{auth-merge } b \ a$

by (*auto simp: auth-merge-def Un-commute*)

show $\bigwedge a \ b \ c. \text{auth-merge } a \ b \ggg \text{auth-merge } c = \text{auth-merge } b \ c \ggg \text{auth-merge } a$

by (*auto simp: auth-merge-def ac-simps split: if-splits*)

show $\bigwedge a \ b. \text{auth-bo } a \ b = (\text{auth-merge } a \ b = \text{Some } b)$

by (*auto simp: auth-bo-def auth-merge-def prod-eq-iff subset-Un-eq*)

qed

The extraction map. *auth-state* $r \ P$ is the global state in which each chain of P holds asset $\emptyset$ in regulatory state r , with nothing locked.

definition *auth-state* :: *reg-state* $\Rightarrow$ *chain-id set* $\Rightarrow$ *global-state* **where**
auth-state $r \ P =$

$\langle \mid \text{gs-chains} = (\lambda c \ a. \text{if } c \in P \wedge a = \emptyset$
 $\text{then } \text{Some } \langle \mid \text{as-reg-state} = r \mid \rangle$
 $\text{else } \text{None}),$
 $\text{gs-locks} = (\lambda a. \text{False}) \mid \rangle$

definition *auth-extract* :: *reg-state* $\times$ *chain-id set* $\Rightarrow$ *global-state option* **where**

auth-extract $a = \text{Some } (\text{auth-state } (\text{fst } a) (\text{snd } a))$

Accessors of the extracted state, read off the definition.

lemma *auth-state-get-reg*:

get-reg-state (*auth-state* r P) c $a = (\text{if } c \in P \wedge a = 0 \text{ then } \text{Some } r \text{ else } \text{None})$
by (*simp add*: *auth-state-def get-reg-state-def get-asset-state-def split: if-splits*)

lemma *auth-state-connected*:

connected-chains (*auth-state* r P) $a = (\text{if } a = 0 \text{ then } P \text{ else } \{\})$
by (*auto simp*: *connected-chains-def asset-exists-def get-asset-state-def auth-state-def split: if-splits*)

Every extracted state is valid: all chains carry the same consensus state r , so the state is consistent, and no asset is locked.

lemma *auth-state-valid*: *valid-state* (*auth-state* r P)

proof –

have *consistent-state* (*auth-state* r P)
unfolding *consistent-state-def* **by** (*auto simp*: *auth-state-get-reg split: if-splits*)
moreover have *no-locked-without-reason* (*auth-state* r P)
by (*simp add*: *no-locked-without-reason-def is-locked-def auth-state-def*)
ultimately show *?thesis* **by** (*simp add*: *valid-state-def*)

qed

A concrete global witness for the multi-domain locale: the two-chain authenticated state instantiates it outright, with no ambient hypotheses.

lemma *reg-multi-domain-concrete*:

multi-domain-preservation $\{0, \text{Suc } 0\}$ *reg-states reg-actions reg-transition*
reg-terminal (*reg-domain-state* (*auth-state* *ACTIVE* $\{0, \text{Suc } 0\}$))
by (*rule reg-multi-domain-instantiation*[*OF - auth-state-valid*]) *simp*

The merge of two views with the same consensus state extracts to the join of the two extracted states: the union of revealed chains, agreeing on r .

lemma *state-join-auth*:

state-join (*auth-state* r Pa) (*auth-state* r Pb) (*auth-state* r ($Pa \cup Pb$))
unfolding *state-join-def*
by (*auto simp*: *auth-state-connected auth-state-get-reg split: if-splits*)

A view over fewer chains refines a view over more chains with the same consensus state.

lemma *state-refines-auth*:

$Pa \subseteq Pb \implies \text{state-refines } (\text{auth-state } r \ Pa) (\text{auth-state } r \ Pb)$
unfolding *state-refines-def*
by (*auto simp*: *auth-state-connected auth-state-get-reg subset-iff split: if-splits*)

The instance. The three coherence obligations are discharged from the lemmas above: merging extracts to the join (*state-join-auth*), blinding extracts to a refinement (*state-refines-auth*), and every extracted state is valid (*auth-state-valid*). The Merkle interface obligation is *merkle-interface-auth*.

interpretation *oss-authenticated*:

authenticated-state auth-hash auth-bo auth-merge auth-extract

proof *unfold-locales*

fix *a b ab sa sb*

assume *hab: auth-hash a = auth-hash b and mab: auth-merge a b = Some ab*

and *ea: auth-extract a = Some sa and eb: auth-extract b = Some sb*

from *hab have rfst: fst a = fst b by (simp add: auth-hash-def)*

with *mab have ab-eq: ab = (fst a, snd a $\cup$ snd b) by (simp add: auth-merge-def)*

have *sa-eq: sa = auth-state (fst a) (snd a) using ea by (simp add: auth-extract-def)*

have *sb-eq: sb = auth-state (fst a) (snd b) using eb rfst by (simp add: auth-extract-def)*

have *auth-extract ab = Some (auth-state (fst a) (snd a $\cup$ snd b))*

by *(simp add: auth-extract-def ab-eq)*

moreover *have state-join sa sb (auth-state (fst a) (snd a $\cup$ snd b))*

unfolding *sa-eq sb-eq by (rule state-join-auth)*

ultimately show $\exists sab. auth-extract ab = Some sab \wedge state-join sa sb sab$

by *blast*

next

fix *a b sb*

assume *bo: auth-bo a b and eb: auth-extract b = Some sb*

from *bo have rfst: fst a = fst b and sub: snd a $\subseteq$ snd b*

by *(simp-all add: auth-bo-def)*

have *sb-eq: sb = auth-state (fst a) (snd b) using eb rfst by (simp add: auth-extract-def)*

have *auth-extract a = Some (auth-state (fst a) (snd a)) by (simp add: auth-extract-def)*

moreover *have state-refines (auth-state (fst a) (snd a)) sb*

unfolding *sb-eq using sub by (rule state-refines-auth)*

ultimately show $\exists sa. auth-extract a = Some sa \wedge state-refines sa sb$

by *blast*

next

fix *a s*

assume *auth-extract a = Some s*

then *have s = auth-state (fst a) (snd a) by (simp add: auth-extract-def)*

then *show valid-state s by (simp add: auth-state-valid)*

qed

The model is non-degenerate: the extraction map returns rich states, and the merging law is exercised by genuinely distinct partial views. The witnesses below record that a two-chain view extracts to a state with two connected chains carrying *ACTIVE*, and that merging the single-chain views over chains *0* and *Suc 0* joins them into the two-chain view.

lemma *oss-authenticated-extract-nontrivial*:

auth-extract (ACTIVE, {0, Suc 0}) = Some (auth-state ACTIVE {0, Suc 0})

connected-chains (auth-state ACTIVE {0, Suc 0}) 0 = {0, Suc 0}

get-reg-state (auth-state ACTIVE {0, Suc 0}) 0 0 = Some ACTIVE

by *(simp-all add: auth-extract-def auth-state-connected auth-state-get-reg)*

lemma *oss-authenticated-merge-nontrivial*:

auth-merge (*ACTIVE*, {0}) (*ACTIVE*, {Suc 0}) = *Some* (*ACTIVE*, {0, Suc 0})
auth-extract (*ACTIVE*, {0}) = *Some* (*auth-state ACTIVE* {0})
auth-extract (*ACTIVE*, {Suc 0}) = *Some* (*auth-state ACTIVE* {Suc 0})
state-join (*auth-state ACTIVE* {0}) (*auth-state ACTIVE* {Suc 0}) (*auth-state ACTIVE* {0, Suc 0})
using *state-join-auth*[of *ACTIVE* {0} {Suc 0}]
by (*simp-all add: auth-merge-def auth-extract-def insert-commute*)

Regression witness for the containment clause of *state-join*. The rogue candidate extends the union of the two single-chain *ACTIVE* views with a *FROZEN* holding on chain 7 — a chain revealed by neither view. The candidate is itself inconsistent, and the containment clause rejects it as a join of the two views.

definition *rogue-join-state* :: *global-state* **where**

rogue-join-state =
 (| *gs-chains* = ($\lambda c a.$ if $a = 0 \wedge (c = 0 \vee c = \text{Suc } 0)$
 then *Some* (| *as-reg-state* = *ACTIVE* |)
 else if $a = 0 \wedge c = 7$
 then *Some* (| *as-reg-state* = *FROZEN* |)
 else *None*),
 gs-locks = ($\lambda a.$ *False*) |)

lemma *rogue-join-state-get-reg*:

get-reg-state *rogue-join-state* *c* *a* =
 (if $a = 0 \wedge (c = 0 \vee c = \text{Suc } 0)$ then *Some ACTIVE*
 else if $a = 0 \wedge c = 7$ then *Some FROZEN* else *None*)
by (*simp add: rogue-join-state-def get-reg-state-def get-asset-state-def*)

lemma *rogue-join-excluded*:

$\neg$ *consistent-state* *rogue-join-state*
 $\neg$ *state-join* (*auth-state ACTIVE* {0}) (*auth-state ACTIVE* {Suc 0})
rogue-join-state

proof —

have *a*: *get-reg-state* *rogue-join-state* 0 0 = *Some ACTIVE*
and *f*: *get-reg-state* *rogue-join-state* 7 0 = *Some FROZEN*
by (*simp-all add: rogue-join-state-get-reg*)

show $\neg$ *consistent-state* *rogue-join-state*

proof

assume *consistent-state* *rogue-join-state*

then have *ACTIVE* = *FROZEN* **using** *a f* **unfolding** *consistent-state-def* **by**

blast

then show *False* **by** *simp*

qed

have *c7*: (7::nat) $\in$ *connected-chains* *rogue-join-state* 0

by (*simp add: connected-chains-def asset-exists-def get-asset-state-def*
rogue-join-state-def)

have (7::nat) $\notin$ *connected-chains* (*auth-state ACTIVE* {0}) 0

```

and ( $\gamma :: \text{nat}$ )  $\notin$  connected-chains (auth-state ACTIVE {Suc 0}) 0
by (simp-all add: auth-state-connected)
then show  $\neg$  state-join (auth-state ACTIVE {0}) (auth-state ACTIVE {Suc
0}) rogue-join-state
using c7 unfolding state-join-def by blast
qed

```

41 Inconsistency Measure on Global States

The convergence argument needs a well-founded progress measure that a synchronization strictly decreases while the global state is inconsistent. We use the number of unordered pairs of connected chains that disagree on the regulatory state of a shared asset. On a state with finitely many asset holdings this count is finite, it is zero exactly when the state is consistent, and a synchronization of an inconsistent asset strictly decreases it.

definition *finite-domain* :: *global-state* $\Rightarrow$ *bool* **where**
finite-domain *gs* $\longleftrightarrow$ *finite* {(*c*, *aid*). *asset-exists* *gs* *c* *aid*}

definition *inconsistency-set* ::
global-state $\Rightarrow$ (*chain-id* $\times$ *chain-id* $\times$ *asset-id*) *set* **where**
inconsistency-set *gs* =
 {(*c1*, *c2*, *aid*). *c1* $\in$ *connected-chains* *gs* *aid* $\wedge$ *c2* $\in$ *connected-chains* *gs* *aid*
 $\wedge$ *c1* < *c2* $\wedge$ *get-reg-state* *gs* *c1* *aid* $\neq$ *get-reg-state* *gs* *c2* *aid*}

definition *inconsistency-pairs* :: *global-state* $\Rightarrow$ *nat* **where**
inconsistency-pairs *gs* = *card* (*inconsistency-set* *gs*)

A connected chain always carries a defined regulatory state for the asset.

lemma *connected-chains-get-reg-state*:
assumes *c* $\in$ *connected-chains* *gs* *aid*
shows $\exists s.$ *get-reg-state* *gs* *c* *aid* = *Some* *s*
using *assms*
unfolding *connected-chains-def* *asset-exists-def* *get-reg-state-def* *get-asset-state-def*
by (*auto split: option.splits*)

On a finite-domain state, the connected-chain set of any asset is finite.

lemma *connected-chains-finite*:
assumes *finite-domain* *gs*
shows *finite* (*connected-chains* *gs* *aid*)
proof –
have *connected-chains* *gs* *aid* $\subseteq$ *fst* ‘ {(*c*, *a*). *asset-exists* *gs* *c* *a*}
by (*auto simp: connected-chains-def image-iff*)
moreover have *finite* (*fst* ‘ {(*c*, *a*). *asset-exists* *gs* *c* *a*})
using *assms* **unfolding** *finite-domain-def* **by** *simp*
ultimately show *?thesis* **by** (*rule finite-subset*)
qed

On a finite-domain state, the inconsistency set is finite.

```

lemma inconsistency-set-finite:
  assumes finite-domain gs
  shows finite (inconsistency-set gs)
proof -
  let  $?E = \{(c, a). \text{asset-exists } gs \ c \ a\}$ 
  have  $\text{inconsistency-set } gs \subseteq (\lambda(p, q). (\text{fst } p, \text{fst } q, \text{snd } p)) \text{ ' } (?E \times ?E)$ 
  proof
    fix  $t$  assume  $t \in \text{inconsistency-set } gs$ 
    then obtain  $c1 \ c2 \ aid$  where  $t: t = (c1, c2, aid)$ 
    and  $m1: c1 \in \text{connected-chains } gs \ aid$  and  $m2: c2 \in \text{connected-chains } gs \ aid$ 
    unfolding inconsistency-set-def by auto
    from  $m1$  have  $e1: (c1, aid) \in ?E$  by (simp add: connected-chains-def)
    from  $m2$  have  $e2: (c2, aid) \in ?E$  by (simp add: connected-chains-def)
    have  $t = (\lambda(p, q). (\text{fst } p, \text{fst } q, \text{snd } p)) ((c1, aid), (c2, aid))$  using  $t$  by simp
    then show  $t \in (\lambda(p, q). (\text{fst } p, \text{fst } q, \text{snd } p)) \text{ ' } (?E \times ?E)$ 
    using  $e1 \ e2$  by blast
  qed
  moreover have  $\text{finite } ((\lambda(p, q). (\text{fst } p, \text{fst } q, \text{snd } p)) \text{ ' } (?E \times ?E))$ 
  using assms unfolding finite-domain-def by simp
  ultimately show ?thesis by (rule finite-subset)
qed

```

The measure is zero exactly on consistent states.

```

lemma inconsistency-pairs-zero-iff-consistent:
  assumes finite-domain gs
  shows  $\text{inconsistency-pairs } gs = 0 \longleftrightarrow \text{consistent-state } gs$ 
proof
  assume  $\text{inconsistency-pairs } gs = 0$ 
  then have empty: inconsistency-set gs = {}
  using inconsistency-set-finite [OF assms] unfolding inconsistency-pairs-def by
simp
  show consistent-state gs
  unfolding consistent-state-def
  proof (intro allI impI)
    fix  $c1 \ c2 \ aid \ s1 \ s2$ 
    assume  $a1: \text{get-reg-state } gs \ c1 \ aid = \text{Some } s1$  and  $a2: \text{get-reg-state } gs \ c2 \ aid$ 
     $= \text{Some } s2$ 
    have  $in1: c1 \in \text{connected-chains } gs \ aid$  and  $in2: c2 \in \text{connected-chains } gs \ aid$ 
    using  $a1 \ a2$  unfolding connected-chains-def asset-exists-def
    get-reg-state-def get-asset-state-def by (auto split: option.splits)
    show  $s1 = s2$ 
    proof (rule ccontr)
      assume  $ne: s1 \neq s2$ 
      then have  $rne: \text{get-reg-state } gs \ c1 \ aid \neq \text{get-reg-state } gs \ c2 \ aid$  using  $a1 \ a2$ 
by simp
      have  $c1 \neq c2$  using  $a1 \ a2 \ ne$  by auto
      then consider  $c1 < c2 \mid c2 < c1$  by linarith
      then show False
      proof cases

```

```

    case 1
    then have (c1, c2, aid) ∈ inconsistency-set gs
      using in1 in2 rne unfolding inconsistency-set-def by simp
    with empty show False by simp
  next
    case 2
    then have (c2, c1, aid) ∈ inconsistency-set gs
      using in1 in2 rne unfolding inconsistency-set-def by auto
    with empty show False by simp
  qed
qed
qed
next
  assume cons: consistent-state gs
  have inconsistency-set gs = {}
  proof (rule ccontr)
    assume inconsistency-set gs ≠ {}
    then obtain c1 c2 aid where
      m1: c1 ∈ connected-chains gs aid and m2: c2 ∈ connected-chains gs aid
      and rne: get-reg-state gs c1 aid ≠ get-reg-state gs c2 aid
      unfolding inconsistency-set-def by auto
    obtain s1 where s1: get-reg-state gs c1 aid = Some s1
      using connected-chains-get-reg-state[OF m1] by blast
    obtain s2 where s2: get-reg-state gs c2 aid = Some s2
      using connected-chains-get-reg-state[OF m2] by blast
    have s1 = s2 using cons s1 s2 unfolding consistent-state-def by blast
    with s1 s2 rne show False by simp
  qed
  then show inconsistency-pairs gs = 0 unfolding inconsistency-pairs-def by
simp
qed

```

42 Structural Effect of Synchronization

The following lemmas isolate the effect of a successful synchronization on the regulatory states and on asset existence. They are proved without assuming the input state is valid (consistent and unlocked), because the convergence argument starts from an arbitrary state. The existing theorem *regulatory-homomorphism* establishes the analogous fact under the *valid-state* hypothesis; here we re-establish the parts needed for convergence with no such hypothesis.

lemma *sync-components*:

```

  assumes sync src act aid0 gs = Some gs'
  shows ∃ current-st new-st gs-locked.
    get-reg-state gs src aid0 = Some current-st
    ∧ reg-transition current-st act = Some new-st
    ∧ acquire-lock gs aid0 = Some gs-locked

```

$\wedge gs' = \text{release-lock}$
 $(\text{update-all-chains } gs\text{-locked } aid0 \text{ new-st } (\text{connected-chains } gs \text{ } aid0))$

aid0

using *assms* **unfolding** *sync-def* **by** (*auto split: option.splits simp: Let-def*)

Synchronizing asset *aid0* leaves the chain map of every other asset unchanged.

lemma *sync-chains-other-asset*:

assumes *synced*: *sync src act aid0 gs = Some gs'* **and** *ne*: *aid ≠ aid0*

shows *gs-chains gs' c aid = gs-chains gs c aid*

proof –

from *sync-components[OF synced]* **obtain** *current-st new-st gs-locked* **where**
lock: *acquire-lock gs aid0 = Some gs-locked*
and *gs'*: *gs' = release-lock*
 $(\text{update-all-chains } gs\text{-locked } aid0 \text{ new-st } (\text{connected-chains } gs \text{ } aid0))$

aid0

by *blast*

have *lc*: *gs-chains gs-locked = gs-chains gs* **using** *acquire-lock-chains[OF lock]* .

have *gs-chains* (*update-all-chains gs-locked aid0 new-st (connected-chains gs aid0)*) *c aid*
 $= gs\text{-chains } gs\text{-locked } c \text{ } aid$

using *update-all-chains-other-asset[OF ne]* **by** *simp*

then show *?thesis* **unfolding** *gs' release-lock-chains* **using** *lc* **by** *simp*

qed

lemma *sync-reg-other-asset*:

assumes *synced*: *sync src act aid0 gs = Some gs'* **and** *ne*: *aid ≠ aid0*

shows *get-reg-state gs' c aid = get-reg-state gs c aid*

unfolding *get-reg-state-def get-asset-state-def*

using *sync-chains-other-asset[OF synced ne]* **by** *simp*

After synchronizing asset *aid0*, all chains connected to *aid0* agree on its new regulatory state.

lemma *sync-synced-asset-uniform*:

assumes *synced*: *sync src act aid0 gs = Some gs'*

shows $\exists ns. \forall c' \in \text{connected-chains } gs \text{ } aid0. \text{get-reg-state } gs' \text{ } c' \text{ } aid0 = \text{Some } ns$

proof –

from *sync-components[OF synced]* **obtain** *current-st new-st gs-locked* **where**
lock: *acquire-lock gs aid0 = Some gs-locked*
and *gs'*: *gs' = release-lock*
 $(\text{update-all-chains } gs\text{-locked } aid0 \text{ new-st } (\text{connected-chains } gs \text{ } aid0))$

aid0

by *blast*

have *lc*: *gs-chains gs-locked = gs-chains gs* **using** *acquire-lock-chains[OF lock]* .

have $\forall c' \in \text{connected-chains } gs \text{ } aid0. \text{get-reg-state } gs' \text{ } c' \text{ } aid0 = \text{Some new-st}$

proof

fix *c'* **assume** *c'conn*: *c' ∈ connected-chains gs aid0*

then have *asset-exists gs c' aid0* **by** (*simp add: connected-chains-def*)

then obtain *ast* **where** *ast*: *get-asset-state gs c' aid0 = Some ast*

```

    unfolding asset-exists-def by auto
  then have astl: get-asset-state gs-locked c' aid0 = Some ast
    unfolding get-asset-state-def using lc by simp
  have get-reg-state
    (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0)) c'
aid0 = Some new-st
    using update-all-chains-reg-state[OF c'conn astl] .
  then show get-reg-state gs' c' aid0 = Some new-st
    unfolding gs' by (simp add: release-lock-reg-state)
qed
then show ?thesis by blast
qed

```

Synchronization preserves asset existence, hence the connected-chain sets.

```

lemma sync-asset-exists-preserved:
  assumes synced: sync src act aid0 gs = Some gs'
  shows asset-exists gs' c aid = asset-exists gs c aid
proof (cases aid = aid0)
  case False
  then have gs-chains gs' c aid = gs-chains gs c aid
    using sync-chains-other-asset[OF synced] by simp
  then show ?thesis unfolding asset-exists-def get-asset-state-def by simp
next
  case True
  show ?thesis
  proof (cases c ∈ connected-chains gs aid0)
    case in-conn: True
    then have e-gs: asset-exists gs c aid0 by (simp add: connected-chains-def)
    obtain ns where ∀ c' ∈ connected-chains gs aid0. get-reg-state gs' c' aid0 =
Some ns
      using sync-synced-asset-uniform[OF synced] by blast
    then have get-reg-state gs' c aid0 = Some ns using in-conn by blast
    then have asset-exists gs' c aid0
      unfolding asset-exists-def get-reg-state-def get-asset-state-def
      by (auto split: option.splits)
    with e-gs True show ?thesis by simp
  next
    case out-conn: False
    then have ngs: ¬ asset-exists gs c aid0 by (simp add: connected-chains-def)
    from sync-components[OF synced] obtain current-st new-st gs-locked where
      lock: acquire-lock gs aid0 = Some gs-locked
      and gs': gs' = release-lock
      (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0))
aid0
      by blast
    have lc: gs-chains gs-locked = gs-chains gs using acquire-lock-chains[OF lock]
    .
    have gs-chains gs' c = gs-chains gs c

```

```

    unfolding gs' release-lock-chains
    using update-all-chains-outside-targets[OF out-conn] lc by simp
  then have asset-exists gs' c aid0 = asset-exists gs c aid0
    unfolding asset-exists-def get-asset-state-def by simp
  with True show ?thesis by simp
qed
qed

```

```

lemma sync-connected-chains-preserved:
  assumes sync src act aid0 gs = Some gs'
  shows connected-chains gs' aid = connected-chains gs aid
  unfolding connected-chains-def using sync-asset-exists-preserved[OF assms] by
  simp

```

```

lemma sync-preserves-finite-domain:
  assumes finite-domain gs and sync src act aid0 gs = Some gs'
  shows finite-domain gs'
proof -
  have {(c, aid). asset-exists gs' c aid} = {(c, aid). asset-exists gs c aid}
    using sync-asset-exists-preserved[OF assms(2)] by simp
  then show ?thesis using assms(1) unfolding finite-domain-def by simp
qed

```

Synchronization preserves the unlocked invariant. This is the lock half of *valid-state-preservation*, re-established without the *consistent-state* hypothesis.

```

lemma sync-preserves-unlocked:
  assumes nl: no-locked-without-reason gs and synced: sync src act aid0 gs = Some
  gs'
  shows no-locked-without-reason gs'
proof -
  from sync-components[OF synced] obtain current-st new-st gs-locked where
    lock: acquire-lock gs aid0 = Some gs-locked
    and gs': gs' = release-lock
    (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0))
  aid0
  by blast
  from lock have locked-locks: gs-locks gs-locked = (gs-locks gs)(aid0 := True)
    unfolding acquire-lock-def is-locked-def by (auto split: if-splits)
  have upd-locks: gs-locks (update-all-chains gs-locked aid0 new-st (connected-chains
  gs aid0))
    = gs-locks gs-locked
    using update-all-chains-locks by simp
  have final-locks: gs-locks gs' = (gs-locks gs-locked)(aid0 := False)
    unfolding gs' release-lock-def using upd-locks by simp
  show ?thesis
  unfolding no-locked-without-reason-def is-locked-def
proof
  fix aid'

```

```

show  $\neg$  gs-locks gs' aid'
proof (cases aid' = aid0)
  case True then show ?thesis using final-locks by simp
next
  case False
  then have gs-locks gs' aid' = gs-locks gs aid'
    using final-locks locked-locks by simp
  also have ... = False
    using nl unfolding no-locked-without-reason-def is-locked-def by simp
  finally show ?thesis by simp
qed
qed
qed

```

43 The Critical Path: Synchronization Reduces Inconsistency

A connected chain witnesses asset existence, so a defined regulatory state places the chain in the connected set.

lemma *get-reg-state-connected*:

```

assumes get-reg-state gs c aid = Some s
shows c  $\in$  connected-chains gs aid
using assms unfolding connected-chains-def asset-exists-def
  get-reg-state-def get-asset-state-def by (auto split: option.splits)

```

The critical convergence step. Synchronizing an asset *aid0* on which two connected chains disagree strictly decreases the inconsistency measure: every inconsistent pair of *aid0* is removed (all its connected chains are driven to one regulatory state), pairs of other assets are untouched, and the witness pair certifies that at least one pair really disappears.

lemma *sync-reduces-inconsistency*:

```

assumes fin: finite-domain gs
  and synced: sync src act aid0 gs = Some gs'
  and ca: ca  $\in$  connected-chains gs aid0
  and cb: cb  $\in$  connected-chains gs aid0
  and dis: get-reg-state gs ca aid0  $\neq$  get-reg-state gs cb aid0
shows inconsistency-pairs gs' < inconsistency-pairs gs
proof —
  obtain ns where uniform:  $\forall c' \in \text{connected-chains } gs \text{ aid0. } \text{get-reg-state } gs' \ c' \text{ aid0}$ 
    = Some ns
    using sync-synced-asset-uniform[OF synced] by blast
  have conn-eq:  $\bigwedge \text{aid. } \text{connected-chains } gs' \text{ aid} = \text{connected-chains } gs \text{ aid}$ 
    using sync-connected-chains-preserved[OF synced] by simp
  — Step 1: the inconsistency set of gs' is contained in that of gs.
  have subset: inconsistency-set gs'  $\subseteq$  inconsistency-set gs
proof
  fix t assume t  $\in$  inconsistency-set gs'

```

then obtain $c1\ c2\ aid$ where $t: t = (c1, c2, aid)$
 and $i1: c1 \in \text{connected-chains } gs' aid$ and $i2: c2 \in \text{connected-chains } gs' aid$
 and $lt: c1 < c2$ and $rne: \text{get-reg-state } gs' c1 aid \neq \text{get-reg-state } gs' c2 aid$
 unfolding *inconsistency-set-def* by *auto*
 have $ane: aid \neq aid0$
 proof
 assume $eq: aid = aid0$
 have $c1 \in \text{connected-chains } gs aid0\ c2 \in \text{connected-chains } gs aid0$
 using $i1\ i2\ \text{conn-eq } eq$ by *auto*
 then have $\text{get-reg-state } gs' c1 aid0 = \text{Some } ns\ \text{get-reg-state } gs' c2 aid0 =$
Some ns
 using *uniform* by *auto*
 with $rne\ eq$ show *False* by *simp*
 qed
 have $c1 \in \text{connected-chains } gs aid\ c2 \in \text{connected-chains } gs aid$
 using $i1\ i2\ \text{conn-eq}$ by *auto*
 moreover have $\text{get-reg-state } gs\ c1 aid \neq \text{get-reg-state } gs\ c2 aid$
 using $rne\ \text{sync-reg-other-asset}[OF\ \text{synced } ane]$ by *simp*
 ultimately have $(c1, c2, aid) \in \text{inconsistency-set } gs$
 using lt unfolding *inconsistency-set-def* by *simp*
 then show $t \in \text{inconsistency-set } gs$ using t by *simp*
 qed
 — Step 2: the witness pair on $aid0$ is in gs but not in gs' .
 have $caneb: ca \neq cb$ using *dis* by *auto*
 define $c1$ where $c1 = \min\ ca\ cb$
 define $c2$ where $c2 = \max\ ca\ cb$
 have $lt: c1 < c2$ using $caneb$ unfolding $c1\text{-def } c2\text{-def}$ by *auto*
 have $c1conn: c1 \in \text{connected-chains } gs aid0$ and $c2conn: c2 \in \text{connected-chains } gs aid0$
 using $ca\ cb$ unfolding $c1\text{-def } c2\text{-def}$ by $(\text{auto } \text{simp: } \text{min-def } \text{max-def})$
 have $wdis: \text{get-reg-state } gs\ c1 aid0 \neq \text{get-reg-state } gs\ c2 aid0$
 using *dis* unfolding $c1\text{-def } c2\text{-def}$ by $(\text{auto } \text{simp: } \text{min-def } \text{max-def})$
 have $wit\text{-in}: (c1, c2, aid0) \in \text{inconsistency-set } gs$
 using $c1conn\ c2conn\ lt\ wdis$ unfolding *inconsistency-set-def* by *simp*
 have $wit\text{-out}: (c1, c2, aid0) \notin \text{inconsistency-set } gs'$
 proof
 assume $(c1, c2, aid0) \in \text{inconsistency-set } gs'$
 then have $rne: \text{get-reg-state } gs' c1 aid0 \neq \text{get-reg-state } gs' c2 aid0$
 and $c1 \in \text{connected-chains } gs' aid0$ and $c2 \in \text{connected-chains } gs' aid0$
 unfolding *inconsistency-set-def* by *auto*
 then have $c1 \in \text{connected-chains } gs aid0\ c2 \in \text{connected-chains } gs aid0$
 using *conn-eq* by *auto*
 then have $\text{get-reg-state } gs' c1 aid0 = \text{Some } ns\ \text{get-reg-state } gs' c2 aid0 = \text{Some } ns$
 using *uniform* by *auto*
 with rne show *False* by *simp*
 qed
 — Step 3: a strict subset of a finite set has strictly smaller cardinality.
 have $psub: \text{inconsistency-set } gs' \subset \text{inconsistency-set } gs$

```

    using subset wit-in wit-out by blast
  have card (inconsistency-set  $gs'$ ) < card (inconsistency-set  $gs$ )
    using psubset-card-mono[OF inconsistency-set-finite[OF fin] psub] .
  then show ?thesis unfolding inconsistency-pairs-def .
qed

```

44 Existence of a Reducing Synchronization

From any inconsistent, finite-domain, unlocked global state there is a synchronization that strictly reduces the inconsistency measure. An inconsistency exhibits two chains disagreeing on a shared asset; at least one carries a non-terminal (non-*CONFISCATED*) regulatory state, from which the *CONFISCATE* action is always enabled (*confiscate-universal*), so a synchronization succeeds and, by *sync-reduces-inconsistency*, reduces the measure.

definition *reducing-sync* ::
 $global_state \Rightarrow (chain_id \times reg_action \times asset_id) \Rightarrow bool$ **where**
 $reducing_sync\ gs\ p \longleftrightarrow (case\ p\ of\ (src,\ act,\ aid) \Rightarrow$
 $(\exists\ gs'.\ sync\ src\ act\ aid\ gs = Some\ gs' \wedge inconsistency_pairs\ gs' < inconsistency_pairs\ gs))$

lemma *inconsistent-has-reducing-sync*:

```

  assumes fin: finite-domain  $gs$  and nl: no-locked-without-reason  $gs$ 
  and inc:  $\neg consistent\_state\ gs$ 
  shows  $\exists p.\ reducing\_sync\ gs\ p$ 

```

proof –

```

  from inc obtain  $c1\ c2\ aid0\ s1\ s2$  where
     $g1: get\_reg\_state\ gs\ c1\ aid0 = Some\ s1$  and  $g2: get\_reg\_state\ gs\ c2\ aid0 = Some\ s2$ 
  and ne:  $s1 \neq s2$ 
  unfolding consistent-state-def by blast
  have conn1:  $c1 \in connected\_chains\ gs\ aid0$  using get-reg-state-connected[OF  $g1$ ]
  .
  have conn2:  $c2 \in connected\_chains\ gs\ aid0$  using get-reg-state-connected[OF  $g2$ ]
  .

```

— One of the two disagreeing chains is non-terminal.

```

obtain  $src\ ssrc$  where  $src\_conn: src \in connected\_chains\ gs\ aid0$ 
  and  $src\_reg: get\_reg\_state\ gs\ src\ aid0 = Some\ ssrc$ 
  and  $src\_nonterm: ssrc \neq CONFISCATED$ 

```

proof (*cases* $s1 = CONFISCATED$)

```

  case True
  then have  $s2 \neq CONFISCATED$  using ne by auto
  then show thesis using that conn2  $g2$  by blast

```

next

```

  case False
  then show thesis using that conn1  $g1$  by blast

```

qed

— *CONFISCATE* is enabled from any non-terminal state, and the asset is unlocked.

```

have ctrans: reg-transition ssrc CONFISCATE = Some CONFISCATED
using confiscate-universal[OF src-nonterm] .
have notlocked:  $\neg$  is-locked gs aid0
using nl unfolding no-locked-without-reason-def by simp
obtain gsl where acq: acquire-lock gs aid0 = Some gsl
using lock-acquire-success[OF notlocked] by blast
have sync src CONFISCATE aid0 gs
    = Some (release-lock
      (update-all-chains gsl aid0 CONFISCATED (connected-chains gs
aid0)) aid0)
unfolding sync-def using src-reg ctrans acq by (simp add: Let-def)
then obtain gs' where synced: sync src CONFISCATE aid0 gs = Some gs' by
blast
have dis: get-reg-state gs c1 aid0  $\neq$  get-reg-state gs c2 aid0 using g1 g2 ne by
simp
have inconsistency-pairs gs' < inconsistency-pairs gs
using sync-reduces-inconsistency[OF fin synced conn1 conn2 dis] .
then have reducing-sync gs (src, CONFISCATE, aid0)
unfolding reducing-sync-def using synced by auto
then show ?thesis by blast
qed

```

45 Terminal-Faithful Safe Recovery

A measure-reducing synchronization alone leaves the realization map two degrees of freedom that no recovery procedure should have. First, it may resolve a mere *ACTIVE/FROZEN* disagreement by broadcasting *CONFISCATED*: an indiscriminate-confiscation implementation would refine the model. Second, in the reverse direction: the broadcast overwrites every connected chain without consulting the target chain's own transition relation, so it may overwrite a recorded *CONFISCATED* holding with a non-terminal value, erasing a confiscation. The predicate *safe-recovery* closes both: a recovery is safe when it reduces the measure *and* uses *CONFISCATE* exactly on the assets that already carry the terminal state on some chain. The equivalence is needed in both directions: left-to-right forbids fresh confiscations, and right-to-left forbids completing an inconsistency on a terminal-bearing asset with anything weaker than the terminal value.

definition *terminal-present* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *bool* **where**
terminal-present *gs* *aid* $\longleftrightarrow$ ($\exists$ *c*. *get-reg-state* *gs* *c* *aid* = *Some* *CONFISCATED*)

definition *safe-recovery* ::
global-state $\Rightarrow$ (*chain-id* $\times$ *reg-action* $\times$ *asset-id*) $\Rightarrow$ *bool* **where**
safe-recovery *gs* *p* $\longleftrightarrow$ *reducing-sync* *gs* *p* $\wedge$
 (*case* *p* *of* (*src*, *act*, *aid*) $\Rightarrow$ (*act* = *CONFISCATE*) = *terminal-present* *gs* *aid*)

lemma *safe-recovery-reducing*:
safe-recovery *gs* *p* $\Longrightarrow$ *reducing-sync* *gs* *p*

by (*simp add: safe-recovery-def*)

The only regulatory action whose result is the terminal state is *CONFISCATE* itself: the transition table admits no other entry to *CONFISCATED*.

lemma *to-confiscated-only-confiscate*:

reg-transition s a = Some CONFISCATED $\implies$ a = CONFISCATE

by (*cases s; cases a*) *simp-all*

From every non-terminal regulatory state some non-*CONFISCATE* action is enabled; the witnesses are the de-escalation returns and, from *ACTIVE*, an escalation that is not a confiscation.

lemma *non-terminal-non-confiscate-step*:

assumes *s $\neq$ CONFISCATED*

shows $\exists a s'. a \neq \text{CONFISCATE} \wedge \text{reg-transition } s \ a = \text{Some } s'$

proof (*cases s*)

case *ACTIVE*

then have *FREEZE $\neq$ CONFISCATE $\wedge$ reg-transition s FREEZE = Some FROZEN* **by** *simp*

then show *?thesis* **by** *blast*

next

case *FROZEN*

then have *UNFREEZE $\neq$ CONFISCATE $\wedge$ reg-transition s UNFREEZE = Some ACTIVE* **by** *simp*

then show *?thesis* **by** *blast*

next

case *SEIZED*

then have *RELEASE $\neq$ CONFISCATE $\wedge$ reg-transition s RELEASE = Some ACTIVE* **by** *simp*

then show *?thesis* **by** *blast*

next

case *CONFISCATED*

then show *?thesis using assms* **by** *contradiction*

next

case *RESTRICTED*

then have *UNRESTRICT $\neq$ CONFISCATE $\wedge$ reg-transition s UNRESTRICT = Some ACTIVE* **by** *simp*

then show *?thesis* **by** *blast*

qed

Existence of a safe recovery from any inconsistent carrier state. The proof refines *inconsistent-has-reducing-sync* by selecting the synchronization action according to the terminal census of the inconsistent asset: if the terminal state is already present on some chain, the recovery completes it by confiscating from a non-terminal disagreeing chain; if it is absent, the recovery synchronizes with a non-*CONFISCATE* action enabled on a disagreeing chain. Either way the synchronization succeeds and strictly reduces the inconsistency measure, and the chosen action satisfies the terminal-faithfulness equivalence.

```

lemma inconsistent-has-safe-recovery:
  assumes fin: finite-domain gs and nl: no-locked-without-reason gs
    and inc:  $\neg$  consistent-state gs
  shows  $\exists p.$  safe-recovery gs p
proof –
  from inc obtain c1 c2 aid0 s1 s2 where
    g1: get-reg-state gs c1 aid0 = Some s1 and g2: get-reg-state gs c2 aid0 = Some
s2
    and ne: s1  $\neq$  s2
    unfolding consistent-state-def by blast
  have conn1: c1  $\in$  connected-chains gs aid0 using get-reg-state-connected[OF g1]
  .
  have conn2: c2  $\in$  connected-chains gs aid0 using get-reg-state-connected[OF g2]
  .
  have dis: get-reg-state gs c1 aid0  $\neq$  get-reg-state gs c2 aid0 using g1 g2 ne by
simp
  have notlocked:  $\neg$  is-locked gs aid0
    using nl unfolding no-locked-without-reason-def by simp
  obtain gsl where acq: acquire-lock gs aid0 = Some gsl
    using lock-acquire-success[OF notlocked] by blast
  show ?thesis
  proof (cases terminal-present gs aid0)
    case True
      — Terminal completion: one of the disagreeing chains is non-terminal, and
      confiscating from it completes the recorded confiscation.
      obtain src ssrc where src-reg: get-reg-state gs src aid0 = Some ssrc
        and src-nonterm: ssrc  $\neq$  CONFISCATED
      proof (cases s1 = CONFISCATED)
        case True
          then have s2  $\neq$  CONFISCATED using ne by simp
          then show thesis using that g2 by blast
        next
          case False
            then show thesis using that g1 by blast
        qed
      have ctrans: reg-transition ssrc CONFISCATE = Some CONFISCATED
        using confiscate-universal[OF src-nonterm] .
      have sync src CONFISCATE aid0 gs
        = Some (release-lock
          (update-all-chains gsl aid0 CONFISCATED (connected-chains gs
aid0)) aid0)
        unfolding sync-def using src-reg ctrans acq by (simp add: Let-def)
      then obtain gs' where synced: sync src CONFISCATE aid0 gs = Some gs'
by blast
      have inconsistency-pairs gs'  $<$  inconsistency-pairs gs
        using sync-reduces-inconsistency[OF fin synced conn1 conn2 dis] .
      then have reducing-sync gs (src, CONFISCATE, aid0)
        unfolding reducing-sync-def using synced by auto
      then have safe-recovery gs (src, CONFISCATE, aid0)

```

```

    using True by (simp add: safe-recovery-def)
  then show ?thesis by blast
next
case False
  — No terminal holding: the first disagreeing chain is non-terminal, and a non-
  CONFISCATE action is enabled there.
  have s1-nonterm: s1 ≠ CONFISCATED
    using False g1 unfolding terminal-present-def by blast
  obtain act ns where act-nc: act ≠ CONFISCATE
    and atrans: reg-transition s1 act = Some ns
    using non-terminal-non-confiscate-step[OF s1-nonterm] by blast
  have sync c1 act aid0 gs
    = Some (release-lock
      (update-all-chains gsl aid0 ns (connected-chains gs aid0)) aid0)
    unfolding sync-def using g1 atrans acq by (simp add: Let-def)
  then obtain gs' where synced: sync c1 act aid0 gs = Some gs' by blast
  have inconsistency-pairs gs' < inconsistency-pairs gs
    using sync-reduces-inconsistency[OF fin synced conn1 conn2 dis] .
  then have reducing-sync gs (c1, act, aid0)
    unfolding reducing-sync-def using synced by auto
  then have safe-recovery gs (c1, act, aid0)
    using False act-nc by (simp add: safe-recovery-def)
  then show ?thesis by blast
qed
qed

```

Terminal completion is definitional: on a terminal-bearing asset a safe recovery can only be the completing confiscation.

corollary *recovery-terminal-completion:*

```

assumes safe-recovery gs (src, act, aid) and terminal-present gs aid
shows act = CONFISCATE
using assms by (simp add: safe-recovery-def)

```

The value a successful synchronization writes onto a connected chain is the output of the regulatory transition fired at the source. This exposes, for the safety argument below, the link between the broadcast value and the transition table that produced it.

lemma *sync-connected-value:*

```

assumes synced: sync src act aid0 gs = Some gs'
and conn: c ∈ connected-chains gs aid0
shows ∃ cur new-st. get-reg-state gs src aid0 = Some cur
  ∧ reg-transition cur act = Some new-st
  ∧ get-reg-state gs' c aid0 = Some new-st

```

proof —

```

from sync-components[OF synced] obtain cur new-st gs-locked where
  cs: get-reg-state gs src aid0 = Some cur
  and tr: reg-transition cur act = Some new-st
  and lock: acquire-lock gs aid0 = Some gs-locked
  and gs': gs' = release-lock

```

```

      (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0))
aid0
  by blast
  have lc: gs-chains gs-locked = gs-chains gs using acquire-lock-chains[OF lock] .
  from conn have asset-exists gs c aid0 by (simp add: connected-chains-def)
  then obtain ast where get-asset-state gs c aid0 = Some ast
    unfolding asset-exists-def by auto
  then have astl: get-asset-state gs-locked c aid0 = Some ast
    unfolding get-asset-state-def using lc by simp
  have get-reg-state
    (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0)) c aid0
    = Some new-st
    using update-all-chains-reg-state[OF conn astl] .
  then have get-reg-state gs' c aid0 = Some new-st
    unfolding gs' by (simp add: release-lock-reg-state)
  with cs tr show ?thesis by blast
qed

```

The single-step safety payload of the terminal-faithfulness equivalence: a safe-recovery synchronization never makes the terminal state appear on an asset that did not already carry it. If the synchronized value were *CONFISCATED*, then by *to-confiscated-only-confiscate* the action was *CONFISCATE*, so by the equivalence the asset already carried the terminal state — a contradiction.

lemma *safe-recovery-sync-no-fresh-terminal*:

```

  assumes sr: safe-recovery gs (src, act, aid0)
    and synced: sync src act aid0 gs = Some gs'
    and nt: ¬ terminal-present gs aid
  shows ¬ terminal-present gs' aid

```

proof

```

  assume terminal-present gs' aid
  then obtain c where c: get-reg-state gs' c aid = Some CONFISCATED
    unfolding terminal-present-def by blast
  show False
  proof (cases aid = aid0)
    case False
    then have get-reg-state gs c aid = Some CONFISCATED
      using c sync-reg-other-asset[OF synced False] by simp
    then show False using nt unfolding terminal-present-def by blast
  next

```

```

    case True
    have c ∈ connected-chains gs' aid0
      using c True get-reg-state-connected by blast
    then have conn: c ∈ connected-chains gs aid0
      using sync-connected-chains-preserved[OF synced] by simp
    obtain cur new-st where
      tr: reg-transition cur act = Some new-st
      and val: get-reg-state gs' c aid0 = Some new-st
      using sync-connected-value[OF synced conn] by blast
  next

```

```

have new-st = CONFISCATED using c val True by simp
then have act = CONFISCATE using to-confiscated-only-confiscate tr by
blast
then have terminal-present gs aid0
using sr by (simp add: safe-recovery-def)
then show False using nt True by simp
qed
qed

```

Machine-checked regression witnesses for the two excluded behaviours. The two-chain builder places the given regulatory values on chains 0 and $\text{Suc } 0$ of asset 0 ; both witnesses evaluate by unfolding.

definition *mixed-pair-state* :: *reg-state* $\Rightarrow$ *reg-state* $\Rightarrow$ *global-state* **where**

```

mixed-pair-state v0 v1 =
  ( $\lambda$  c a. if a =  $0 \wedge c = 0$ 
    then Some ( $\lambda$  as-reg-state = v0)
    else if a =  $0 \wedge c = \text{Suc } 0$ 
    then Some ( $\lambda$  as-reg-state = v1)
    else None),
  gs-locks = ( $\lambda$  a. False)

```

lemma *mixed-pair-get-reg*:

```

get-reg-state (mixed-pair-state v0 v1) c a =
  (if a =  $0 \wedge c = 0$  then Some v0 else if a =  $0 \wedge c = \text{Suc } 0$  then Some v1 else
None)
by (simp add: mixed-pair-state-def get-reg-state-def get-asset-state-def)

```

Indiscriminate confiscation is excluded: on a plain *ACTIVE/FROZEN* disagreement with no terminal holding, no *CONFISCATE* propagation is a safe recovery, from any source chain.

lemma *blind-confiscate-excluded*:

```

 $\neg$  terminal-present (mixed-pair-state ACTIVE FROZEN) 0
 $\neg$  safe-recovery (mixed-pair-state ACTIVE FROZEN) (c, CONFISCATE, 0)
by (auto simp: terminal-present-def safe-recovery-def mixed-pair-get-reg split: if-splits)

```

Confiscation erasure is excluded: on a *CONFISCATED/ACTIVE* disagreement the terminal state is present, so no non-*CONFISCATE* broadcast is a safe recovery — a recorded confiscation cannot be revived by overwriting it with a non-terminal value.

lemma *terminal-overwrite-excluded*:

```

terminal-present (mixed-pair-state CONFISCATED ACTIVE) 0
act  $\neq$  CONFISCATE  $\implies \neg$  safe-recovery (mixed-pair-state CONFISCATED
ACTIVE) (c, act, 0)

```

proof —

```

have get-reg-state (mixed-pair-state CONFISCATED ACTIVE) 0 0 = Some
CONFISCATED
by (simp add: mixed-pair-get-reg)

```

```

then show terminal-present (mixed-pair-state CONFISCATED ACTIVE) 0
  unfolding terminal-present-def by blast
then show  $act \neq \text{CONFISCATE} \implies \neg \text{safe-recovery}$  (mixed-pair-state CONFISCATED ACTIVE) (c, act, 0)
  by (simp add: safe-recovery-def)
qed

```

46 The Oraclizer as a Converging Composition

We package the oraclizer synchronization protocol as the data of a *converging-composition*: the carrier is the set of finite-domain, unlocked global states; the operations are synchronization triples; the invariant is cross-chain consistency; the progress measure is the inconsistency count; and the realization map picks, in any inconsistent state, a terminal-faithful safe recovery — a synchronization that reduces the measure and uses *CONFISCATE* exactly on terminal-bearing assets (existence guaranteed by *inconsistent-has-safe-recovery*).

definition *oss-carrier* :: *global-state set* **where**
oss-carrier = {*gs. finite-domain gs* $\wedge$ *no-locked-without-reason gs*}

definition *oss-step* ::
global-state $\Rightarrow$ (*chain-id* $\times$ *reg-action* $\times$ *asset-id*) $\Rightarrow$ *global-state option* **where**
oss-step gs p = (*case p of* (*src*, *act*, *aid*) $\Rightarrow$ *sync src act aid gs*)

definition *oss-ops* :: (*chain-id* $\times$ *reg-action* $\times$ *asset-id*) *set* **where**
oss-ops = *UNIV*

definition *oss-guard* ::
global-state $\Rightarrow$ (*chain-id* $\times$ *reg-action* $\times$ *asset-id*) $\Rightarrow$ *bool* **where**
oss-guard gs p = (*case p of* (*src*, *act*, *aid*) $\Rightarrow$ ($\exists gs'. \text{sync src act aid gs} = \text{Some } gs'$))

definition *oss-realize* ::
node-info $\Rightarrow$ *global-state* $\Rightarrow$ (*chain-id* $\times$ *reg-action* $\times$ *asset-id*) *option* **where**
oss-realize ev gs =
 (*if consistent-state gs then* *None* *else* *Some (SOME p. safe-recovery gs p)*)

The carrier is inhabited: every authenticated extraction with a finite revealed-chain set lies in it, so the convergence development below ranges over a non-empty carrier.

lemma *auth-state-in-oss-carrier*:
assumes *finite P*
shows *auth-state r P* $\in$ *oss-carrier*
proof –
have {(*c*, *aid*). *asset-exists* (*auth-state r P*) *c aid*} = *P* $\times$ {0}
by (*auto simp: asset-exists-def get-asset-state-def auth-state-def split: if-splits*)
then have *finite-domain* (*auth-state r P*)

unfolding *finite-domain-def* using *assms* by *simp*
 moreover have *no-locked-without-reason* (*auth-state* *r* *P*)
 by (*simp* *add*: *no-locked-without-reason-def is-locked-def auth-state-def*)
 ultimately show ?thesis by (*simp* *add*: *oss-carrier-def*)
 qed

lemma *oss-step-closed*:

assumes $gs \in \text{oss-carrier}$ and $\text{oss-step } gs \ p = \text{Some } gs'$
 shows $gs' \in \text{oss-carrier}$
 proof –
 obtain *src act aid* where $p: p = (\text{src}, \text{act}, \text{aid})$ by (*cases* *p*)
 with *assms*(2) have *synced*: $\text{sync } \text{src } \text{act } \text{aid } gs = \text{Some } gs'$ by (*simp* *add*:
oss-step-def)
 from *assms*(1) have *finite-domain* *gs* and *no-locked-without-reason* *gs*
 by (*simp-all* *add*: *oss-carrier-def*)
 then have *finite-domain* *gs'* and *no-locked-without-reason* *gs'*
 using *sync-preserves-finite-domain*[*OF* - *synced*] *sync-preserves-unlocked*[*OF* -
synced] by *auto*
 then show ?thesis by (*simp* *add*: *oss-carrier-def*)
 qed

lemma *oss-guarded-preservation*:

assumes *carr*: $gs \in \text{oss-carrier}$ and *cons*: *consistent-state* *gs*
 and *synced*: $\text{oss-step } gs \ p = \text{Some } gs'$
 shows *consistent-state* *gs'*
 proof –
 obtain *src act aid* where $p: p = (\text{src}, \text{act}, \text{aid})$ by (*cases* *p*)
 with *synced* have *s*: $\text{sync } \text{src } \text{act } \text{aid } gs = \text{Some } gs'$ by (*simp* *add*: *oss-step-def*)
 from *carr* have *nl*: *no-locked-without-reason* *gs*
 by (*simp* *add*: *oss-carrier-def*)
 have *valid*: *valid-state* *gs* using *cons* *nl* by (*simp* *add*: *valid-state-def*)
 from *sync-components*[*OF* *s*] obtain *current-st new-st* *gs-locked* where
cur: *get-reg-state* *gs* *src* *aid* = *Some* *current-st*
 and *tr*: *reg-transition* *current-st* *act* = *Some* *new-st* by *blast*
 show ?thesis
 using *sync-preserves-consistent-state*[*OF* *valid cur tr s*] .
 qed

lemma *oss-realize-discharges*:

assumes *carr*: $gs \in \text{oss-carrier}$ and *r*: $\text{oss-realize } ev \ gs = \text{Some } \text{opn}$
 shows $\text{opn} \in \text{oss-ops} \wedge \text{oss-guard } gs \ \text{opn}$
 proof –
 from *r* have *ninc*: $\neg \text{consistent-state } gs$ and *opn*: $\text{opn} = (\text{SOME } p. \text{safe-recovery } gs \ p)$
 by (*auto* *simp*: *oss-realize-def split*: *if-splits*)
 from *carr* have *fd*: *finite-domain* *gs* and *nl*: *no-locked-without-reason* *gs*
 by (*simp-all* *add*: *oss-carrier-def*)
 have $\exists p. \text{safe-recovery } gs \ p$ using *inconsistent-has-safe-recovery*[*OF* *fd nl ninc*]
 .

```

then have safe-recovery gs opn using opn by (metis someI-ex)
then have reducing-sync gs opn by (rule safe-recovery-reducing)
then obtain src act aid gs' where opn = (src, act, aid) and sync src act aid
gs = Some gs'
  unfolding reducing-sync-def by (auto split: prod.splits)
  then have oss-guard gs opn by (auto simp: oss-guard-def)
  then show ?thesis by (simp add: oss-ops-def)
qed

```

lemma *oss-discharge-progresses*:

```

assumes carr: gs ∈ oss-carrier and ninc:  $\neg$  consistent-state gs
shows  $\exists$  opn gs'. oss-realize ev gs = Some opn  $\wedge$  oss-step gs opn = Some gs'
   $\wedge$  inconsistency-pairs gs' < inconsistency-pairs gs

```

proof –

```

from carr have fd: finite-domain gs and nl: no-locked-without-reason gs
  by (simp-all add: oss-carrier-def)
have ex:  $\exists$  p. safe-recovery gs p using inconsistent-has-safe-recovery[OF fd nl
ninc] .
define opn where opn = (SOME p. safe-recovery gs p)
have sr: safe-recovery gs opn unfolding opn-def using ex by (rule someI-ex)
have rs: reducing-sync gs opn using sr by (rule safe-recovery-reducing)
have realize: oss-realize ev gs = Some opn
  using ninc unfolding oss-realize-def opn-def by simp
from rs obtain src act aid gs' where
  p: opn = (src, act, aid) and s: sync src act aid gs = Some gs'
  and dec: inconsistency-pairs gs' < inconsistency-pairs gs
  unfolding reducing-sync-def by (auto split: prod.splits)
have oss-step gs opn = Some gs' using p s by (simp add: oss-step-def)
with realize dec show ?thesis by blast
qed

```

47 Guarded Bounded Convergence for the Oraclizer

Inside the D-quencer liveness context (a fair-leader assumption over a Byzantine-threshold cardinality bound), the oraclizer data above forms a *converging-composition*: the honest leader within every fairness window discharges a measure-reducing synchronization. The fairness assumption is exactly the *fair-leader* assumption of *dquencer-liveness*; window positivity is derived from bounded occurrence inside the composition locale (*window-positive*), so no positivity obligation remains here.

The fairness assumption of *dquencer-liveness* is satisfiable in every D-quencer system: the BFT threshold yields an honest node (*dquencer-system.honest-nonempty*), and the constant schedule on that node meets every fairness window. This grounds the assume-guarantee abstraction: the fair-leader hypothesis is the deterministic residue of VRF-

based election, and the system's own threshold already supplies a witness schedule, so the hypothesis is satisfiable rather than merely plausible.

context *dquencer-system*
begin

lemma *fair-schedule-exists*:

$\exists \text{ sched} :: \text{nat} \Rightarrow \text{node-info}.$
 $\forall \text{ epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$
 $\text{ni-behavior} (\text{sched } e) = \text{Honest}$

proof –

obtain *h* **where** $h \in \text{honest-nodes nodes}$
using *honest-nonempty* **by** *blast*
then have *hb*: $\text{ni-behavior } h = \text{Honest}$ **by** (*simp add: honest-nodes-def*)
define *sched* :: $\text{nat} \Rightarrow \text{node-info}$ **where** *sched* = ($\lambda \cdot. h$)
have *body*: $\forall \text{ epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound}$
 $\wedge \text{ni-behavior} (\text{sched } e) = \text{Honest}$

proof

fix *epoch* :: *nat*
have $\text{epoch} \leq \text{epoch} \wedge \text{epoch} < \text{epoch} + \text{fairness-bound}$
 $\wedge \text{ni-behavior} (\text{sched } \text{epoch}) = \text{Honest}$
using *fairness-positive hb* **by** (*simp add: sched-def*)
then show $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound}$
 $\wedge \text{ni-behavior} (\text{sched } e) = \text{Honest}$
by *blast*

qed

show *?thesis*

by (*rule exI*[**where** $P = \lambda s :: \text{nat} \Rightarrow \text{node-info}.$
 $\forall \text{ epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound}$
 $\wedge \text{ni-behavior} (s \ e) = \text{Honest}$
and $x = \text{sched}, \text{OF } \text{body}$])

qed

The Byzantine threshold is load-bearing. The fair schedule witnessed above is not idle: it instantiates the assume-guarantee liveness locale. For *every* D-quencer system the threshold therefore guarantees that *dquencer-liveness* is inhabitable — the fair-leader assumption is satisfiable rather than merely postulated. This is the downstream consumer that makes the entire Byzantine chain load-bearing: deleting any of *bft-threshold*, *honest-majority*, *honest-nonempty*, or *fair-schedule-exists* breaks the proof below. This load-bearing role is for the inhabitability result (*liveness_inhabitable*, a satisfiability witness for the locale, terminal); the headline bounded-convergence theorem does not route through it, being driven instead by the cross-chain inconsistency measure.

theorem *liveness-inhabitable*:

$\exists (ls :: \text{nat} \Rightarrow \text{node-info}) (pc :: \text{nat} \Rightarrow \text{nat}).$
 $\text{dquencer-liveness nodes } f\text{-max } \text{fairness-bound } ls \ pc$

proof –

obtain *sched* :: $\text{nat} \Rightarrow \text{node-info}$ **where** *fair*:

```

     $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$ 
     $\text{ni-behavior } (\text{sched } e) = \text{Honest}$ 
    using fair-schedule-exists by metis
  have dquencer-liveness nodes f-max fairness-bound sched ( $\lambda-. 0$ )
  proof unfold-locales
    show  $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$ 
     $\text{ni-behavior } (\text{sched } e) = \text{Honest}$ 
    using fair .
  next
    fix e :: nat
    assume  $0 < (\lambda::\text{nat}. 0::\text{nat}) e$ 
    then show  $(\lambda::\text{nat}. 0::\text{nat}) (\text{Suc } e) < (\lambda::\text{nat}. 0::\text{nat}) e$  by simp
  next
    fix e :: nat
    show  $(\lambda::\text{nat}. 0::\text{nat}) (\text{Suc } e) \leq (\lambda::\text{nat}. 0::\text{nat}) e$  by simp
  qed
  thus ?thesis by blast
qed

end

context dquencer-liveness
begin

interpretation oss: converging-composition
  oss-carrier oss-step oss-ops consistent-state oss-guard
  leader-schedule  $\lambda n. \text{ni-behavior } n = \text{Honest fairness-bound oss-realize inconsistency-pairs}$ 
proof unfold-locales
  show  $\bigwedge s \text{ opn } s'. s \in \text{oss-carrier} \implies \text{opn} \in \text{oss-ops} \implies \text{oss-step } s \text{ opn} = \text{Some } s'$ 
   $\implies s' \in \text{oss-carrier}$ 
  using oss-step-closed by blast
next
  show  $\bigwedge s \text{ opn } s'. s \in \text{oss-carrier} \implies \text{opn} \in \text{oss-ops} \implies \text{consistent-state } s$ 
   $\implies \text{oss-guard } s \text{ opn} \implies \text{oss-step } s \text{ opn} = \text{Some } s' \implies \text{consistent-state } s'$ 
  using oss-guarded-preservation by blast
next
  show  $\forall t. \exists t'. t \leq t' \wedge t' < t + \text{fairness-bound} \wedge \text{ni-behavior } (\text{leader-schedule } t') = \text{Honest}$ 
  using fair-leader .
next
  show  $\bigwedge s \text{ ev } \text{opn}. s \in \text{oss-carrier} \implies \text{ni-behavior } \text{ev} = \text{Honest} \implies \text{oss-realize } \text{ev}$ 
   $s = \text{Some } \text{opn}$ 
   $\implies \text{opn} \in \text{oss-ops} \wedge \text{oss-guard } s \text{ opn}$ 
  using oss-realize-discharges by blast
next
  show  $\bigwedge s \text{ ev}. s \in \text{oss-carrier} \implies \neg \text{consistent-state } s \implies \text{ni-behavior } \text{ev} = \text{Honest}$ 
   $\implies \exists \text{opn } s'. \text{oss-realize } \text{ev } s = \text{Some } \text{opn} \wedge \text{oss-step } s \text{ opn} = \text{Some } s'$ 

```

$\wedge \text{inconsistency-pairs } s' < \text{inconsistency-pairs } s$
using *oss-discharge-progresses* **by** *blast*
qed

The headline of this entry’s convergence layer. From an *arbitrary* finite-domain, unlocked global state — in particular with *no* assumption that the cross-chain consistency invariant holds initially — the oraclizer reaches, within *inconsistency-pairs* $gs_0 * \text{fairness-bound}$ evolution steps, a state that is valid (consistent and unlocked). Contrast *conditional-safety-preservation*, which assumes *valid-state* gs : the convergence layer frees the safety guarantee of its initial-validity hypothesis, at the price of the fair-leader context.

theorem *oraclizer-guarded-bounded-convergence*:

assumes *fin*: *finite-domain* gs_0
and *unlocked*: *no-locked-without-reason* gs_0
shows $\exists t \leq \text{oss.convergence-bound } gs_0. \exists gs_t.$
 $\text{oss.evokes-to } gs_0 \ t \ gs_t \wedge \text{valid-state } gs_t$

proof —

have $carr: gs_0 \in \text{oss-carrier}$ **using** *fin* *unlocked* **by** (*simp* *add*: *oss-carrier-def*)
obtain $t \ gs_t$ **where** $tb: t \leq \text{oss.convergence-bound } gs_0$
and $ev: \text{oss.evokes-to } gs_0 \ t \ gs_t$ **and** $cons: \text{consistent-state } gs_t$
using *oss.bounded-convergence-from-arbitrary*[*OF carr*] **by** *blast*
— The reached state remains in the carrier, hence is unlocked, hence valid.
have $gs_t = \text{oss.run-from } 0 \ gs_0 \ t$
using *ev* **by** (*simp* *add*: *oss.evokes-to-def* *oss.run-def*)
then have $gs_t \in \text{oss-carrier}$ **using** *oss.run-from-carrier*[*OF carr*] **by** *simp*
then have *no-locked-without-reason* gs_t **by** (*simp* *add*: *oss-carrier-def*)
then have *valid-state* gs_t **using** *cons* **by** (*simp* *add*: *valid-state-def*)
with $tb \ ev$ **show** *?thesis* **by** *blast*

qed

Terminal faithfulness along whole recovery runs. Every evolution step either stutters or applies a realized safe recovery, and a safe recovery never confiscates an asset that carried no terminal holding (*safe-recovery-sync-no-fresh-terminal*); by induction the guarantee extends to the entire trajectory: an asset that starts with no recorded confiscation never acquires one, no matter how the run is scheduled. Together with *recovery-terminal-completion* — on a terminal-bearing asset a safe recovery can only complete the confiscation — this pins the recovery layer, on the terminal axis, to the two behaviours the regulatory semantics admits for *CONFISCATED*: indiscriminate confiscation and confiscation erasure are both excluded. On the non-terminal axis the broadcast value is constrained by the source chain’s transition table, not by the target chains’.

lemma *oss-evolve-step-no-fresh-terminal*:

assumes $carr: gs \in \text{oss-carrier}$
and $nt: \neg \text{terminal-present } gs \text{ aid}$
shows $\neg \text{terminal-present } (\text{oss.evolve-step } t \ gs) \text{ aid}$
proof (*cases* *ni-behavior* (*leader-schedule* t) = *Honest*)

```

    case False
    then show ?thesis using nt by (simp add: oss.evolve-step-def)
next
    case True
    show ?thesis
    proof (cases oss-realize (leader-schedule t) gs)
      case None
      with True show ?thesis using nt by (simp add: oss.evolve-step-def)
    next
      case (Some opn)
      note r = this
      show ?thesis
      proof (cases oss-step gs opn)
        case None
        with True r show ?thesis using nt by (simp add: oss.evolve-step-def)
      next
        case (Some gs')
        note st = this
        from r have ninc:  $\neg$  consistent-state gs
          and opn-eq:  $\text{opn} = (\text{SOME } p. \text{ safe-recovery } gs \ p)$ 
          by (auto simp: oss-realize-def split: if-splits)
        from carr have fd: finite-domain gs and nl: no-locked-without-reason gs
          by (simp-all add: oss-carrier-def)
        have  $\exists p. \text{ safe-recovery } gs \ p$  using inconsistent-has-safe-recovery[OF fd nl
ninc] .
        then have sr: safe-recovery gs opn using opn-eq by (metis someI-ex)
        obtain src act aid0 where p:  $\text{opn} = (\text{src}, \text{act}, \text{aid0})$  by (cases opn)
        have synced:  $\text{sync } \text{src } \text{act } \text{aid0 } gs = \text{Some } gs'$  using st p by (simp add:
oss-step-def)
        have  $\neg$  terminal-present gs' aid
          using safe-recovery-sync-no-fresh-terminal[OF sr[unfolded p] synced nt] .
        with True r st show ?thesis by (simp add: oss.evolve-step-def)
      qed
    qed
  qed

lemma oss-run-from-no-fresh-terminal:
  assumes carr:  $gs_0 \in \text{oss-carrier}$ 
  and nt:  $\neg$  terminal-present  $gs_0$  aid
  shows  $\neg$  terminal-present (oss.run-from k  $gs_0$  n) aid
proof (induction n)
  case 0
  show ?case using nt by simp
next
  case (Suc n)
  have c: oss.run-from k  $gs_0$  n  $\in$  oss-carrier using oss.run-from-carrier[OF carr]
  .
  show ?case using oss-evolve-step-no-fresh-terminal[OF c Suc.IH] by simp
qed

```

```

theorem recovery-no-fresh-terminal:
  assumes carr:  $gs_0 \in \text{oss-carrier}$ 
    and nt:  $\neg \text{terminal-present } gs_0 \text{ aid}$ 
    and ev:  $\text{oss.evokes-to } gs_0 \ t \ gs_t$ 
  shows  $\neg \text{terminal-present } gs_t \text{ aid}$ 
proof –
  have  $gs_t = \text{oss.run-from } 0 \ gs_0 \ t$ 
    using ev by (simp add: oss.evokes-to-def oss.run-def)
  then show ?thesis using oss-run-from-no-fresh-terminal[OF carr nt] by simp
qed

end

```

48 Global Model Witnesses for the Liveness Hosts

The liveness interpretations above live inside *dquencer-liveness*, so they establish the consistency of their conclusions only relative to the host locale’s assumptions. The interpretations below discharge those assumptions globally, with no ambient hypotheses: a single-honest-node system with zero Byzantine budget and a unit fairness window, scheduled constantly on its one node, with no pending requests. Every assumption of *dquencer-system* and *dquencer-liveness* holds outright (the BFT threshold reads $1 \geq 3 \cdot 0 + 1$, the Byzantine census is empty, and the constant schedule meets every unit fairness window), so the assumption sets of the liveness hosts are consistent absolutely, not merely relative to a hypothetical deployment.

definition *singleton-node* :: *node-info* **where**
singleton-node = $\langle \text{ni-id} = 0, \text{ni-behavior} = \text{Honest} \rangle$

lemma *singleton-fair-leader*:
 $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + (1::\text{nat})$
 $\wedge \text{ni-behavior } ((\lambda-. \text{singleton-node}) \ e) = \text{Honest}$

proof
fix *epoch* :: *nat*
have $\text{epoch} \leq \text{epoch} \wedge \text{epoch} < \text{epoch} + 1$
 $\wedge \text{ni-behavior } ((\lambda-. \text{singleton-node}) \ \text{epoch}) = \text{Honest}$
by (*simp add: singleton-node-def*)
then show $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + 1$
 $\wedge \text{ni-behavior } ((\lambda-. \text{singleton-node}) \ e) = \text{Honest}$
by *blast*
qed

interpretation *singleton-dq: dquencer-liveness*
 $\{ \text{singleton-node} \} \ 0 \ 1 \ 0 \ 0 \ \lambda-. \text{singleton-node} \ \lambda-. \ 0$
by *unfold-locales*
(auto simp: byzantine-nodes-def singleton-node-def intro: singleton-fair-leader)

The priority sublocale receives the analogous witness: the one-message

set over the same singleton system. The well-formedness bounds hold with both maxima at 0, and priority-key distinctness on a one-element set is immediate.

definition *singleton-msg* :: *dq-message* **where**

singleton-msg =
 $\langle \langle \text{msg-action} = \text{FREEZE}, \text{msg-timestamp} = 0,$
 $\text{dqm-authority-level} = \text{National}, \text{dqm-source-node} = 0 \rangle \rangle$

interpretation *singleton-dq-priority*: *dquencer-priority-concrete*

$\{\text{singleton-node}\} \ 0 \ 1 \ 0 \ 0 \ \{\text{singleton-msg}\}$

by *unfold-locales*

(*auto simp*: *byzantine-nodes-def singleton-node-def singleton-msg-def*
intro: *singleton-fair-leader*)

A non-degenerate Byzantine witness. The singleton system above carries a zero Byzantine budget, so it meets the threshold vacuously ($1 \geq 3 \cdot 0 + 1$). The four-node system below exercises the threshold at its tight bound with a *real* fault: one Byzantine node among four ($4 \geq 3 \cdot 1 + 1$ and $\text{card}(\text{byzantine-nodes}) = 1 \leq 1$), an honest super-majority of three, scheduled constantly on one honest node. It discharges every assumption of *dquencer-liveness* with $f\text{-max} = 1$, so the Byzantine threshold is exercised at its tight bound by a real Byzantine-tagged node (a cardinality witness), not only the degenerate zero-fault case.

definition *quorum-nodes* :: *node-info set* **where**

quorum-nodes =
 $\{\langle \langle \text{ni-id} = 0, \text{ni-behavior} = \text{Byzantine} \rangle, \langle \text{ni-id} = 1, \text{ni-behavior} = \text{Honest} \rangle, \langle \text{ni-id} = 2, \text{ni-behavior} = \text{Honest} \rangle, \langle \text{ni-id} = 3, \text{ni-behavior} = \text{Honest} \rangle\}$

definition *quorum-leader* :: *nat* $\Rightarrow$ *node-info* **where**

quorum-leader = ($\lambda\cdot. \langle \langle \text{ni-id} = 1, \text{ni-behavior} = \text{Honest} \rangle \rangle$)

lemma *quorum-finite*: *finite quorum-nodes*

by (*simp add*: *quorum-nodes-def*)

lemma *quorum-nonempty*: *quorum-nodes* $\neq \{\}$

by (*simp add*: *quorum-nodes-def*)

lemma *quorum-card*: *card quorum-nodes* = 4

by (*simp add*: *quorum-nodes-def*)

lemma *quorum-byzantine-eq*:

byzantine-nodes quorum-nodes = $\{\langle \langle \text{ni-id} = 0, \text{ni-behavior} = \text{Byzantine} \rangle \rangle\}$
by (*auto simp*: *quorum-nodes-def byzantine-nodes-def*)

lemma *quorum-byz-card*: *card (byzantine-nodes quorum-nodes)* = 1

by (*simp add*: *quorum-byzantine-eq*)

interpretation *bft-quorum: dquencer-liveness*
quorum-nodes 1 1 0 0 quorum-leader $\lambda e. 1 - e$
by *unfold-locales*
(auto simp: quorum-card quorum-byz-card quorum-finite quorum-nonempty
quorum-leader-def)

Deterministic selection is exercised on a genuine conflict: two well-formed messages from different authorities over the four-node quorum system. The international confiscation outranks the regional unfreeze in the lexicographic priority order, and selection picks it deterministically — the conflict-resolution reading of Property 2 on a non-degenerate candidate set.

definition *conflict-msg-intl :: dq-message where*
conflict-msg-intl =
(| msg-action = CONFISCATE, msg-timestamp = 0,
dqm-authority-level = International, dqm-source-node = 0 |)

definition *conflict-msg-regional :: dq-message where*
conflict-msg-regional =
(| msg-action = UNFREEZE, msg-timestamp = 0,
dqm-authority-level = Regional, dqm-source-node = 0 |)

interpretation *conflict-dq-priority: dquencer-priority-concrete*
quorum-nodes 1 1 0 0 {conflict-msg-intl, conflict-msg-regional}
by *unfold-locales*
(auto simp: quorum-card quorum-byz-card quorum-finite quorum-nonempty
conflict-msg-intl-def conflict-msg-regional-def
make-priority-key-def)

lemma *conflict-selection-deterministic:*
dq-priority.select-highest conflict-dq-priority.msg-priority-keys
= Some (make-priority-key 0 0 conflict-msg-intl)

proof —

have *keys: conflict-dq-priority.msg-priority-keys*
= {make-priority-key 0 0 conflict-msg-intl,
make-priority-key 0 0 conflict-msg-regional}
by *(simp add: conflict-dq-priority.msg-priority-keys-def)*
have *ki: make-priority-key 0 0 conflict-msg-intl = (3, 0, 7, 0)*
by *(simp add: conflict-msg-intl-def make-priority-key-def)*
have *kr: make-priority-key 0 0 conflict-msg-regional = (1, 0, 2, 0)*
by *(simp add: conflict-msg-regional-def make-priority-key-def)*
show *?thesis*
unfolding *dq-priority.select-highest-def keys ki kr*
by *(auto intro!: the-equality simp: less-eq-prod-def)*

qed

end

```

theory Hierarchy
  imports Functor-Laws
begin

```

49 Sync Degree Stratification

A degree- k broadcast drives every chain $c \leq k$ that currently holds the asset aid to the regulatory value v , leaving every other chain and every other asset untouched. It is the state-level effect of synchronizing the asset across the chains coupled at degree k .

definition *broadcast-le* ::

```

nat  $\Rightarrow$  asset-id  $\Rightarrow$  reg-state  $\Rightarrow$  global-state  $\Rightarrow$  global-state where
broadcast-le k aid v gs =
  gs[| gs-chains :=
    ( $\lambda c$ . if  $c \leq k$ 
      then ( $\lambda aid'$ . if  $aid' = aid$ 
        then map-option ( $\lambda ast$ . ast[| as-reg-state := v |])
          (gs-chains gs c aid)
        else gs-chains gs c aid')
      else gs-chains gs c) |]

```

lemma *broadcast-le-locks* [*simp*]:

```

gs-locks (broadcast-le k aid v gs) = gs-locks gs
by (simp add: broadcast-le-def)

```

lemma *broadcast-le-exists* [*simp*]:

```

asset-exists (broadcast-le k aid v gs) c aid' = asset-exists gs c aid'
by (auto simp: broadcast-le-def asset-exists-def get-asset-state-def
  split: option.splits)

```

lemma *broadcast-le-reg-other-asset*:

```

aid'  $\neq$  aid  $\implies$ 
  get-reg-state (broadcast-le k aid v gs) c aid' = get-reg-state gs c aid'
by (simp add: broadcast-le-def get-reg-state-def get-asset-state-def)

```

lemma *broadcast-le-reg-out*:

```

 $\neg c \leq k \implies$ 
  get-reg-state (broadcast-le k aid v gs) c aid' = get-reg-state gs c aid'
by (simp add: broadcast-le-def get-reg-state-def get-asset-state-def)

```

lemma *broadcast-le-reg-in*:

```

c  $\leq k \implies$ 
  get-reg-state (broadcast-le k aid v gs) c aid =
    (case get-reg-state gs c aid of None  $\Rightarrow$  None | Some -  $\Rightarrow$  Some v)
by (simp add: broadcast-le-def get-reg-state-def get-asset-state-def
  map-option-case split: option.splits)

```

Pointwise reading of the chain map after a broadcast: a clean rewrite that avoids unfolding the raw record update inside nested terms.

lemma *broadcast-le-chains*:

```

gs-chains (broadcast-le k aid v gs) c =
  (if c ≤ k
   then (λaid'. if aid' = aid
                then map-option (λast. ast(| as-reg-state := v |)) (gs-chains gs c aid)
                else gs-chains gs c aid')
   else gs-chains gs c)
by (simp add: broadcast-le-def)

```

Degree demotion: *degree-forget j* drops the holdings of chain *j*, the top chain coupled one degree up. It keeps locks and every other chain.

definition *degree-forget* :: *nat* ⇒ *global-state* ⇒ *global-state* **where**

```

degree-forget j gs = gs(| gs-chains := (gs-chains gs)(j := (λaid. None)) |)

```

lemma *degree-forget-locks* [simp]:

```

gs-locks (degree-forget j gs) = gs-locks gs
by (simp add: degree-forget-def)

```

lemma *degree-forget-reg-eq*:

```

c ≠ j ⇒ get-reg-state (degree-forget j gs) c aid = get-reg-state gs c aid
by (simp add: degree-forget-def get-reg-state-def get-asset-state-def)

```

lemma *degree-forget-exists*:

```

asset-exists (degree-forget j gs) c aid =
  (c ≠ j ∧ asset-exists gs c aid)
by (simp add: degree-forget-def asset-exists-def get-asset-state-def)

```

lemma *degree-forget-chains*:

```

gs-chains (degree-forget j gs) c =
  (if c = j then (λaid. None) else gs-chains gs c)
by (simp add: degree-forget-def)

```

Demotion forgets, never invents: *degree-forget j gs* is a blinding refinement of *gs* in the sense of the glue layer of *Cross-Domain-State-Preservation.Functor-Laws*. Every chain still known after demotion carries exactly the regulatory state it had before. This is the state-level reading of the ADS blinding order *bo* (*state-refines*): degree demotion lands inside the partial-view preorder on which the authenticated layer is built, so the hierarchy sits on top of the functor laws rather than beside them.

lemma *degree-forget-refines*: *state-refines* (*degree-forget j gs*) *gs*

unfolding *state-refines-def*

proof (*intro allI impI*)

fix *aid c*

assume *c* ∈ *connected-chains* (*degree-forget j gs*) *aid*

then have *asset-exists* (*degree-forget j gs*) *c aid*

by (*simp add: connected-chains-def*)

```

then have cne:  $c \neq j$  and ce: asset-exists gs c aid
  by (simp-all add: degree-forget-exists)
from ce have  $c \in \text{connected-chains } gs \text{ } aid$  by (simp add: connected-chains-def)
moreover have get-reg-state (degree-forget j gs) c aid = get-reg-state gs c aid
  using cne by (rule degree-forget-reg-eq)
ultimately show
   $c \in \text{connected-chains } gs \text{ } aid \wedge$ 
   $\text{get-reg-state } (\text{degree-forget } j \text{ } gs) \text{ } c \text{ } aid = \text{get-reg-state } gs \text{ } c \text{ } aid$ 
by blast
qed

```

The degree- k transition step. An operation (act, aid) reads the hub chain's regulatory state for aid , fires the regulatory transition, and broadcasts the result across the chains coupled at degree k . It is undefined exactly when the hub does not hold the asset or the regulatory action is not enabled there.

definition *deg-step* ::
 $nat \Rightarrow (reg-action \times asset-id) \Rightarrow global-state \Rightarrow global-state \text{ option } \mathbf{where}$
deg-step $k \text{ } opn \text{ } gs =$
 (*case* *opn* *of* (*act*, *aid*) $\Rightarrow$
 (*case* *get-reg-state* *gs* 0 *aid* *of*
 None $\Rightarrow$ *None*
 | *Some* *s* $\Rightarrow$ (*case* *reg-transition* *s* *act* *of*
 None $\Rightarrow$ *None*
 | *Some* *s'* $\Rightarrow$ *Some* (*broadcast-le* $k \text{ } aid \text{ } s' \text{ } gs$))))

The carrier of the degree- k functor: states supported on chains $0..k$, i.e. only chains within the degree's coupling reach hold assets.

definition *deg-carrier* :: $nat \Rightarrow global-state \text{ set } \mathbf{where}$
deg-carrier $k = \{gs. \forall c \text{ } aid. \text{asset-exists } gs \text{ } c \text{ } aid \longrightarrow c \leq k\}$

50 Degree Functors and Their Natural Transformations

We package each degree as the data of a transition functor: a set of states and an action-indexed family of partial transitions. Reading the one-object action category into *Set*, a natural transformation between two such functors is a single state-component map whose naturality squares commute. The conditions below mirror, on *global-state*, the well-definedness and naturality conditions of *state-preservation* (with the action map the identity, since the regulatory action vocabulary is shared across degrees); this is the same commuting square that defines the morphisms of the functor in *Cross-Domain-State-Preservation.Functor-Laws*.

```

record gobj =
  obj-states :: global-state set
  obj-step   :: (reg-action  $\times$  asset-id)  $\Rightarrow$  global-state  $\Rightarrow$  global-state option

```

definition $F :: nat \Rightarrow gobj$ **where**

$F\ k = (\mid obj\text{-}states = deg\text{-}carrier\ k, obj\text{-}step = deg\text{-}step\ k \mid)$

definition $natural\text{-}transformation ::$

$gobj \Rightarrow gobj \Rightarrow (global\text{-}state \Rightarrow global\text{-}state) \Rightarrow bool$ **where**

$natural\text{-}transformation\ Fhi\ Flo\ \eta \longleftrightarrow$

$(\forall gs. gs \in obj\text{-}states\ Fhi \longrightarrow \eta\ gs \in obj\text{-}states\ Flo)$

$\wedge (\forall opn\ gs\ gs'. gs \in obj\text{-}states\ Fhi \longrightarrow obj\text{-}step\ Fhi\ opn\ gs = Some\ gs'$
 $\longrightarrow obj\text{-}step\ Flo\ opn\ (\eta\ gs) = Some\ (\eta\ gs'))$

$\wedge (\forall opn\ gs. gs \in obj\text{-}states\ Fhi \longrightarrow obj\text{-}step\ Fhi\ opn\ gs = None$
 $\longrightarrow obj\text{-}step\ Flo\ opn\ (\eta\ gs) = None)$

Vertical composition of natural transformations is a natural transformation. This is the *global-state*-level analogue of *preservation-compose*: the naturality square of the lower transformation is stacked on that of the upper one, exactly as the second morphism's square is stacked on the first there. It is the structural fact that makes the whole degree ladder cohere.

lemma *nt-vertical-compose*:

assumes hi : $natural\text{-}transformation\ A\ B\ \eta$

and lo : $natural\text{-}transformation\ B\ C\ \vartheta$

shows $natural\text{-}transformation\ A\ C\ (\vartheta \circ \eta)$

proof –

from hi **have**

$hi1$: $\bigwedge gs. gs \in obj\text{-}states\ A \Longrightarrow \eta\ gs \in obj\text{-}states\ B$

and $hi2$: $\bigwedge opn\ gs\ gs'. gs \in obj\text{-}states\ A \Longrightarrow obj\text{-}step\ A\ opn\ gs = Some\ gs'$
 $\Longrightarrow obj\text{-}step\ B\ opn\ (\eta\ gs) = Some\ (\eta\ gs')$

and $hi3$: $\bigwedge opn\ gs. gs \in obj\text{-}states\ A \Longrightarrow obj\text{-}step\ A\ opn\ gs = None$
 $\Longrightarrow obj\text{-}step\ B\ opn\ (\eta\ gs) = None$

unfolding $natural\text{-}transformation\text{-}def$ **by** $blast+$

from lo **have**

$lo1$: $\bigwedge gs. gs \in obj\text{-}states\ B \Longrightarrow \vartheta\ gs \in obj\text{-}states\ C$

and $lo2$: $\bigwedge opn\ gs\ gs'. gs \in obj\text{-}states\ B \Longrightarrow obj\text{-}step\ B\ opn\ gs = Some\ gs'$
 $\Longrightarrow obj\text{-}step\ C\ opn\ (\vartheta\ gs) = Some\ (\vartheta\ gs')$

and $lo3$: $\bigwedge opn\ gs. gs \in obj\text{-}states\ B \Longrightarrow obj\text{-}step\ B\ opn\ gs = None$
 $\Longrightarrow obj\text{-}step\ C\ opn\ (\vartheta\ gs) = None$

unfolding $natural\text{-}transformation\text{-}def$ **by** $blast+$

show *?thesis*

unfolding $natural\text{-}transformation\text{-}def$

proof (*intro conjI allI impI*)

fix gs **assume** $gs \in obj\text{-}states\ A$

then show $(\vartheta \circ \eta)\ gs \in obj\text{-}states\ C$

using $hi1\ lo1$ **by** (*simp add: comp-def*)

next

fix $opn\ gs\ gs'$

assume a : $gs \in obj\text{-}states\ A$ **and** s : $obj\text{-}step\ A\ opn\ gs = Some\ gs'$

have $obj\text{-}step\ C\ opn\ (\vartheta\ (\eta\ gs)) = Some\ (\vartheta\ (\eta\ gs'))$

using $lo2[OF\ hi1[OF\ a]\ hi2[OF\ a\ s]]$.

then show $obj\text{-}step\ C\ opn\ ((\vartheta \circ \eta)\ gs) = Some\ ((\vartheta \circ \eta)\ gs')$

```

    by (simp add: comp-def)
  next
    fix opn gs
    assume a: gs ∈ obj-states A and s: obj-step A opn gs = None
    have obj-step C opn ( $\vartheta$  ( $\eta$  gs)) = None
      using lo3[OF hi1[OF a] hi3[OF a s]] .
    then show obj-step C opn ( $(\vartheta \circ \eta)$  gs) = None
      by (simp add: comp-def)
  qed
qed

```

The key commutation: broadcasting at degree k after forgetting chain $Suc\ k$ equals forgetting chain $Suc\ k$ after broadcasting at degree $Suc\ k$. This is the heart of the naturality square: on chains $c \leq k$ both sides broadcast, on chain $Suc\ k$ both sides clear, and on higher chains both sides leave gs alone.

lemma *broadcast-forget-commute*:

```

  broadcast-le k aid v (degree-forget (Suc k) gs)
    = degree-forget (Suc k) (broadcast-le (Suc k) aid v gs)

```

proof (*rule global-state.equality, goal-cases chains locks more*)

case *chains*

show ?*case*

proof (*intro ext*)

fix *c aid'*

consider (*le*) $c \leq k$ | (*top*) $c = Suc\ k$ | (*gt*) $Suc\ k < c$ by *linarith*

```

  then show gs-chains (broadcast-le k aid v (degree-forget (Suc k) gs)) c aid'
    = gs-chains (degree-forget (Suc k) (broadcast-le (Suc k) aid v gs)) c

```

aid'

proof *cases*

case *le*

then have $c \neq Suc\ k$ and $c \leq Suc\ k$ by *linarith+*

with *le* show ?*thesis*

by (simp add: broadcast-le-chains degree-forget-chains)

next

case *top*

then have $\neg c \leq k$ by *linarith*

with *top* show ?*thesis*

by (simp add: broadcast-le-chains degree-forget-chains)

next

case *gt*

then have $\neg c \leq k$ and $\neg c \leq Suc\ k$ and $c \neq Suc\ k$ by *linarith+*

then show ?*thesis*

by (simp add: broadcast-le-chains degree-forget-chains)

qed

qed

next

case *locks* then show ?*case* by *simp*

next

case *more* then show ?*case* by *simp*

qed

The degree demotion *degree-forget* (*Suc k*) is a natural transformation $F (Suc\ k) \Rightarrow F\ k$: it maps degree-*Suc k* states to degree-*k* states, and its naturality square commutes (forget-after-step equals step-after-forget, in both the enabled and the disabled case). Because the hub chain *0* is never the dropped chain *Suc k*, demotion leaves the transition's enabling condition — the hub's regulatory state — untouched.

theorem *degree-natural-transformation*:

natural-transformation ($F (Suc\ k)$) ($F\ k$) (*degree-forget* (*Suc k*))

unfolding *natural-transformation-def*

proof (*intro conjI allI impI*)

fix *gs* **assume** *gsin*: $gs \in \text{obj-states } (F (Suc\ k))$

show *degree-forget* (*Suc k*) $gs \in \text{obj-states } (F\ k)$

using *gsin* **by** (*fastforce simp: F-def deg-carrier-def degree-forget-exists*)

next

fix *opn gs gs'*

assume $gs \in \text{obj-states } (F (Suc\ k))$

and *step*: *obj-step* ($F (Suc\ k)$) *opn gs* = *Some gs'*

obtain *act aid* **where** *opn*: *opn* = (*act, aid*) **by** (*cases opn*)

from *step opn* **have** *deg-step* (*Suc k*) (*act, aid*) *gs* = *Some gs'*

by (*simp add: F-def*)

then obtain *s s'* **where**

hub: *get-reg-state* *gs 0 aid* = *Some s*

and *tr*: *reg-transition* *s act* = *Some s'*

and *gs'*: $gs' = \text{broadcast-le } (Suc\ k) \text{ aid } s' \text{ gs}$

by (*auto simp: deg-step-def split: option.splits*)

have *hub'*: *get-reg-state* (*degree-forget* (*Suc k*) *gs*) *0 aid* = *Some s*

using *hub* **by** (*simp add: degree-forget-reg-eq*)

have *obj-step* ($F\ k$) *opn* (*degree-forget* (*Suc k*) *gs*)

= *Some* (*broadcast-le* *k aid s' (degree-forget (Suc k) gs)*)

using *hub' tr* **by** (*simp add: F-def deg-step-def opn*)

also have *broadcast-le* *k aid s' (degree-forget (Suc k) gs)*

= *degree-forget* (*Suc k*) (*broadcast-le* (*Suc k*) *aid s' gs*)

by (*rule broadcast-forget-commute*)

also have ... = *degree-forget* (*Suc k*) *gs'* **using** *gs'* **by** *simp*

finally show *obj-step* ($F\ k$) *opn* (*degree-forget* (*Suc k*) *gs*)

= *Some* (*degree-forget* (*Suc k*) *gs'*) .

next

fix *opn gs*

assume $gs \in \text{obj-states } (F (Suc\ k))$

and *step*: *obj-step* ($F (Suc\ k)$) *opn gs* = *None*

obtain *act aid* **where** *opn*: *opn* = (*act, aid*) **by** (*cases opn*)

from *step opn* **have** *none*: *deg-step* (*Suc k*) (*act, aid*) *gs* = *None*

by (*simp add: F-def*)

have *key*: *get-reg-state* (*degree-forget* (*Suc k*) *gs*) *0 aid* = *get-reg-state* *gs 0 aid*

by (*simp add: degree-forget-reg-eq*)

show *obj-step* ($F\ k$) *opn* (*degree-forget* (*Suc k*) *gs*) = *None*

proof (*cases get-reg-state gs 0 aid*)

```

    case None
    then show ?thesis using key by (simp add: F-def deg-step-def opn)
next
case (Some s)
with none have reg-transition s act = None
  by (simp add: deg-step-def split: option.splits)
with key Some show ?thesis by (simp add: F-def deg-step-def opn)
qed
qed

```

Natural transformations of degree functors compose. The two-step demotion $F(k+2) \Rightarrow Fk$ is a natural transformation, obtained by vertically composing $F(k+2) \Rightarrow F(k+1)$ with $F(k+1) \Rightarrow Fk$. This is what lifts the degree ladder from a point-to-point collection of forgetful maps to a structurally coherent tower: the whole chain $S_3 \Rightarrow S_2 \Rightarrow S_1 \Rightarrow S_0$ is natural at once. The degree functors share one action category as common source, so vertical composition is the only composition that arises for this tower; horizontal composition and whiskering do not typecheck here and are not claimed. This is a layer that, to our knowledge, **ADS_Functor** does not develop: it closes functors under composition but does not build a hierarchy of functors with natural transformations between them.

theorem *nt-compose:*

```

  natural-transformation (F (Suc (Suc k))) (F k)
    (degree-forget (Suc k)  $\circ$  degree-forget (Suc (Suc k)))

```

proof (*rule nt-vertical-compose*)

```

  show natural-transformation (F (Suc (Suc k))) (F (Suc k)) (degree-forget (Suc
(Suc k)))

```

```

    using degree-natural-transformation[of Suc k] .

```

next

```

  show natural-transformation (F (Suc k)) (F k) (degree-forget (Suc k))

```

```

    using degree-natural-transformation[of k] .

```

qed

51 Functoriality of the Degree Transition Functor on Action Words

The natural-transformation results above operate on single regulatory operations. To make F a functor in the genuine sense — and the degree demotion a natural transformation on the whole one-object action category, not only on its generators — we must extend the transition step to action *words* and prove it preserves the category structure (identity word and word concatenation). The action category is the free monoid of regulatory action sequences; its non-degenerate content is supplied by the regulatory transition relation, which composes the sequence into a single composite state.

fun *reg-run* :: *reg-state* $\Rightarrow$ *reg-action list* $\Rightarrow$ *reg-state option* **where**

```

  reg-run s [] = Some s

```

| $\text{reg-run } s \ (a \ \# \ w) =$
 $(\text{case } \text{reg-transition } s \ a \text{ of } \text{None} \Rightarrow \text{None} \mid \text{Some } s' \Rightarrow \text{reg-run } s' \ w)$

Regulatory composition: running a concatenated action word is running the first part and then, from the resulting state, the second. This is the monoid structure the degree functor must transport.

lemma *reg-run-append*:

$\text{reg-run } s \ (v \ @ \ w) = (\text{case } \text{reg-run } s \ v \text{ of } \text{None} \Rightarrow \text{None} \mid \text{Some } s' \Rightarrow \text{reg-run } s' \ w)$

by (*induction* v arbitrary: s) (*auto split: option.splits*)

fun *deg-run* ::

$\text{nat} \Rightarrow \text{asset-id} \Rightarrow \text{reg-action list} \Rightarrow \text{global-state} \Rightarrow \text{global-state option}$ **where**

$\text{deg-run } k \ \text{aid} \ [] \ gs = \text{Some } gs$

| $\text{deg-run } k \ \text{aid} \ (a \ \# \ w) \ gs =$

$(\text{case } \text{deg-step } k \ (a, \ \text{aid}) \ gs \text{ of } \text{None} \Rightarrow \text{None} \mid \text{Some } gs' \Rightarrow \text{deg-run } k \ \text{aid } w \ gs')$

Broadcast overwrite: a second degree- k broadcast of the same asset obliterates the first. This is not a free-monoid triviality — it holds because the broadcast set is the chains that *already* hold the asset (which the first broadcast does not change) so the second writes over exactly the same chains. It is the structural reason a word of synchronizations collapses to a single one.

lemma *broadcast-le-overwrite*:

$\text{broadcast-le } k \ \text{aid} \ v2 \ (\text{broadcast-le } k \ \text{aid} \ v1 \ gs) = \text{broadcast-le } k \ \text{aid} \ v2 \ gs$

proof (*rule global-state.equality, goal-cases chains locks more*)

case *chains*

show *?case*

proof (*intro ext*)

fix $c \ \text{aid}'$

show $gs\text{-chains } (\text{broadcast-le } k \ \text{aid} \ v2 \ (\text{broadcast-le } k \ \text{aid} \ v1 \ gs)) \ c \ \text{aid}'$

$= gs\text{-chains } (\text{broadcast-le } k \ \text{aid} \ v2 \ gs) \ c \ \text{aid}'$

proof (*cases* $c \leq k$)

case *True*

show *?thesis*

by (*cases* $\text{aid}' = \text{aid}$; *cases* $gs\text{-chains } gs \ c \ \text{aid}$)

(*simp-all add: broadcast-le-chains True*)

next

case *False*

then show *?thesis* **by** (*simp add: broadcast-le-chains*)

qed

qed

next

case *locks* **then show** *?case* **by** (*simp add: broadcast-le-locks*)

next

case *more* **then show** *?case* **by** *simp*

qed

Hub tracking: because the hub chain 0 is always within the coupling reach ($0 \leq k$), a broadcast that finds the asset at the hub leaves the hub

carrying exactly the broadcast value, so the next step in a word reads the updated state.

lemma *broadcast-le-hub*:

assumes *get-reg-state gs 0 aid = Some s0*
shows *get-reg-state (broadcast-le k aid v gs) 0 aid = Some v*
using *assms broadcast-le-reg-in[of 0 k aid v gs]* **by** *simp*

The collapse, on an already-broadcast state. Running an action word at degree k from a state of the form *broadcast-le k aid v gs* reduces to the regulatory composite *reg-run v w* broadcast once. The induction is where the degree/broadcast structure does real work: each successive step reads the hub value left by the previous broadcast (*broadcast-le-hub*) and overwrites it (*broadcast-le-overwrite*).

lemma *deg-run-on-broadcast*:

assumes *asset-exists gs 0 aid*
shows *deg-run k aid w (broadcast-le k aid v gs)*
 $= (\text{case } \text{reg-run } v \ w \text{ of } \text{None} \Rightarrow \text{None}$
 $\quad | \text{Some } sn \Rightarrow \text{Some } (\text{broadcast-le } k \text{ aid } sn \ gs))$
using *assms*
proof (*induction w arbitrary: v*)
case *Nil*
show *?case* **by** *simp*
next
case (*Cons a w*)
have *hub: get-reg-state (broadcast-le k aid v gs) 0 aid = Some v*
proof –
from *Cons.prem*s **obtain** *ast* **where** *get-asset-state gs 0 aid = Some ast*
by (*auto simp: asset-exists-def*)
then have *get-reg-state gs 0 aid = Some (as-reg-state ast)*
by (*simp add: get-reg-state-def*)
then show *?thesis* **by** (*rule broadcast-le-hub*)
qed
show *?case*
proof (*cases reg-transition v a*)
case *None*
then show *?thesis* **using** *hub* **by** (*simp add: deg-step-def*)
next
case (*Some v'*)
have *step: deg-step k (a, aid) (broadcast-le k aid v gs)*
 $= \text{Some } (\text{broadcast-le } k \text{ aid } v' \ gs)$
using *hub Some* **by** (*simp add: deg-step-def broadcast-le-overwrite*)
show *?thesis* **using** *Cons.IH[OF Cons.prem*s, *of v'*] *Some* **by** (*simp add: step*)
qed
qed

The general collapse: a non-empty word of degree- k synchronizations on an asset held at the hub equals broadcasting the composite regulatory state once.

```

lemma deg-run-collapse:
  assumes get-reg-state gs 0 aid = Some s0
  shows deg-run k aid (a # w) gs
    = (case reg-run s0 (a # w) of None  $\Rightarrow$  None
      | Some sn  $\Rightarrow$  Some (broadcast-le k aid sn gs))
proof -
  have ex: asset-exists gs 0 aid
  using assms by (auto simp: asset-exists-def get-reg-state-def get-asset-state-def
    split: option.splits)
  show ?thesis
  proof (cases reg-transition s0 a)
    case None
    then show ?thesis using assms by (simp add: deg-step-def)
  next
    case (Some s1)
    have deg-step k (a, aid) gs = Some (broadcast-le k aid s1 gs)
    using assms Some by (simp add: deg-step-def)
    then show ?thesis
    using deg-run-on-broadcast[OF ex, of k w s1] Some by simp
  qed
qed

```

The degree- k step keeps a state inside the degree- k carrier: a broadcast rewrites regulatory values but never changes which chains hold the asset, so the support bound $c \leq k$ is preserved.

```

lemma deg-step-carrier:
  assumes gs  $\in$  deg-carrier k and deg-step k (a, aid) gs = Some gs'
  shows gs'  $\in$  deg-carrier k
proof -
  from assms(2) obtain s s' where
    get-reg-state gs 0 aid = Some s and reg-transition s a = Some s'
    and gs': gs' = broadcast-le k aid s' gs
    by (auto simp: deg-step-def split: option.splits)
  have  $\bigwedge c \text{ aid''}. \text{asset-exists } gs' \ c \text{ aid''} = \text{asset-exists } gs \ c \text{ aid''}$ 
    unfolding gs' by (simp add: broadcast-le-exists)
  with assms(1) show ?thesis by (auto simp: deg-carrier-def)
qed

```

Functor laws on action words. The identity word acts as the identity, and a concatenation of words is the Kleisli composition of the two segment maps: *deg-run* is a functor from the free monoid of regulatory action sequences into the Kleisli category of the option monad on *global-state*. These two laws are the categorical skeleton; the semantic content that the functor is non-degenerate — that its action on a word is the regulatory composite broadcast once, not an arbitrary fold — is *deg-run-collapse*.

```

lemma deg-run-Nil: deg-run k aid [] = Some
  by (simp add: fun-eq-iff)

```

lemma *deg-run-append*:

deg-run k aid (v @ w) gs = Option.bind (deg-run k aid v gs) (deg-run k aid w)
by (*induction v arbitrary: gs*) (*auto split: option.splits*)

Non-degeneracy witness. A genuine two-action regulatory word — *FREEZE* then *CONFISCATE* — composes to *CONFISCATED* through the intermediate *FROZEN*, and at the degree level its second synchronization overwrites the first: the word $[FREEZE, CONFISCATE]$ broadcasts *CONFISCATED* while its prefix $[FREEZE]$ broadcasts *FROZEN*. The two differ, so the composition is not idempotent padding; the functor transports real regulatory content.

lemma *deg-run-nondegenerate*:

assumes *get-reg-state gs 0 0 = Some ACTIVE*

shows *deg-run k 0 [FREEZE, CONFISCATE] gs = Some (broadcast-le k 0 CONFISCATED gs)*

and *deg-run k 0 [FREEZE] gs = Some (broadcast-le k 0 FROZEN gs)*

and *broadcast-le k 0 CONFISCATED gs ≠ broadcast-le k 0 FROZEN gs*

proof —

have *f1: reg-transition ACTIVE FREEZE = Some FROZEN* **by** *simp*

have *f2: reg-transition FROZEN CONFISCATE = Some CONFISCATED* **by** *simp*

have *rc: reg-run ACTIVE [FREEZE, CONFISCATE] = Some CONFISCATED*
using *f1 f2* **by** *simp*

have *rf: reg-run ACTIVE [FREEZE] = Some FROZEN* **using** *f1* **by** *simp*

show *deg-run k 0 [FREEZE, CONFISCATE] gs = Some (broadcast-le k 0 CONFISCATED gs)*

using *deg-run-collapse[OF assms, of k FREEZE [CONFISCATE]] rc* **by** *simp*

show *deg-run k 0 [FREEZE] gs = Some (broadcast-le k 0 FROZEN gs)*

using *deg-run-collapse[OF assms, of k FREEZE [] rf]* **by** *simp*

have *c1: get-reg-state (broadcast-le k 0 CONFISCATED gs) 0 0 = Some CONFISCATED*

using *assms* **by** (*simp add: broadcast-le-reg-in*)

have *c2: get-reg-state (broadcast-le k 0 FROZEN gs) 0 0 = Some FROZEN*

using *assms* **by** (*simp add: broadcast-le-reg-in*)

show *broadcast-le k 0 CONFISCATED gs ≠ broadcast-le k 0 FROZEN gs*

proof

assume *broadcast-le k 0 CONFISCATED gs = broadcast-le k 0 FROZEN gs*

then have *get-reg-state (broadcast-le k 0 CONFISCATED gs) 0 0*

= get-reg-state (broadcast-le k 0 FROZEN gs) 0 0 **by** *simp*

with *c1 c2* **show** *False* **by** *simp*

qed

qed

The single-step naturality square of *degree-natural-transformation*, re-exposed at the level of *deg-step* for use inside the word induction below.

lemma *deg-step-forget-some*:

assumes *gs ∈ deg-carrier (Suc k)* **and** *deg-step (Suc k) opn gs = Some gs'*

shows *deg-step k opn (degree-forget (Suc k) gs) = Some (degree-forget (Suc k) gs')*

proof –
 obtain $a\ b$ where $opn: opn = (a, b)$ by (cases opn)
 show ?thesis
 using degree-natural-transformation[of k] *assms*
 unfolding opn natural-transformation-def F -def by auto
qed

lemma *deg-step-forget-none*:
 assumes $gs \in \text{deg-carrier } (Suc\ k)$ and $\text{deg-step } (Suc\ k)\ opn\ gs = \text{None}$
 shows $\text{deg-step } k\ opn\ (\text{degree-forget } (Suc\ k)\ gs) = \text{None}$
proof –
 obtain $a\ b$ where $opn: opn = (a, b)$ by (cases opn)
 show ?thesis
 using degree-natural-transformation[of k] *assms*
 unfolding opn natural-transformation-def F -def by auto
qed

Naturality on the whole action category. The single-step square above lifts, by induction on the action word, to the statement that degree demotion commutes with running an *entire* word: forgetting the top chain and then running at degree k equals running at degree $Suc\ k$ and then forgetting. Together with the functor laws *deg-run-Nil* and *deg-run-append*, this promotes *degree-natural-transformation* from a square on generators to a genuine natural transformation between the degree functors $F\ (Suc\ k)$ and $F\ k$ on the free action category — the functor-tower structure the entry claims.

theorem *degree-forget-natural-run*:
 assumes $gs \in \text{deg-carrier } (Suc\ k)$
 shows $\text{deg-run } k\ aid\ w\ (\text{degree-forget } (Suc\ k)\ gs)$
 $= \text{map-option } (\text{degree-forget } (Suc\ k))\ (\text{deg-run } (Suc\ k)\ aid\ w\ gs)$
 using *assms*
proof (induction w arbitrary: gs)
 case Nil
 show ?case by simp
next
 case (Cons $a\ w$)
 show ?case
proof (cases $\text{deg-step } (Suc\ k)\ (a, aid)\ gs$)
 case None
 then have $\text{deg-step } k\ (a, aid)\ (\text{degree-forget } (Suc\ k)\ gs) = \text{None}$
 using *deg-step-forget-none*[OF *Cons.prem*s] by simp
 then show ?thesis using None by simp
next
 case (Some gs')
 have $\text{deg-step } k\ (a, aid)\ (\text{degree-forget } (Suc\ k)\ gs) = \text{Some } (\text{degree-forget } (Suc\ k)\ gs')$
 using *deg-step-forget-some*[OF *Cons.prem*s *Some*] .
 moreover have $gs' \in \text{deg-carrier } (Suc\ k)$
 using *deg-step-carrier*[OF *Cons.prem*s *Some*] .
 ultimately show ?thesis using *Cons.IH Some* by simp

qed
qed

52 Degree Classes and Hierarchy Monotonicity

Each asset carries a required coupling strength, its *degree*, ranging over the four classes $S_0..S_3$. An asset belongs to the degree- k class when its required strength is at least k ; since a requirement of at least k' entails a requirement of at least any $k \leq k'$, the classes are nested (*reduction-containment*). A system, in turn, has a capability degree; it is the comparison of the asset's required degree with the system's capability degree that decides over- and under-provisioning.

The assignment of degrees to assets is operational data, not structure: the whole degree-class layer below is therefore parametric in an arbitrary assignment $asset-degree :: asset-id \Rightarrow nat$, fixed in an anonymous context. Every theorem of the layer holds for every assignment; a concrete example assignment witnesses non-vacuity at the end of the section.

Processing an asset at system degree d reconciles it across the chains the system couples (those with index $\leq d$): it broadcasts the hub chain's regulatory value over that reach. Reconciling to the hub value is the state-level action of a degree- d synchronization. The processing machinery is independent of the degree assignment, so it lives outside the parametric context.

definition $process-at-degree :: nat \Rightarrow global-state \Rightarrow asset-id \Rightarrow global-state$ **where**
 $process-at-degree\ d\ gs\ aid =$
 $(case\ get-reg-state\ gs\ 0\ aid\ of$
 $\quad None \Rightarrow gs$
 $\quad | Some\ v \Rightarrow broadcast-le\ d\ aid\ v\ gs)$

On a consistent state every chain holding the asset already agrees with the hub, so broadcasting the hub value over any reach leaves the state consistent.

lemma *broadcast-le-consistent:*

assumes *cons:* $consistent-state\ gs$ **and** *hub:* $get-reg-state\ gs\ 0\ aid = Some\ v$

shows $consistent-state\ (broadcast-le\ d\ aid\ v\ gs)$

unfolding *consistent-state-def*

proof (*intro allI impI*)

fix $c1\ c2\ aid''\ s1\ s2$

assume $g1: get-reg-state\ (broadcast-le\ d\ aid\ v\ gs)\ c1\ aid'' = Some\ s1$

and $g2: get-reg-state\ (broadcast-le\ d\ aid\ v\ gs)\ c2\ aid'' = Some\ s2$

have $val: \bigwedge c\ s. get-reg-state\ (broadcast-le\ d\ aid\ v\ gs)\ c\ aid'' = Some\ s \implies aid'' = aid \implies s = v$

proof –

fix $c\ s$

assume $gc: get-reg-state\ (broadcast-le\ d\ aid\ v\ gs)\ c\ aid'' = Some\ s$ **and** *eqaid:* $aid'' = aid$

```

show  $s = v$ 
proof (cases  $c \leq d$ )
  case True
    with gc eqaid have (case get-reg-state gs c aid of None  $\Rightarrow$  None | Some -  $\Rightarrow$ 
Some v) = Some s
      by (simp add: broadcast-le-reg-in)
      then show  $s = v$  by (auto split: option.splits)
  next
    case False
      with gc eqaid have get-reg-state gs c aid = Some s by (simp add: broadcast-le-reg-out)
      with cons hub show  $s = v$  unfolding consistent-state-def by blast
    qed
  qed
show  $s1 = s2$ 
proof (cases  $aid'' = aid$ )
  case True
    from val[OF g1 True] val[OF g2 True] show ?thesis by simp
  next
    case False
      then have get-reg-state gs c1 aid'' = Some s1 and get-reg-state gs c2 aid'' =
Some s2
        using g1 g2 by (simp-all add: broadcast-le-reg-other-asset)
      then show ?thesis using cons unfolding consistent-state-def by blast
    qed
  qed

```

Validity preservation under processing holds with no reference to degrees at all: on a consistent state the broadcast value agrees with every chain it overwrites, and the broadcast touches no lock. Isolating this as its own lemma records that the degree hypothesis of the over-provisioning headline below is not load-bearing for validity — the regime-sensitive content, where over- and under-provisioning genuinely diverge, is the pair *over-provisioning-guarantees / no-downward-safety*.

lemma *processing-preserves-validity*:

```

assumes valid-state gs
shows valid-state (process-at-degree d gs aid)
proof (cases get-reg-state gs 0 aid)
  case None
    then show ?thesis using assms by (simp add: process-at-degree-def)
  next
    case (Some v)
      from assms have c: consistent-state gs and nl: no-locked-without-reason gs
        by (simp-all add: valid-state-def)
      have consistent-state (broadcast-le d aid v gs)
        using broadcast-le-consistent[OF c Some] .
      moreover have no-locked-without-reason (broadcast-le d aid v gs)
        using nl by (simp add: no-locked-without-reason-def is-locked-def)
      ultimately show ?thesis using Some by (simp add: process-at-degree-def)

```

valid-state-def)
qed

The timestamp order `lamport_hb` underlying the causal boundary is likewise independent of the degree assignment; it is a plain strict order on integer timestamps, not a distributed causal order.

definition *lamport-hb* :: *timestamp* $\Rightarrow$ *timestamp* $\Rightarrow$ *bool* **where**
lamport-hb *t1* *t2* $\longleftrightarrow$ *t1* < *t2*

The degree-class layer, parametric in the degree assignment. Everything proved inside this context holds for an *arbitrary* assignment *asset-degree*; on export each constant and theorem generalises over the assignment.

context
fixes *asset-degree* :: *asset-id* $\Rightarrow$ *nat*
begin

definition *degree-capable* :: *nat* $\Rightarrow$ *asset-id* $\Rightarrow$ *bool* **where**
degree-capable *k* *aid* $\longleftrightarrow$ *k* $\leq$ *asset-degree* *aid*

Reduction containment: the degree classes form a descending chain, so membership in a higher class entails membership in every lower one. This is the partial-order closure of the hierarchy.

theorem *reduction-containment*:
 $k \leq k' \implies \text{degree-capable } k' \text{ aid} \longrightarrow \text{degree-capable } k \text{ aid}$
unfolding *degree-capable-def* **by** *simp*

Validity preservation in the over-provisioning regime (auxiliary). A system preserves global validity when it processes an asset. This is a *degree-free* fact: its proof is exactly *processing-preserves-validity*, and it carries no provisioning hypothesis. It is therefore recorded as an auxiliary lemma aliasing that fact, *not* the headline monotonicity result. The genuinely degree-sensitive, one-directional monotonicity is carried by *over-provisioning-guarantees* together with *no-downward-safety* below, where the provisioning hypothesis is load-bearing.

lemma *hierarchy-monotonicity*:
assumes *valid-state* *gs*
shows *valid-state* (*process-at-degree* *system-degree* *gs* *aid*)
using *assms* **by** (*rule* *processing-preserves-validity*)

The coupling guarantee a degree-*d* system actually delivers: after processing, all of the asset's *required* chains — those within its degree class, index $\leq$ *asset-degree* *aid* — that hold the asset agree on its regulatory state.

definition *guarantees-preservation* :: *nat* $\Rightarrow$ *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *bool* **where**
guarantees-preservation *d* *gs* *aid* $\longleftrightarrow$
 $(\forall c1\ c2. c1 \leq \text{asset-degree } \text{aid} \longrightarrow c2 \leq \text{asset-degree } \text{aid}$
 $\longrightarrow \text{get-reg-state } (\text{process-at-degree } d\ \text{gs}\ \text{aid})\ c1\ \text{aid} \neq \text{None}$
 $\longrightarrow \text{get-reg-state } (\text{process-at-degree } d\ \text{gs}\ \text{aid})\ c2\ \text{aid} \neq \text{None})$

$$\begin{aligned} &\longrightarrow \text{get-reg-state } (\text{process-at-degree } d \text{ } gs \text{ } aid) \text{ } c1 \text{ } aid \\ &= \text{get-reg-state } (\text{process-at-degree } d \text{ } gs \text{ } aid) \text{ } c2 \text{ } aid \end{aligned}$$

Under over-provisioning the guarantee is met: every required chain lies within the system's coupling reach ($\leq \text{asset-degree } aid \leq d$), so all are driven to the single hub value.

theorem *over-provisioning-guarantees:*

assumes *deg: asset-degree aid $\leq d$ and val: valid-state gs*

shows *guarantees-preservation d gs aid*

proof (*cases get-reg-state gs 0 aid*)

case *None*

then have *proc: process-at-degree d gs aid = gs* **by** (*simp add: process-at-degree-def*)

show *?thesis*

unfolding *guarantees-preservation-def proc*

proof (*intro allI impI*)

fix *c1 c2*

assume *c1 $\leq$ asset-degree aid and c2 $\leq$ asset-degree aid*

and *get-reg-state gs c1 aid $\neq$ None and get-reg-state gs c2 aid $\neq$ None*

then obtain *s1 s2 where*

s1: get-reg-state gs c1 aid = Some s1 and s2: get-reg-state gs c2 aid = Some

s2

by *auto*

have *s1 = s2* **using** *val s1 s2 unfolding valid-state-def consistent-state-def*

by *blast*

with *s1 s2* **show** *get-reg-state gs c1 aid = get-reg-state gs c2 aid* **by** *simp*

qed

next

case (*Some v*)

then have *proc: process-at-degree d gs aid = broadcast-le d aid v gs*

by (*simp add: process-at-degree-def*)

show *?thesis*

unfolding *guarantees-preservation-def*

proof (*intro allI impI*)

fix *c1 c2*

assume *c1d: c1 $\leq$ asset-degree aid and c2d: c2 $\leq$ asset-degree aid*

and *d1: get-reg-state (process-at-degree d gs aid) c1 aid $\neq$ None*

and *d2: get-reg-state (process-at-degree d gs aid) c2 aid $\neq$ None*

from *c1d deg* **have** *l1: c1 $\leq$ d* **by** *simp*

from *c2d deg* **have** *l2: c2 $\leq$ d* **by** *simp*

have *get-reg-state (process-at-degree d gs aid) c1 aid = Some v*

using *l1 d1 unfolding proc* **by** (*auto simp: broadcast-le-reg-in split: option.splits*)

moreover have *get-reg-state (process-at-degree d gs aid) c2 aid = Some v*

using *l2 d2 unfolding proc* **by** (*auto simp: broadcast-le-reg-in split: option.splits*)

ultimately show *get-reg-state (process-at-degree d gs aid) c1 aid*

= get-reg-state (process-at-degree d gs aid) c2 aid **by** *simp*

qed

qed

The degree hypothesis is load-bearing in its own right: from an *arbitrary* hub-defined state — with no validity assumption, so the chains may well disagree beforehand — over-provisioning alone drives every required chain to the hub value. Without it, *no-downward-safety* below exhibits the failure on exactly such a disagreeing state, so the contrast pair is matched: both directions speak about arbitrary states.

theorem *over-provisioning-reconciles*:

assumes *deg*: *asset-degree aid* $\leq d$
and *hub*: *get-reg-state gs 0 aid* = *Some v*
shows *guarantees-preservation d gs aid*

proof —

have *proc*: *process-at-degree d gs aid* = *broadcast-le d aid v gs*
using *hub* **by** (*simp add*: *process-at-degree-def*)

show *?thesis*

unfolding *guarantees-preservation-def*

proof (*intro allI impI*)

fix *c1 c2*

assume *c1d*: *c1* $\leq$ *asset-degree aid* **and** *c2d*: *c2* $\leq$ *asset-degree aid*

and *d1*: *get-reg-state (process-at-degree d gs aid) c1 aid* $\neq$ *None*

and *d2*: *get-reg-state (process-at-degree d gs aid) c2 aid* $\neq$ *None*

from *c1d deg* **have** *l1*: *c1* $\leq d$ **by** *simp*

from *c2d deg* **have** *l2*: *c2* $\leq d$ **by** *simp*

have *get-reg-state (process-at-degree d gs aid) c1 aid* = *Some v*

using *l1 d1* **unfolding** *proc* **by** (*auto simp*: *broadcast-le-reg-in split: option.splits*)

moreover **have** *get-reg-state (process-at-degree d gs aid) c2 aid* = *Some v*

using *l2 d2* **unfolding** *proc* **by** (*auto simp*: *broadcast-le-reg-in split: option.splits*)

ultimately **show** *get-reg-state (process-at-degree d gs aid) c1 aid*
 $=$ *get-reg-state (process-at-degree d gs aid) c2 aid* **by** *simp*

qed

qed

No downward safety. Under-provisioning carries no preservation guarantee: if the asset's required degree exceeds the system's capability, then some state defeats every preservation claim. The witness places the hub at *ACTIVE* and a required chain just beyond the system's reach (index *Suc system-degree*) at *FROZEN*; processing reconciles only the chains within reach, leaving the out-of-reach required chain in disagreement. This is the formal counterpart of over-provisioning safety: the monotonicity is genuinely one-directional.

theorem *no-downward-safety*:

assumes *asset-degree aid* $>$ *system-degree*

shows $\neg (\forall gs. \text{guarantees-preservation } \text{system-degree } gs \text{ aid})$

proof —

define *ast0* **where** *ast0* = $\langle \langle \text{as-reg-state} = \text{ACTIVE} \rangle \rangle$

```

define ast1 where ast1 = ( $\mid$  as-reg-state = FROZEN  $\mid$ )
define gs0 where gs0 =
  ( $\mid$  gs-chains = ( $\lambda c$ . if c = 0 then ( $\lambda a$ . if a = aid then Some ast0 else None)
    else if c = Suc system-degree then ( $\lambda a$ . if a = aid then Some
ast1 else None)
    else ( $\lambda a$ . None)),
  gs-locks = ( $\lambda a$ . False)  $\mid$ )
have c0: get-reg-state gs0 0 aid = Some ACTIVE
by (simp add: gs0-def get-reg-state-def get-asset-state-def ast0-def)
have c1: get-reg-state gs0 (Suc system-degree) aid = Some FROZEN
by (simp add: gs0-def get-reg-state-def get-asset-state-def ast1-def)
have proc: process-at-degree system-degree gs0 aid = broadcast-le system-degree
aid ACTIVE gs0
using c0 by (simp add: process-at-degree-def)
have p0: get-reg-state (process-at-degree system-degree gs0 aid) 0 aid = Some
ACTIVE
using proc c0 by (simp add: broadcast-le-reg-in)
have p1: get-reg-state (process-at-degree system-degree gs0 aid) (Suc sys-
tem-degree) aid = Some FROZEN
using proc c1 by (simp add: broadcast-le-reg-out)
have le0: ( $0::nat$ )  $\leq$  asset-degree aid by simp
have le1: Suc system-degree  $\leq$  asset-degree aid using assms by simp
have  $\neg$  guarantees-preservation system-degree gs0 aid
proof
  assume guarantees-preservation system-degree gs0 aid
  then have get-reg-state (process-at-degree system-degree gs0 aid) 0 aid
    = get-reg-state (process-at-degree system-degree gs0 aid) (Suc sys-
tem-degree) aid
  unfolding guarantees-preservation-def using le0 le1 p0 p1 by blast
with p0 p1 show False by simp
qed
then show ?thesis by blast
qed

```

The causal-consistency boundary. The classes S_1 and S_2 are separated by whether the asset requires *causal* consistency, in the sense of Lamport's happened-before order: assets at degree ≥ 2 do, lower ones do not. We model happened-before by the strict timestamp order and show the boundary is well-defined on $(asset, system)$ pairs: the requirement is a single-valued function of the asset's degree, it is met whenever the system over-provisions the asset, and the underlying happened-before relation is a strict partial order (irreflexive and asymmetric; in fact linear, as timestamps are linearly ordered), so the causal boundary it induces is itself well-formed.

The degree index corresponds to the product's synchronization-degree hierarchy $S_0..S_3$: $F k$ formalizes the *coupling breadth* of degree k — the containment and monotonicity structure of the ladder — while the per-degree synchronization semantics itself (static, unidirectional observation, bidirectional coupling, atomic state binding) is abstracted. The causal boundary

sits at $k = 2$, the S1/S2 transition below.

definition *requires-causal* :: *asset-id* $\Rightarrow$ *bool* **where**
requires-causal aid $\longleftrightarrow 2 \leq \text{asset-degree aid}$

definition *causal-consistent-at* :: *asset-id* $\Rightarrow$ *nat* $\Rightarrow$ *bool* **where**
causal-consistent-at aid d $\longleftrightarrow (\text{requires-causal aid} \longrightarrow 2 \leq d)$

theorem *boundary-well-defined*:

(*causal-consistent-at aid d* $\longleftrightarrow (2 \leq \text{asset-degree aid} \longrightarrow 2 \leq d)$)
 $\wedge (\text{asset-degree aid} \leq d \longrightarrow \text{causal-consistent-at aid d})$
 $\wedge (\forall t1\ t2. \text{lamport-hb } t1\ t2 \longrightarrow \neg \text{lamport-hb } t2\ t1)$
 $\wedge (\forall t. \neg \text{lamport-hb } t\ t)$
 $\wedge (\forall t1\ t2\ t3. \text{lamport-hb } t1\ t2 \longrightarrow \text{lamport-hb } t2\ t3 \longrightarrow \text{lamport-hb } t1\ t3)$
by (*auto simp: causal-consistent-at-def requires-causal-def lamport-hb-def*)

end

Non-vacuity of the parametric layer: a concrete example assignment, cycling the four degree classes over asset identifiers. The over-provisioning guarantee instantiates to it directly.

definition *example-degree* :: *asset-id* $\Rightarrow$ *nat* **where**
example-degree aid = *aid mod 4*

corollary *example-degree-over-provisioning*:

example-degree aid $\leq d \implies \text{valid-state gs}$
 $\implies \text{guarantees-preservation example-degree d gs aid}$
by (*rule over-provisioning-guarantees*)

Static promotion safety. A promotion re-assigns an asset's required degree; statically — between synchronizations — the change is just a different assignment parameter, so as long as the promoted degree stays within the system's capability, the over-provisioning guarantee transfers verbatim to the new assignment. In-flight promotion (a re-assignment crossing a live synchronization) is out of scope for this entry; subsequent work on retry-queue semantics covers it.

corollary *static-promotion-safety*:

assumes *asset-degree' aid* $\leq$ *system-degree* **and** *valid-state gs*
shows *guarantees-preservation asset-degree' system-degree gs aid*
using *assms* **by** (*rule over-provisioning-guarantees*)

end

theory *External-Instance*
imports *Proof-Automation*
begin

53 The TCP Endpoint State Machine (RFC 793 Subset)

A single-endpoint view of the TCP connection lifecycle, restricted to six states and five packet events. The transitions follow RFC 793: a passive open receives a *SYN* in *listen*; an active open in *SYN-sent* is completed by the peer's *SYN+ACK*; the passive side completes the handshake on the final *ACK*; a *FIN* begins the teardown, and the acknowledgement of the *FIN* closes the connection (the two *FIN-WAIT* stages of RFC 793 are folded into one, a standard simplification of the one-endpoint view). A *RST* aborts any connection in flight.

Two modelling notes. First, *CLOSED* is terminal: a re-connection is a new connection identity (a new tracker entry), not a transition of the old one. Second, a *RST* in *listen* is ignored by RFC 793; the partial transition function returns *None* there, as for every other state/event pair the subset does not admit.

```
datatype tcp-state =  
  LISTEN | SYN-SENT | SYN-RCVD | ESTABLISHED | FIN-WAIT | CLOSED
```

```
datatype tcp-event = EvSyn | EvSynAck | EvAck | EvFin | EvRst
```

```
fun tcp-transition :: tcp-state  $\Rightarrow$  tcp-event  $\Rightarrow$  tcp-state option where  
  tcp-transition LISTEN    EvSyn    = Some SYN-RCVD  
| tcp-transition SYN-SENT  EvSynAck = Some ESTABLISHED  
| tcp-transition SYN-RCVD  EvAck    = Some ESTABLISHED  
| tcp-transition ESTABLISHED EvFin   = Some FIN-WAIT  
| tcp-transition FIN-WAIT  EvAck    = Some CLOSED  
| tcp-transition SYN-SENT  EvRst    = Some CLOSED  
| tcp-transition SYN-RCVD  EvRst    = Some CLOSED  
| tcp-transition ESTABLISHED EvRst  = Some CLOSED  
| tcp-transition FIN-WAIT  EvRst    = Some CLOSED  
| tcp-transition -        -        = None
```

```
definition tcp-states :: tcp-state set where  
  tcp-states = {LISTEN, SYN-SENT, SYN-RCVD, ESTABLISHED, FIN-WAIT,  
  CLOSED}
```

```
definition tcp-events :: tcp-event set where  
  tcp-events = {EvSyn, EvSynAck, EvAck, EvFin, EvRst}
```

```
definition tcp-terminal :: tcp-state set where  
  tcp-terminal = {CLOSED}
```

```
lemma tcp-states-UNIV: tcp-states = UNIV  
  unfolding tcp-states-def by (auto, case-tac x, auto)
```

```
lemma finite-tcp-states: finite tcp-states
```

unfolding *tcp-states-def* **by** *simp*

lemma *finite-tcp-state-UNIV*: *finite* (*UNIV* :: *tcp-state set*)
using *finite-tcp-states* **by** (*simp add: tcp-states-UNIV*)

lemma *closed-terminal*: *tcp-transition CLOSED e = None*
by (*cases e*) *simp-all*

54 The Connection-Tracker Representation

The tracker entry is a structured record: a phase tag together with the peer-endpoint metadata the tracker keeps for a connection in flight. The well-formedness invariant ties the auxiliary field to the tag — the peer is recorded exactly while a connection attempt or connection exists — mirroring the layer-crossing pattern of the regulatory development. The tracker’s state space is the image of the representation map, and its transition function is the lift of the endpoint transition through that map, guarded by the image domain.

datatype *ct-phase* =
CtNew | *CtSynSent* | *CtSynRecv* | *CtEstablished* | *CtFinWait* | *CtClosed*

record *ct-entry* =
ct-state :: *ct-phase*
ct-peer :: *nat option*

definition *default-peer* :: *nat* **where**
default-peer = 0

fun *tcp-to-ct* :: *tcp-state* $\Rightarrow$ *ct-entry* **where**
tcp-to-ct LISTEN = ($\lfloor$ *ct-state* = *CtNew*, *ct-peer* = *None* $\rfloor$)
| *tcp-to-ct SYN-SENT* = ($\lfloor$ *ct-state* = *CtSynSent*, *ct-peer* = *Some default-peer*
 $\rfloor$)
| *tcp-to-ct SYN-RCVD* = ($\lfloor$ *ct-state* = *CtSynRecv*, *ct-peer* = *Some default-peer*
 $\rfloor$)
| *tcp-to-ct ESTABLISHED* = ($\lfloor$ *ct-state* = *CtEstablished*, *ct-peer* = *Some default-peer*
 $\rfloor$)
| *tcp-to-ct FIN-WAIT* = ($\lfloor$ *ct-state* = *CtFinWait*, *ct-peer* = *Some default-peer*
 $\rfloor$)
| *tcp-to-ct CLOSED* = ($\lfloor$ *ct-state* = *CtClosed*, *ct-peer* = *None* $\rfloor$)

fun *ct-to-tcp* :: *ct-entry* $\Rightarrow$ *tcp-state* **where**
ct-to-tcp p = (case *ct-state p* of
CtNew $\Rightarrow$ *LISTEN*
| *CtSynSent* $\Rightarrow$ *SYN-SENT*
| *CtSynRecv* $\Rightarrow$ *SYN-RCVD*
| *CtEstablished* $\Rightarrow$ *ESTABLISHED*
| *CtFinWait* $\Rightarrow$ *FIN-WAIT*

$$| \text{ CtClosed} \quad \Rightarrow \text{ CLOSED})$$

definition *valid-ct-entry* :: *ct-entry* $\Rightarrow$ *bool* **where**
valid-ct-entry *p* $\longleftrightarrow$ (*ct-state* *p* $\in$ {*CtNew*, *CtClosed*}) = (*ct-peer* *p* = *None*)

lemma *tcp-to-ct-valid*: *valid-ct-entry* (*tcp-to-ct* *s*)
by (*cases* *s*) (*auto simp: valid-ct-entry-def*)

lemma *ct-to-tcp-to-ct-id*: *ct-to-tcp* (*tcp-to-ct* *s*) = *s*
by (*cases* *s*) *simp-all*

definition *ct-states* :: *ct-entry* *set* **where**
ct-states = *range tcp-to-ct*

definition *ct-terminal* :: *ct-entry* *set* **where**
ct-terminal = {*tcp-to-ct CLOSED*}

55 The Tracked Event Subset and the Tracker Transition

The tracker observes the wire-visible handshake and teardown packets and does not track *RST*: an aborted connection is expired by the tracker's time-out machinery rather than by following the reset packet, the standard con-track behaviour. The source action set of the preservation morphism is therefore the strict subset of tracked events — the action-scoping pattern — and the tracker has its own event alphabet, mapped one-to-one from that subset.

datatype *ct-event* = *CT-SYN* | *CT-SYNACK* | *CT-ACK* | *CT-FIN*

definition *tracked-events* :: *tcp-event* *set* **where**
tracked-events = {*EvSyn*, *EvSynAck*, *EvAck*, *EvFin*}

definition *ct-events* :: *ct-event* *set* **where**
ct-events = {*CT-SYN*, *CT-SYNACK*, *CT-ACK*, *CT-FIN*}

fun *tcp-to-ct-event* :: *tcp-event* $\Rightarrow$ *ct-event* **where**
tcp-to-ct-event *EvSyn* = *CT-SYN*
| *tcp-to-ct-event* *EvSynAck* = *CT-SYNACK*
| *tcp-to-ct-event* *EvAck* = *CT-ACK*
| *tcp-to-ct-event* *EvFin* = *CT-FIN*
| *tcp-to-ct-event* *EvRst* = *CT-SYN*

— The last clause is unused by the morphism, since *actions_s* is the tracked subset, which excludes *EvRst*; it only makes the function total, exactly as in the escalation action map.

fun *ct-event-to-tcp* :: *ct-event* $\Rightarrow$ *tcp-event* **where**
ct-event-to-tcp *CT-SYN* = *EvSyn*

```

| ct-event-to-tcp CT-SYNACK = EvSynAck
| ct-event-to-tcp CT-ACK    = EvAck
| ct-event-to-tcp CT-FIN    = EvFin

```

definition *ct-transition* :: *ct-entry* $\Rightarrow$ *ct-event* $\Rightarrow$ *ct-entry option* **where**

```

ct-transition p e =
  (if p ∈ ct-states
   then map-option tcp-to-ct (tcp-transition (ct-to-tcp p) (ct-event-to-tcp e))
   else None)

```

lemma *ct-event-roundtrip*:

```

e ∈ tracked-events  $\implies$  ct-event-to-tcp (tcp-to-ct-event e) = e
by (auto simp: tracked-events-def)

```

56 Discharging the Instance with the Generic Methods

The helper lemmas mirror, point for point, those of the regulatory development: finiteness in the simplifier's normal form, terminal absorption, closure, domain, the naturality of the representation map on the tracked subset, and the terminal/well-definedness facts of the map. They are declared into the discharge collections, after which the machine and preservation instances close by one method invocation each.

lemma *ct-states-finite*: *finite ct-states*

```

by (auto simp: ct-states-def finite-tcp-state-UNIV intro: finite-imageI)

```

lemma *ct-terminal-subset*: *ct-terminal* $\subseteq$ *ct-states*

```

unfolding ct-terminal-def ct-states-def by blast

```

lemma *ct-closed-absorbing*:

```

[[ p ∈ ct-terminal; e ∈ ct-events ]]  $\implies$  ct-transition p e = None
by (auto simp: ct-terminal-def ct-transition-def ct-to-tcp-to-ct-id closed-terminal)

```

lemma *ct-transition-closed*:

```

[[ p ∈ ct-states; e ∈ ct-events; ct-transition p e = Some p' ]]  $\implies$  p' ∈ ct-states
unfolding ct-transition-def ct-states-def
by (auto split: if-splits option.splits)

```

lemma *ct-transition-outside-states*:

```

p ∉ ct-states  $\implies$  ct-transition p e = None
by (simp add: ct-transition-def)

```

lemma *tcp-to-ct-in-states*: *tcp-to-ct s* $\in$ *ct-states*

```

by (simp add: ct-states-def)

```

lemma *tcp-to-ct-terminal*:

```

s ∈ tcp-terminal  $\implies$  tcp-to-ct s ∈ ct-terminal

```

by (*auto simp: tcp-terminal-def ct-terminal-def*)

lemma *tcp-to-ct-naturality-some*:

$\llbracket s \in \text{tcp-states}; e \in \text{tracked-events}; \text{tcp-transition } s \ e = \text{Some } s' \rrbracket$
 $\implies \text{ct-transition } (\text{tcp-to-ct } s) (\text{tcp-to-ct-event } e) = \text{Some } (\text{tcp-to-ct } s')$
by (*simp add: ct-transition-def tcp-to-ct-in-states ct-to-tcp-to-ct-id*
ct-event-roundtrip
del: ct-to-tcp.simps tcp-to-ct.simps)

lemma *tcp-to-ct-naturality-none*:

$\llbracket s \in \text{tcp-states}; e \in \text{tracked-events}; \text{tcp-transition } s \ e = \text{None} \rrbracket$
 $\implies \text{ct-transition } (\text{tcp-to-ct } s) (\text{tcp-to-ct-event } e) = \text{None}$
by (*simp add: ct-transition-def tcp-to-ct-in-states ct-to-tcp-to-ct-id*
ct-event-roundtrip
del: ct-to-tcp.simps tcp-to-ct.simps)

lemmas [*discharge-simps*] =

tcp-states-UNIV finite-tcp-state-UNIV
tcp-events-def tracked-events-def ct-events-def
tcp-terminal-def closed-terminal
ct-states-finite ct-terminal-subset
ct-closed-absorbing ct-transition-outside-states
tcp-to-ct-naturality-some tcp-to-ct-naturality-none
tcp-to-ct-terminal tcp-to-ct-in-states

lemmas [*discharge-intros*] =

ct-transition-closed

lemmas [*discharge-dels*] =

tcp-to-ct.simps ct-to-tcp.simps

The two state machines and the preservation morphism, each discharged by the corresponding generic method. No obligation is weakened: the state-ments are the full locale predicates.

theorem *tcp-state-machine*:

state-machine tcp-states tcp-events tcp-transition tcp-terminal
by *discharge-state-machine*

theorem *conntrack-state-machine*:

state-machine ct-states ct-events ct-transition ct-terminal
by *discharge-state-machine*

theorem *tcp-conntrack-preservation*:

state-preservation tcp-states tracked-events tcp-transition tcp-terminal
ct-states ct-events ct-transition ct-terminal
tcp-to-ct tcp-to-ct-event
by *discharge-preservation*

Registered form of the morphism, giving access to the locale's sequential payload: every tracked packet sequence accepted by the endpoint is mirrored,

packet for packet, by the tracker, ending in the corresponding tracker entry.

interpretation *tcp-ct*:

state-preservation tcp-states tracked-events tcp-transition tcp-terminal
ct-states ct-events ct-transition ct-terminal
tcp-to-ct tcp-to-ct-event
by (*rule tcp-conntrack-preservation*)

corollary *tracked-sequence-mirrored*:

assumes $s \in \text{tcp-states}$
and $\forall e \in \text{set } es. e \in \text{tracked-events}$
and $\text{tcp-ct.source.apply-actions } s \text{ es} = \text{Some } s'$
shows $\text{tcp-ct.target.apply-actions } (\text{tcp-to-ct } s) (\text{map tcp-to-ct-event } es)$
 $= \text{Some } (\text{tcp-to-ct } s')$
using *assms* **by** (*rule tcp-ct.sequential-preservation*)

Non-vacuity witnesses: the three-way handshake is accepted by the end-point machine, and its tracked image steps the tracker entry from the fresh entry to the established entry.

lemma *three-way-handshake-endpoint*:

tcp-transition LISTEN EvSyn = Some SYN-RCVD
tcp-transition SYN-RCVD EvAck = Some ESTABLISHED
by *simp-all*

lemma *three-way-handshake-tracked*:

ct-transition (tcp-to-ct LISTEN) CT-SYN = Some (tcp-to-ct SYN-RCVD)
ct-transition (tcp-to-ct SYN-RCVD) CT-ACK = Some (tcp-to-ct ESTABLISHED)
by (*simp-all add: ct-transition-def tcp-to-ct-in-states ct-to-tcp-to-ct-id*
 $\text{del: ct-to-tcp.simps tcp-to-ct.simps}$)

end

theory *Canton-Bridge*

imports

Functor-Laws
ADS-Constructor.ADS-Construction
ADS-Constructor.Canton-Transaction-Tree

begin

57 Auxiliary list predicate for synchronization sequences

A blinding path: each element of the list is a blinding of its successor. As blindings narrow what is revealed, the last element is the most-revealed view of the sequence and every earlier element is a partial view of it.

definition *blinding-path* :: $('d \Rightarrow 'd \Rightarrow \text{bool}) \Rightarrow 'd \text{ list} \Rightarrow \text{bool}$ **where**

$$\text{blinding-path } bl \text{ } xs \longleftrightarrow (\forall i. \text{Suc } i < \text{length } xs \longrightarrow bl \text{ } (xs ! i) \text{ } (xs ! \text{Suc } i))$$

58 Composite functor laws: cross-domain preservation after the ADS Merkle functor

Inside *authenticated-state* the ADS layer supplies a thin category: objects are authenticated values, and the unique morphism $a \rightarrow b$ is the blinding relation $bo \ a \ b$, a preorder (reflexive and transitive by the Merkle interface). The cross-domain layer supplies another thin category: objects are global states and the morphism $s \rightarrow s'$ is *state-refines* $s \ s'$, also a preorder. The extraction map *extract-map* is the composite functor from the first category to the second: it sends a value to the state it extracts to, and a blinding morphism to a refinement morphism. The next four results are the category axioms of this composite functor together with the associativity of the lifted merge (join) algebra.

context *authenticated-state*
begin

Identity is preserved: the identity (blinding) morphism on a , which exists by reflexivity of bo , is sent to the identity (refinement) morphism on the extracted state.

lemma *cdsp-ads-id*:

assumes *extract-map* $a = \text{Some } sa$
shows $bo \ a \ a \wedge \text{valid-state } sa \wedge \text{state-refines } sa \ sa$
using *mk.reflp extract-preserves-validity[OF assms] state-refines-refl*
by (*simp add: reflp-def*)

The functor's action on a single morphism: a blinding $a \rightarrow b$ is sent to a refinement $\text{extract } a \rightarrow \text{extract } b$, with both extracted states valid. This is *blinded-view-preserves-validity* with the validity hypothesis on b discharged internally from *extract-preserves-validity*, so the morphism map is total on the blinding preorder.

lemma *cdsp-ads-morphism*:

assumes $bo: bo \ a \ b$ **and** $eb: \text{extract-map } b = \text{Some } sb$
shows $\exists sa. \text{extract-map } a = \text{Some } sa \wedge \text{valid-state } sa \wedge \text{valid-state } sb \wedge \text{state-refines } sa \ sb$
proof –
have $hab: h \ a = h \ b$ **using** *bo-hash-eq[OF bo]* .
obtain sa **where** $ea: \text{extract-map } a = \text{Some } sa$ **and** $sr: \text{state-refines } sa \ sb$
using *extract-under-blinding[OF hab bo eb]* **by** *blast*
have $\text{valid-state } sa$ **using** *extract-preserves-validity[OF ea]* .
moreover **have** $\text{valid-state } sb$ **using** *extract-preserves-validity[OF eb]* .
ultimately show *?thesis* **using** $ea \ sr$ **by** *blast*
qed

Composition is preserved (closed). Two composable blinding morphisms $a \rightarrow b$ and $b \rightarrow c$ compose in the source category to $a \rightarrow c$ (transitivity

of bo), and the functor sends the composite to the composite of the images $extract\ a \rightarrow extract\ b \rightarrow extract\ c$ (transitivity of *state-refines*). This is the cross-domain, authenticated analogue of *preservation-compose*.

theorem *cdsp-ads-compose*:

assumes $ab: bo\ a\ b$ **and** $bc: bo\ b\ c$ **and** $ec: extract\text{-}map\ c = Some\ sc$
shows $bo\ a\ c$
 $\wedge (\exists sa\ sb. extract\text{-}map\ a = Some\ sa \wedge extract\text{-}map\ b = Some\ sb$
 $\wedge valid\text{-}state\ sa \wedge valid\text{-}state\ sb \wedge valid\text{-}state\ sc$
 $\wedge state\text{-}refines\ sa\ sb \wedge state\text{-}refines\ sb\ sc \wedge state\text{-}refines\ sa\ sc)$

proof –

have $boac: bo\ a\ c$ **using** $mk.transp\ ab\ bc$ **by** ($metis\ transpD$)
obtain sb **where** $eb: extract\text{-}map\ b = Some\ sb$
and $srbc: state\text{-}refines\ sb\ sc$ **and** $vsb: valid\text{-}state\ sb$ **and** $vsc: valid\text{-}state\ sc$
using $cdsp\text{-}ads\text{-}morphisms[OF\ bc\ ec]$ **by** *blast*
obtain sa **where** $ea: extract\text{-}map\ a = Some\ sa$
and $srab: state\text{-}refines\ sa\ sb$ **and** $vsa: valid\text{-}state\ sa$
using $cdsp\text{-}ads\text{-}morphisms[OF\ ab\ eb]$ **by** *blast*
have $srac: state\text{-}refines\ sa\ sc$ **using** $state\text{-}refines\text{-}trans[OF\ srab\ srbc]$.
show *?thesis* **using** $boac\ ea\ eb\ vsa\ vsb\ vsc\ srab\ srbc\ srac$ **by** *blast*
qed

Associativity. Three composable blinding morphisms yield the same composite $a \rightarrow d$ regardless of the bracketing; in this thin category the composite morphism is the relation $bo\ a\ d$ (and its image *state-refines* $sa\ sd$), which is bracketing-independent because the relations are propositional. The substantive associativity of the lifted merge appears in the merge-associativity result below.

theorem *cdsp-ads-assoc*:

assumes $ab: bo\ a\ b$ **and** $bc: bo\ b\ c$ **and** $cd: bo\ c\ d$
and $ed: extract\text{-}map\ d = Some\ sd$
shows $bo\ a\ d \wedge (\exists sa. extract\text{-}map\ a = Some\ sa \wedge valid\text{-}state\ sa \wedge state\text{-}refines\ sa\ sd)$

proof –

have $bo\ b\ d$ **using** $mk.transp\ bc\ cd$ **by** ($metis\ transpD$)
hence $boad: bo\ a\ d$ **using** $mk.transp\ ab$ **by** ($metis\ transpD$)
obtain sa **where** $extract\text{-}map\ a = Some\ sa$ $valid\text{-}state\ sa$ $state\text{-}refines\ sa\ sd$
using $cdsp\text{-}ads\text{-}morphisms[OF\ boad\ ed]$ **by** *blast*
thus *?thesis* **using** $boad$ **by** *blast*
qed

The functor’s action on the merge (join) product: a merge of two views is sent to a join of the two extracted states, with the joined state valid and refining each input. This lifts *extract-respects-merging* to a theorem carrying validity, and is exactly *authenticated-preservation-soundness* read as the product map of the composite functor.

lemma *cdsp-ads-merge*:

assumes $mab: m\ a\ b = Some\ ab$
and $ea: extract\text{-}map\ a = Some\ sa$ **and** $eb: extract\text{-}map\ b = Some\ sb$

shows $\exists sab. \text{extract-map } ab = \text{Some } sab \wedge \text{valid-state } sab$
 $\wedge \text{state-refines } sa \text{ } sab \wedge \text{state-refines } sb \text{ } sab$
using *authenticated-preservation-soundness*[*OF mab ea eb*]
by *blast*

Associativity of the lifted merge/join algebra. The ADS merge is associative (Merkle-interface axiom *mk.assoc*), so the three-way combination of authenticated views is independent of the bracketing: both $(a \cdot b) \cdot c$ and $a \cdot (b \cdot c)$ land on the same merged value, hence (the extraction being a function) on the same joined state. This is the substantive “associativity preserved” of the composite functor. Commutativity and idempotence of the lifted combination descend directly from the interface laws (*mk.commute*, *mk.idem*) with the extraction a function; associativity is the one law that needs the rearrangement below, so it is the law recorded as a theorem.

theorem *cdsp-ads-merge-assoc*:

assumes *mab*: $m \ a \ b = \text{Some } ab$ **and** *mabc*: $m \ ab \ c = \text{Some } abc$
shows $\exists bc. m \ b \ c = \text{Some } bc \wedge m \ a \ bc = \text{Some } abc$

proof –

have $m \ b \ c \gg m \ a = m \ a \ b \gg m \ c$ **by** (rule *mk.assoc[symmetric]*)

also have $m \ a \ b \gg m \ c = m \ c \ ab$ **using** *mab* **by** *simp*

also have $m \ c \ ab = m \ ab \ c$ **by** (rule *mk.commute*)

also have $\dots = \text{Some } abc$ **using** *mabc* **by** *simp*

finally have $m \ b \ c \gg m \ a = \text{Some } abc$.

thus *?thesis* **by** (cases *m b c*) *auto*

qed

end

59 Authenticity preserved along whole synchronization sequences

Four authenticity properties are shown to hold not across a single blinding or merge step but along the *whole* synchronization sequence:

1. **validity** — every view in the sequence extracts to a valid state, along a blinding path (theorem *sequence-authenticity-preservation*) and along a merge fold (theorem *sequence-merge-soundness*);
2. **need-to-know (refinement)** — every view is a partial view refining the most-revealed endpoint (theorem *sequence-authenticity-preservation*, extended to arbitrary blinding-step reachability by corollary *reachable-view-authenticity*); dually, every contributor to a merge fold refines the combined view (theorem *sequence-merge-soundness*);
3. **hash soundness** — the whole sequence commits to one authenticating root, in both the blinding direction (theorem

blinding-path-hash-soundness) and the merge direction (theorem *merge-seq-hash*);

4. inclusion integrity — theorem *sequence-inclusion-integrity*.

Scope qualifier (read carefully). The inclusion result is *state-level*: it is about *revealed holdings* — a holding exhibited by any view in the sequence is genuinely present in the endpoint with the same regulatory state, and no view fabricates a chain the endpoint does not authenticate. This is *not* a claim about the *concrete Merkle inclusion path* (the witness list authenticating a leaf against a root hash); that construction lives in *ADS-Functor.Inclusion-Proof-Construction* and belongs to the concrete recursive-tree layer, not to this abstract layer.

context *authenticated-state*
begin

59.1 Need-to-know blinding along a sequence

Along a blinding path, an earlier (more blinded) element blinds any later element: the blinding relation is monotone along the path by transitivity.

lemma *blinding-path-mono*:

assumes *path*: *blinding-path bo xs* **and** *ij*: $i \leq j$ **and** *j*: $j < \text{length } xs$
shows $bo \ (xs \ ! \ i) \ (xs \ ! \ j)$

proof —

from *ij* **obtain** *d* **where** *jd*: $j = i + d$ **using** *le-Suc-ex* **by** *blast*

from *j jd* **show** *?thesis* **unfolding** *jd*

proof (*induction d arbitrary: j*)

case 0

show *?case* **using** *mk.reflp* **by** (*simp add: reflp-def*)

next

case (*Suc d*)

have *lt*: $i + d < \text{length } xs$ **using** *Suc.prem*s **by** *simp*

have *ih*: $bo \ (xs \ ! \ i) \ (xs \ ! \ (i + d))$ **using** *Suc.IH lt* **by** *simp*

have *step*: $bo \ (xs \ ! \ (i + d)) \ (xs \ ! \ \text{Suc } (i + d))$

using *path Suc.prem*s **unfolding** *blinding-path-def* **by** *simp*

show *?case* **using** *mk.transp ih step* **by** (*metis transpD add-Suc-right*)

qed

qed

Sequence-level need-to-know preservation. Given a synchronization sequence presented as a blinding path whose most-revealed endpoint (the last element) extracts to a valid cross-domain state, *every* intermediate view extracts to a valid state that refines the endpoint. Authenticity (validity together with the refinement guarantee of a partial view) is preserved along the whole sequence, not merely across one blinding step. This generalises *blinded-view-preserves-validity*.

theorem *sequence-authenticity-preservation*:

```

assumes path: blinding-path bo xs and ne: xs ≠ []
and elast: extract-map (last xs) = Some s-last
and i: i < length xs
shows ∃ si. extract-map (xs ! i) = Some si ∧ valid-state si ∧ state-refines si s-last
proof –
  have le: i ≤ length xs – 1 using i by simp
  have lt: length xs – 1 < length xs using ne by (cases xs) auto
  have bo (xs ! i) (xs ! (length xs – 1))
    using blinding-path-mono[OF path le lt] .
  also have xs ! (length xs – 1) = last xs using ne by (simp add: last-conv-nth)
  finally have bo (xs ! i) (last xs) .
  from cdsp-ads-morphism[OF this elast] show ?thesis by blast
qed

```

Hash soundness along the sequence. Every view in a blinding path commits to the same root hash as the most-revealed endpoint; a blinded view cannot present a different root. This is the sequence-level form of the ADS guarantee *bo-hash-eq*: along the whole synchronization sequence there is a single authenticating root, so any inclusion claim made anywhere along the sequence is checked against the same commitment.

theorem *blinding-path-hash-soundness*:

```

assumes path: blinding-path bo xs and ne: xs ≠ [] and i: i < length xs
shows h (xs ! i) = h (last xs)
proof –
  have le: i ≤ length xs – 1 using i by simp
  have lt: length xs – 1 < length xs using ne by (cases xs) auto
  have bo (xs ! i) (xs ! (length xs – 1)) using blinding-path-mono[OF path le lt] .
  moreover have xs ! (length xs – 1) = last xs using ne by (simp add: last-conv-nth)
  ultimately show ?thesis using bo-hash-eq by simp
qed

```

Inclusion-proof integrity along the sequence. Every holding revealed by any view in the sequence is genuinely included in the most-revealed endpoint with the same regulatory state, and the revealing chain is connected in the endpoint. No view along the sequence can therefore exhibit a holding that the endpoint does not authenticate. Together with *blinding-path-hash-soundness* (one root for the whole sequence) this is the state-level reading of inclusion-proof integrity preserved across the synchronization sequence.

At this abstract layer inclusion reduces to the refinement guarantee (*sequence-authenticity-preservation*) read through *state-refines-def*: a partial view’s revealed leaves are exactly the leaves it shares with the endpoint. The concrete Merkle inclusion-path machinery lives one level down in *ADS-Functor.Inclusion-Proof-Construction*.

theorem *sequence-inclusion-integrity*:

```

assumes path: blinding-path bo xs and ne: xs ≠ []

```

```

and elast: extract-map (last xs) = Some s-last
and i: i < length xs and ei: extract-map (xs ! i) = Some si
and rev: get-reg-state si c aid = Some r
shows get-reg-state s-last c aid = Some r ∧ c ∈ connected-chains s-last aid
proof –
obtain s' where es': extract-map (xs ! i) = Some s' and sr: state-refines s' s-last
using sequence-authenticity-preservation[OF path ne elast i] by blast
from es' ei have eq: s' = si by simp
have conn: c ∈ connected-chains si aid
using rev unfolding connected-chains-def asset-exists-def
get-reg-state-def get-asset-state-def by (auto split: option.splits)
have c ∈ connected-chains s-last aid
  ∧ get-reg-state si c aid = get-reg-state s-last c aid
using sr eq conn unfolding state-refines-def by metis
thus ?thesis using rev by simp
qed

```

The same guarantee phrased over the reflexive-transitive closure: any view reachable from *b* by any number of blinding steps extracts to a valid state refining *b*. (The closure collapses because *bo* is already a preorder, which is precisely why arbitrary-length sequences add no new obligation.)

corollary *reachable-view-authenticity*:

```

assumes reach: bo** a b and eb: extract-map b = Some sb
shows ∃ sa. extract-map a = Some sa ∧ valid-state sa ∧ state-refines sa sb
proof –
from reach have bo a b
proof (induction rule: rtranclp-induct)
  case base
    show ?case using mk.reflp by (simp add: reflp-def)
  next
    case (step y z)
    show ?case using mk.transp step.IH step.hyps(2) by (metis transpD)
qed
from cdsp-ads-morphism[OF this eb] show ?thesis by blast
qed

```

59.2 Re-revealing merge along a sequence

The *n*-ary right fold of the (locale) merge along a non-empty sequence of authenticated views. It is defined with the locale merge *m* fixed, so its induction principle keeps *m* in place (a theory-level fold parameterised by the merge would generalise it away under induction).

```

fun merge-seq :: 'd list ⇒ 'd option where
  merge-seq [] = None
| merge-seq [x] = Some x
| merge-seq (x # y # ys) =
  (case merge-seq (y # ys) of None ⇒ None | Some z ⇒ m x z)

```

A single merge preserves the common hash, since each input blinds to the merge and blinding respects hashes.

lemma *merge-preserves-hash*:

assumes $m\ a\ b = \text{Some}\ ab$

shows $h\ ab = h\ a$

proof –

have $bo\ a\ ab$ **using** *mk.join assms* **by** *blast*

thus *?thesis* **using** *bo-hash-eq* **by** *simp*

qed

Sequence-level hash preservation for the merge fold. Combining a whole hash-compatible sequence of partial views yields a value with that same common hash: re-revealing along the sequence never changes the authenticating root. This is the merge-direction companion of *blinding-path-hash-soundness* (which covers the blinding direction), so along either form of synchronization sequence the root hash is preserved.

theorem *merge-seq-hash*:

assumes $\text{merge-seq}\ xs = \text{Some}\ r$ **and** $x\ s \neq []$

and $\forall x \in \text{set}\ xs.\ h\ x = h\ (\text{hd}\ xs)$

shows $h\ r = h\ (\text{hd}\ xs)$

using *assms*

proof (*induction xs rule: merge-seq.induct*)

case $(\exists x\ y\ ys)$

obtain z **where** $z: \text{merge-seq}\ (y \# ys) = \text{Some}\ z$ **and** $m\ x\ z = \text{Some}\ r$

using *3.prem1* **by** (*auto split: option.splits*)

have $h\ r = h\ x$ **using** *merge-preserves-hash[OF m x z]* .

thus *?case* **by** *simp*

qed *simp-all*

Sequence-level merging soundness. A whole synchronization sequence of partial authenticated views over the same committed object (all sharing a hash) folds into a single combined view that extracts to a valid cross-domain state refining every contributor. This is the n-ary generalisation of *authenticated-preservation-soundness*: re-revealing along the entire sequence preserves validity and reconstructs a consistent joint view.

theorem *sequence-merge-soundness*:

$[[\ xs \neq []; \forall x \in \text{set}\ xs.\ \exists s.\ \text{extract-map}\ x = \text{Some}\ s;$

$\forall x \in \text{set}\ xs.\ h\ x = h\ (\text{hd}\ xs)]]$

$\implies \exists ab\ s.\ \text{merge-seq}\ xs = \text{Some}\ ab \wedge \text{extract-map}\ ab = \text{Some}\ s \wedge \text{valid-state}\ s$

$\wedge (\forall x \in \text{set}\ xs.\ \exists sx.\ \text{extract-map}\ x = \text{Some}\ sx \wedge \text{state-refines}\ sx\ s)$

proof (*induction xs rule: merge-seq.induct*)

case *1*

then show *?case* **by** *simp*

next

case $(\exists x)$

obtain sx **where** $ex: \text{extract-map}\ x = \text{Some}\ sx$ **using** *2.prem1* **by** *auto*

have $\text{valid-state}\ sx$ **using** *extract-preserves-validity[OF ex]* .

thus *?case* **using** *ex state-refines-refl* **by** *auto*

```

next
  case ( $\exists x y ys$ )
  have ne-rest:  $(y \# ys) \neq []$  by simp
  have hx:  $\bigwedge w. w \in \text{set } (y \# ys) \implies h w = h x$ 
    using  $\exists$ .prems(3) by auto
  have hrest:  $\forall w \in \text{set } (y \# ys). h w = h (hd (y \# ys))$  using hx by auto
  have erest:  $\forall w \in \text{set } (y \# ys). \exists s. \text{extract-map } w = \text{Some } s$ 
    using  $\exists$ .prems(2) by auto
  obtain z sz where mz:  $\text{merge-seq } (y \# ys) = \text{Some } z$ 
    and ez:  $\text{extract-map } z = \text{Some } sz$  and vz:  $\text{valid-state } sz$ 
    and refz:  $\forall w \in \text{set } (y \# ys). \exists sw. \text{extract-map } w = \text{Some } sw \wedge \text{state-refines } sw$ 
    sz
  using  $\exists$ .IH[OF ne-rest erest hrest] by blast
  obtain sx where ex:  $\text{extract-map } x = \text{Some } sx$  using  $\exists$ .prems(2) by auto
  have hz:  $h z = h x$ 
  proof -
    have  $h z = h (hd (y \# ys))$  using merge-seq-hash[OF mz ne-rest hrest] .
    thus ?thesis using hx by simp
  qed
  have  $\exists ab. m x z = \text{Some } ab$ 
    using hz mk.merge-respects-hashes[of x z] by simp
  then obtain xz where mxz:  $m x z = \text{Some } xz$  by blast
  have mseq:  $\text{merge-seq } (x \# y \# ys) = \text{Some } xz$  using mz mxz by simp
  obtain sxz where exz:  $\text{extract-map } xz = \text{Some } sxz$  and vxz:  $\text{valid-state } sxz$ 
    and rx:  $\text{state-refines } sx sxz$  and rz:  $\text{state-refines } sz sxz$ 
    using authenticated-preservation-soundness[OF mxz ex ez] by blast
  have  $\forall w \in \text{set } (x \# y \# ys). \exists sw. \text{extract-map } w = \text{Some } sw \wedge \text{state-refines } sw$ 
    sxz
  proof
    fix w assume  $w \in \text{set } (x \# y \# ys)$ 
    then consider  $w = x \mid w \in \text{set } (y \# ys)$  by auto
    then show  $\exists sw. \text{extract-map } w = \text{Some } sw \wedge \text{state-refines } sw$  sxz
    proof cases
      case 1
      then show ?thesis using ex rx by blast
    next
      case 2
      then obtain sw where  $\text{extract-map } w = \text{Some } sw$  and  $\text{state-refines } sw$  sz
        using refz by blast
      then show ?thesis using state-refines-trans[OF - rz] by blast
    qed
  qed
  then show ?case using mseq exz vxz by blast
qed
end

```

60 Instantiation on the ADS blindable-position functor

We instantiate *authenticated-state* on a concrete ADS construction: the blindable-position functor of *ADS-Functor.ADS-Construction* applied to the Oraclizer Merkle interface *merkle-interface-auth*. The carrier is $(\text{reg-state} \times \text{chain-id set}, \text{reg-state})$ *blindable_m*: an authenticated datum is either *Unblinded* (r, P) (the fully revealed consensus state r with revealed-chain set P) or *Blinded* x (a hash-only commitment that reveals nothing). The blindable hash, blinding order and merge are inherited from the ADS construction; the Merkle-interface obligation is discharged by *merkle-blindable* from *merkle-interface-auth*, so no new interface proof is needed.

The empty extracted view does not depend on the (unreachable) consensus label, so all hash-only commitments extract to the same state.

lemma *auth-state-empty-eq*: $\text{auth-state } r \{\} = \text{auth-state } r' \{\}$
by (*simp add: auth-state-def*)

Extraction: a revealed datum yields its cross-domain state; a hash-only commitment yields the empty (no-chains) view, which is valid and refines every state. The commitment must still extract to a state – a blinding of a value must itself be extractable – so it maps to the empty view rather than to *None*.

fun *bl-extract* :: $(\text{reg-state} \times \text{chain-id set}, \text{reg-state})$ *blindable_m* $\Rightarrow$ *global-state option* **where**
bl-extract (*Unblinded* a) = *Some* (*auth-state* (*fst* a) (*snd* a))
| *bl-extract* (*Blinded* $-$) = *Some* (*auth-state* *ACTIVE* $\{\}$)

interpretation *oss-blindable*:

authenticated-state
hash-blindable *auth-hash*
blinding-of-blindable *auth-hash* *auth-bo*
merge-blindable *auth-hash* *auth-merge*
bl-extract

proof (*rule authenticated-state.intro*)

show *merkle-interface* (*hash-blindable* *auth-hash*)

(*blinding-of-blindable* *auth-hash* *auth-bo*) (*merge-blindable* *auth-hash* *auth-merge*)

by (*rule merkle-blindable[OF merkle-interface-auth]*)

next

show *authenticated-state-axioms* (*hash-blindable* *auth-hash*)

(*blinding-of-blindable* *auth-hash* *auth-bo*) (*merge-blindable* *auth-hash* *auth-merge*) *bl-extract*

proof

— *extract* respects merging

fix $a \ b \ ab \ sa \ sb$

```

assume Hab: hash-blindable auth-hash a = hash-blindable auth-hash b
and Mab: merge-blindable auth-hash auth-merge a b = Some ab
and ea: bl-extract a = Some sa and eb: bl-extract b = Some sb
show  $\exists sab. bl\text{-}extract\ ab = Some\ sab \wedge state\text{-}join\ sa\ sb\ sab$ 
proof (cases a)
  case (Unblinded p) note A = this
  show ?thesis
  proof (cases b)
    case (Unblinded q) note B = this
    have fp: fst p = fst q using Hab by (simp add: A B auth-hash-def)
    have ab-eq: ab = Unblinded (fst p, snd p  $\cup$  snd q)
      using Mab fp by (simp add: A B auth-merge-def)
    have sa-eq: sa = auth-state (fst p) (snd p) using ea by (simp add: A)
    have sb-eq: sb = auth-state (fst p) (snd q) using eb fp by (simp add: B)
    have bl-extract ab = Some (auth-state (fst p) (snd p  $\cup$  snd q)) by (simp
add: ab-eq)
    moreover have state-join sa sb (auth-state (fst p) (snd p  $\cup$  snd q))
      unfolding sa-eq sb-eq by (rule state-join-auth)
    ultimately show ?thesis by blast
  next
    case (Blinded y) note B = this
    have ab-eq: ab = Unblinded p using Mab Hab by (simp add: A B
auth-hash-def)
    have sa-eq: sa = auth-state (fst p) (snd p) using ea by (simp add: A)
    have sb-eq: sb = auth-state ACTIVE {} using eb by (simp add: B)
    have bl-extract ab = Some (auth-state (fst p) (snd p)) by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state (fst p) (snd p))
      unfolding sa-eq sb-eq
      using state-join-auth[of fst p snd p {}] auth-state-empty-eq[of ACTIVE fst
p]
      by simp
    ultimately show ?thesis by blast
  qed
next
  case (Blinded x) note A = this
  show ?thesis
  proof (cases b)
    case (Unblinded q) note B = this
    have ab-eq: ab = Unblinded q using Mab Hab by (simp add: A B
auth-hash-def)
    have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
    have sb-eq: sb = auth-state (fst q) (snd q) using eb by (simp add: B)
    have bl-extract ab = Some (auth-state (fst q) (snd q)) by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state (fst q) (snd q))
      unfolding sa-eq sb-eq
      using state-join-auth[of fst q {} snd q] auth-state-empty-eq[of ACTIVE fst
q]
      by simp
    ultimately show ?thesis by blast

```

```

next
  case (Blinded y) note  $B = \text{this}$ 
  have ab-eq:  $ab = \text{Blinded } y$  using Mab Hab by (simp add: A B)
  have sa-eq:  $sa = \text{auth-state ACTIVE } \{\}$  using ea by (simp add: A)
  have sb-eq:  $sb = \text{auth-state ACTIVE } \{\}$  using eb by (simp add: B)
  have bl-extract  $ab = \text{Some } (\text{auth-state ACTIVE } \{\})$  by (simp add: ab-eq)
  moreover have state-join  $sa\ sb\ (\text{auth-state ACTIVE } \{\})$ 
  unfolding sa-eq sb-eq using state-join-auth[of ACTIVE {} {}] by simp
  ultimately show ?thesis by blast
qed
qed
next
  — extract under blinding
  fix a b sb
  assume Hab: hash-blindable auth-hash a = hash-blindable auth-hash b
  and BOab: blinding-of-blindable auth-hash auth-bo a b
  and eb: bl-extract b = Some sb
  show  $\exists sa. \text{bl-extract } a = \text{Some } sa \wedge \text{state-refines } sa\ sb$ 
  proof (cases b)
  case (Unblinded q) note  $B = \text{this}$ 
  show ?thesis
  proof (cases a)
  case (Unblinded p) note  $A = \text{this}$ 
  have bo:  $\text{auth-bo } p\ q$  using BOab by (simp add: A B)
  have rfst:  $\text{fst } p = \text{fst } q$  and sub:  $\text{snd } p \subseteq \text{snd } q$ 
  using bo by (simp-all add: auth-bo-def)
  have sr:  $\text{state-refines } (\text{auth-state } (\text{fst } p)\ (\text{snd } p))\ (\text{auth-state } (\text{fst } p)\ (\text{snd } q))$ 
  using sub by (rule state-refines-auth)
  have bl-extract  $a = \text{Some } (\text{auth-state } (\text{fst } p)\ (\text{snd } p))$  by (simp add: A)
  moreover have state-refines  $(\text{auth-state } (\text{fst } p)\ (\text{snd } p))\ sb$ 
  using sr rfst eb by (simp add: B)
  ultimately show ?thesis by blast
next
  case (Blinded x) note  $A = \text{this}$ 
  have sr:  $\text{state-refines } (\text{auth-state } (\text{fst } q)\ \{\})\ (\text{auth-state } (\text{fst } q)\ (\text{snd } q))$ 
  using empty-subsetI by (rule state-refines-auth)
  have bl-extract  $a = \text{Some } (\text{auth-state } (\text{fst } q)\ \{\})$ 
  by (simp add: A auth-state-empty-eq[of ACTIVE fst q])
  moreover have state-refines  $(\text{auth-state } (\text{fst } q)\ \{\})\ sb$ 
  using sr eb by (simp add: B)
  ultimately show ?thesis by blast
qed
next
  case (Blinded y) note  $B = \text{this}$ 
  — Only a blinded value can blind to a blinded value, and then they are equal.
  have  $a = \text{Blinded } y$  using BOab by (cases a) (simp-all add: B)
  then have bl-extract  $a = \text{Some } (\text{auth-state ACTIVE } \{\})$  by simp
  moreover have  $sb = \text{auth-state ACTIVE } \{\}$  using eb by (simp add: B)
  ultimately show ?thesis using state-refines-refl by auto

```

```

    qed
  next
    — extract preserves validity
    fix a s
    assume bl-extract a = Some s
    then show valid-state s
      by (cases a) (auto simp: auth-state-valid)
    qed
  qed

```

Non-degeneracy of the blindable instance: revealed two-chain views merge to their union, and a hash-only commitment merges back to the revealed view.

lemma *oss-blindable-nontrivial*:

```

  merge-blindable auth-hash auth-merge (Unblinded (ACTIVE, {0})) (Unblinded
    (ACTIVE, {Suc 0}))
    = Some (Unblinded (ACTIVE, {0, Suc 0}))
  bl-extract (Unblinded (ACTIVE, {0, Suc 0})) = Some (auth-state ACTIVE {0,
    Suc 0})
  merge-blindable auth-hash auth-merge (Blinded (Content ACTIVE)) (Unblinded
    (ACTIVE, {0}))
    = Some (Unblinded (ACTIVE, {0}))
  by (simp-all add: auth-merge-def auth-hash-def insert-commute)

```

Regression witness exhibiting a non-vacuous instance for the sequence-level results above. The generalised predicate *blinding-path* is satisfied by a genuinely increasing need-to-know sequence at this instance: a hash-only commitment, then a one-chain revealed view, then a two-chain revealed view, each blinding its successor. The most-revealed endpoint extracts to a non-degenerate two-chain state. This pins *authenticated-state.sequence-authenticity-preservation*, *authenticated-state.blinding-path-hash-soundness* and *authenticated-state.sequence-inclusion-integrity* to a non-vacuous instance.

lemma *blinding-path-witness*:

```

  blinding-path (blinding-of-blindable auth-hash auth-bo)
    [Blinded (Content ACTIVE),
     Unblinded (ACTIVE, {0::nat}),
     Unblinded (ACTIVE, {0::nat, Suc 0})]
  unfolding blinding-path-def
  by (auto simp: nth-Cons' auth-hash-def auth-bo-def)

```

lemma *blinding-path-witness-endpoint*:

```

  bl-extract (Unblinded (ACTIVE, {0::nat, Suc 0})) = Some (auth-state ACTIVE
    {0, Suc 0})
  connected-chains (auth-state ACTIVE {0::nat, Suc 0}) (0::nat) = {0, Suc 0}
  by (simp-all add: auth-state-connected)

```

61 Instantiation on the top-level Canton transaction commitment

Theory *ADS-Functor.Canton-Transaction-Tree* formalises the production Canton transaction tree as an authenticated data structure, but over abstract content: *view-data*, *view-metadata*, *common-metadata* and *participant-metadata* are introduced by **typeddecl** and carry no regulatory meaning, so a cross-domain state cannot be read out of *transaction_m* without an inadmissible axiom relating those opaque types to the regulatory model. We therefore model the same public structure with concrete content. A Canton transaction is *Transaction_m* applied to one top-level blindable position over a payload of metadata and a view list; we mirror exactly this one-constructor-over-a-blindable shape, taking the payload to be the cross-domain regulatory datum — the consensus regulatory state (in the role of the common metadata) together with the set of chains whose views attest it (the per-participant view list, abstracted to its attesting-chain set). Hash, blinding and merge are the ADS blindable-position operations on the payload: precisely the top-level authenticated operation Canton performs, since *Transaction_m* wraps a single *blindable_m*. The Merkle interface and the three coherence obligations are transported from the blindable instance *oss-blindable* through the constructor bijection, so no new axiom and no residual proof obligation arise. The one open item is model fidelity: whether this payload faithfully abstracts Canton’s metadata-and-view-list datatype. That is a sanity check of the model against the Canton specification, not a proof gap.

datatype *reg-transaction* =

RegTransaction (*the-reg-tx*: (*reg-state* × *chain-id set*, *reg-state*) *blindable_m*)

definition *rtx-hash* :: *reg-transaction* ⇒ *reg-state blindable_h* **where**

rtx-hash *t* = *hash-blindable auth-hash* (*the-reg-tx* *t*)

definition *rtx-bo* :: *reg-transaction* ⇒ *reg-transaction* ⇒ *bool* **where**

rtx-bo *s* *t* = *blinding-of-blindable auth-hash auth-bo* (*the-reg-tx* *s*) (*the-reg-tx* *t*)

definition *rtx-merge* :: *reg-transaction* ⇒ *reg-transaction* ⇒ *reg-transaction option* **where**

rtx-merge *s* *t* =
map-option RegTransaction
(*merge-blindable auth-hash auth-merge* (*the-reg-tx* *s*) (*the-reg-tx* *t*))

definition *rtx-extract* :: *reg-transaction* ⇒ *global-state option* **where**

rtx-extract *t* = *bl-extract* (*the-reg-tx* *t*)

The single constructor is a bijection, so the Merkle interface of the payload blindable transports to the transaction wrapper.

lemma *merkle-reg-transaction*: *merkle-interface rtx-hash rtx-bo rtx-merge*

```

proof –
  have bind-simp: rtx-merge a b  $\gg$  rtx-merge c
    = map-option RegTransaction
      (merge-blindable auth-hash auth-merge (the-reg-tx a) (the-reg-tx b)
         $\gg$  merge-blindable auth-hash auth-merge (the-reg-tx c)) for a b c
  by (cases merge-blindable auth-hash auth-merge (the-reg-tx a) (the-reg-tx b))
    (simp-all add: rtx-merge-def)
  have mrh: (rtx-hash a = rtx-hash b) = ( $\exists$  ab. rtx-merge a b = Some ab) for a b
  by (cases merge-blindable auth-hash auth-merge (the-reg-tx a) (the-reg-tx b))
    (auto simp: rtx-hash-def rtx-merge-def oss-blindable.mk.merge-respects-hashes)
  have midem: rtx-merge a a = Some a for a
  by (simp add: rtx-merge-def oss-blindable.mk.idem)
  have mcomm: rtx-merge a b = rtx-merge b a for a b
  by (simp add: rtx-merge-def oss-blindable.mk.commute)
  have massoc: rtx-merge a b  $\gg$  rtx-merge c = rtx-merge b c  $\gg$  rtx-merge a for
a b c
  using bind-simp[of a b c] bind-simp[of b c a]
    oss-blindable.mk.assoc[of the-reg-tx a the-reg-tx b the-reg-tx c] by simp
  have mbod: rtx-bo a b = (rtx-merge a b = Some b) for a b
  by (cases merge-blindable auth-hash auth-merge (the-reg-tx a) (the-reg-tx b))
    (auto simp: rtx-bo-def rtx-merge-def oss-blindable.mk.bo-def-reg-transaction.expand)
  show ?thesis
  by (rule merkle-interface.intro[OF mrh midem mcomm massoc mbod])
qed

```

Extraction reads the revealed cross-domain datum out of the transaction's top-level commitment; the three coherence obligations transport from the blindable instance *oss-blindable*.

interpretation *canton-authenticated*:

authenticated-state *rtx-hash* *rtx-bo* *rtx-merge* *rtx-extract*

proof (*rule* *authenticated-state.intro*)

show *merkle-interface* *rtx-hash* *rtx-bo* *rtx-merge* **by** (*rule* *merkle-reg-transaction*)

next

show *authenticated-state-axioms* *rtx-hash* *rtx-bo* *rtx-merge* *rtx-extract*

proof

— *extract* respects merging, transported through the constructor

fix *a b ab sa sb*

assume *H*: *rtx-hash* *a* = *rtx-hash* *b* **and** *M*: *rtx-merge* *a b* = *Some* *ab*

and *ea*: *rtx-extract* *a* = *Some* *sa* **and** *eb*: *rtx-extract* *b* = *Some* *sb*

have *h'*: *hash-blindable* *auth-hash* (*the-reg-tx* *a*) = *hash-blindable* *auth-hash* (*the-reg-tx* *b*)

using *H* **by** (*simp* *add*: *rtx-hash-def*)

have *m'*: *merge-blindable* *auth-hash* *auth-merge* (*the-reg-tx* *a*) (*the-reg-tx* *b*)
= *Some* (*the-reg-tx* *ab*)

using *M* **by** (*auto* *simp*: *rtx-merge-def* *split*: *option.splits*)

have *ea'*: *bl-extract* (*the-reg-tx* *a*) = *Some* *sa* **using** *ea* **by** (*simp* *add*:
rtx-extract-def)

have *eb'*: *bl-extract* (*the-reg-tx* *b*) = *Some* *sb* **using** *eb* **by** (*simp* *add*:
rtx-extract-def)

```

obtain sab where bl-extract (the-reg-tx ab) = Some sab state-join sa sb sab
using oss-blindable.extract-respects-merging[OF h' m' ea' eb'] by blast
then show  $\exists sab. rtx-extract\ ab = Some\ sab \wedge state-join\ sa\ sb\ sab$ 
by (auto simp: rtx-extract-def)
next
— extract under blinding, transported through the constructor
fix a b sb
assume H: rtx-hash a = rtx-hash b and BO: rtx-bo a b
and eb: rtx-extract b = Some sb
have h': hash-blindable auth-hash (the-reg-tx a) = hash-blindable auth-hash
(the-reg-tx b)
using H by (simp add: rtx-hash-def)
have bo': blinding-of-blindable auth-hash auth-bo (the-reg-tx a) (the-reg-tx b)
using BO by (simp add: rtx-bo-def)
have eb': bl-extract (the-reg-tx b) = Some sb using eb by (simp add:
rtx-extract-def)
obtain sa where bl-extract (the-reg-tx a) = Some sa state-refines sa sb
using oss-blindable.extract-under-blinding[OF h' bo' eb'] by blast
then show  $\exists sa. rtx-extract\ a = Some\ sa \wedge state-refines\ sa\ sb$ 
by (auto simp: rtx-extract-def)
next
— extract preserves validity
fix a s
assume rtx-extract a = Some s
then have bl-extract (the-reg-tx a) = Some s by (simp add: rtx-extract-def)
then show valid-state s using oss-blindable.extract-preserves-validity by blast
qed
qed

```

Non-degeneracy: a revealed two-chain transaction joins with a hash-only commitment to the same chains, re-revealing the cross-domain view.

lemma *canton-authenticated-nontrivial*:

```

rtx-merge (RegTransaction (Unblinded (ACTIVE, {0})))
  (RegTransaction (Unblinded (ACTIVE, {Suc 0})))
= Some (RegTransaction (Unblinded (ACTIVE, {0, Suc 0})))
rtx-extract (RegTransaction (Unblinded (ACTIVE, {0, Suc 0})))
= Some (auth-state ACTIVE {0, Suc 0})
rtx-merge (RegTransaction (Blinded (Content ACTIVE)))
  (RegTransaction (Unblinded (ACTIVE, {0})))
= Some (RegTransaction (Unblinded (ACTIVE, {0})))
by (simp-all add: rtx-merge-def rtx-extract-def auth-merge-def auth-hash-def in-
sert-commute)

```

62 Recursive instantiation on the Canton transaction tree

The coarse instance *canton-authenticated* above models a Canton transaction as one top-level blindable over an abstract payload. Here we

keep the public *recursive* shape: a view is the rose tree rose-tree_m of $\text{ADS-Functor.ADS-Construction}$ — the very datatype the public Canton formalisation uses to build view_m ($\text{view}_m \cong \text{view-content } \text{rose-tree}_m$, with $\text{merge-view} = \text{merge-tree} \dots$). We instantiate that content-agnostic machine with concrete regulatory content: each node carries the *chain-id* it attests, and the children are its subviews. No new datatype, no new Merkle proof: the interface is inherited from *merkle-tree*.

type-synonym $\text{reg-view} = (\text{chain-id}, \text{chain-id}) \text{ rose-tree}_m$

abbreviation $\text{rv-hash} :: (\text{reg-view}, \text{chain-id } \text{rose-tree}_h) \text{ hash}$ **where**

$\text{rv-hash} \equiv \text{hash-tree } (\text{hash-discrete} :: \text{chain-id} \Rightarrow \text{chain-id})$

abbreviation $\text{rv-bo} :: \text{reg-view} \text{ blinding-of}$ **where**

$\text{rv-bo} \equiv \text{blinding-of-tree } \text{hash-discrete } (\text{blinding-of-discrete} :: \text{chain-id} \text{ blinding-of})$

abbreviation $\text{rv-merge} :: \text{reg-view} \text{ merge}$ **where**

$\text{rv-merge} \equiv \text{merge-tree } \text{hash-discrete } (\text{merge-discrete} :: \text{chain-id} \text{ merge})$

lemma *merkle-reg-view: merkle-interface rv-hash rv-bo rv-merge*

by (rule *merkle-tree*[OF *merkle-discrete*])

Extraction collects the attested chains from every revealed node, recursing into subviews; a blinded node contributes nothing. By the Merkle property a subview is reachable only through its (revealed) parent. Lemma *reg-chains-blinding* below makes this monotone: blinding a node can only shrink the attested-chain set.

fun *reg-chains* :: $\text{reg-view} \Rightarrow \text{chain-id set}$ **where**

$\text{reg-chains } (\text{Tree}_m (\text{Unblinded } (c, \text{kids}))) = \text{insert } c \ (\bigcup x \in \text{set kids. reg-chains } x)$
 $\mid \text{reg-chains } (\text{Tree}_m (\text{Blinded } -)) = \{\}$

62.1 Extraction commutes with blinding and merge

A pointwise containment along a list lifts to the union of the collected chains.

lemma *UN-reg-chains-mono:*

assumes *list-all2* ($\lambda x y. \text{reg-chains } x \subseteq \text{reg-chains } y$) *xs ys*

shows $(\bigcup x \in \text{set } xs. \text{reg-chains } x) \subseteq (\bigcup y \in \text{set } ys. \text{reg-chains } y)$

using *assms* **by** (induction *xs ys* rule: *list-all2-induct*) *auto*

Positional characterisation of the ADS list merge (used for the children lists of a rose-tree node). The length alignment is *derived*, not assumed: the lemmas *merge-list-NC* / *merge-list-CN* below reject lists of different length, exactly because the list hash (*map*) commits to the length.

context **begin**

interpretation *list-R1* .

lemma *merge-list-Cons:*

$\text{merge-list } m \ (x \# xs) \ (y \# ys) =$
 $(\text{case } m \ x \ y \text{ of } \text{None} \Rightarrow \text{None} \mid \text{Some } z \Rightarrow \text{map-option } ((\#) \ z) \ (\text{merge-list } m \ xs \ ys))$

by (*simp add: merge-list-def merge-F-def merge-R1.simps*
list-to-list-R1.simps list-R1-to-list-simps merge-sum.simps
merge-prod.simps
merge-discrete-def option.map-comp o-def split: option.splits)

lemma *merge-list-NN* [*simp*]: *merge-list m [] [] = Some []*
by (*simp add: merge-list-def merge-F-def merge-R1.simps*
list-to-list-R1.simps list-R1-to-list-simps merge-sum.simps
merge-discrete-def)

lemma *merge-list-CN* [*simp*]: *merge-list m (x # xs) [] = None*
by (*simp add: merge-list-def merge-F-def merge-R1.simps list-to-list-R1.simps*
merge-sum.simps)

lemma *merge-list-NC* [*simp*]: *merge-list m [] (y # ys) = None*
by (*simp add: merge-list-def merge-F-def merge-R1.simps list-to-list-R1.simps*
merge-sum.simps)

end

The collected chains of a list-merge are the union of the contributors' chains, given that the elementwise merge has that property.

lemma *reg-chains-merge-list*:
assumes *merge-list m k1 k2 = Some k*
and $\bigwedge s t u. s \in \text{set } k1 \implies m s t = \text{Some } u \implies \text{reg-chains } u = \text{reg-chains } s \cup \text{reg-chains } t$
shows $(\bigcup_{s \in \text{set } k. \text{reg-chains } s} = (\bigcup_{s \in \text{set } k1. \text{reg-chains } s} \cup \bigcup_{t \in \text{set } k2. \text{reg-chains } t})$
using *assms*
proof (*induction k1 arbitrary: k2 k*)
case Nil
then show *?case* **by** (*cases k2*) *auto*
next
case (Cons x xs)
show *?case*
proof (*cases k2*)
case Nil **then show** *?thesis* **using** *Cons.prem1* **by** *simp*
next
case (Cons y ys)
from *Cons.prem1* **obtain** *z k'* **where** *z: m x y = Some z*
and *k': merge-list m xs ys = Some k'* **and** *keq: k = z # k'*
by (*auto simp: Cons merge-list-Cons split: option.splits*)
have *hz: reg-chains z = reg-chains x $\cup$ reg-chains y*
using *Cons.prem2* [*of x y z*] *z* **by** *simp*
have *ih: ($\bigcup_{s \in \text{set } k'. \text{reg-chains } s} = (\bigcup_{s \in \text{set } xs. \text{reg-chains } s} \cup \bigcup_{t \in \text{set } ys. \text{reg-chains } t})$*
using *k'* **by** (*intro Cons.IH*) (*use Cons.prem2*) **in** *auto*
show *?thesis* **using** *keq hz ih* **by** (*simp add: Cons*) *blast*
qed

qed

Blinding shrinks the revealed chains, recursively through subviews: a blinded view's chains are contained in those of the view it blinds.

lemma *reg-chains-blinding*:

rv-bo a b $\implies$ reg-chains a $\subseteq$ reg-chains b

proof (*induction a arbitrary: b rule: rose-tree_m.induct*)

case (*Tree_m x b*)

obtain *t2 where bt2: b = Tree_m t2 by (cases b)*

note *rt = Tree_m.prems[unfolded bt2 blinding-of-tree-simps]*

show *?case*

proof (*cases x*)

case (*Blinded h*) **then show** *?thesis by simp*

next

case (*Unblinded p*)

obtain *c kids1 where p: p = (c, kids1) by (cases p)*

from *rt obtain kids2 where t2eq: t2 = Unblinded (c, kids2) and*

la: list-all2 rv-bo kids1 kids2

by (*auto simp: Unblinded p rel-prod-inject*)

have *la2: list-all2 ($\lambda s t. \text{reg-chains } s \subseteq \text{reg-chains } t$) kids1 kids2*

proof (*rule list-all2-all-nthI*)

show *length kids1 = length kids2 using la list-all2-lengthD by blast*

fix *n assume n: n < length kids1*

have *m1: (c, kids1) $\in$ set1-blindable_m x by (simp add: Unblinded p)*

have *sm: kids1 $\in$ snds (c, kids1) by (simp add: prod-set-simps)*

have *memn: kids1 ! n $\in$ set kids1 using n by simp*

have *rvn: rv-bo (kids1 ! n) (kids2 ! n) using la n list-all2-nthD by blast*

show *reg-chains (kids1 ! n) $\subseteq$ reg-chains (kids2 ! n)*

using *Tree_m.IH[OF m1 sm memn rvn] .*

qed

from *UN-reg-chains-mono[OF la2] show ?thesis*

by (*auto simp: Unblinded p bt2 t2eq*)

qed

qed

Merging re-reveals exactly the union of the two contributors' chains, recursively through subviews.

lemma *reg-chains-merge*:

rv-merge a b = Some ab $\implies$ reg-chains ab = reg-chains a $\cup$ reg-chains b

proof (*induction a arbitrary: b ab rule: rose-tree_m.induct*)

case (*Tree_m x b ab*)

obtain *y where bY: b = Tree_m y by (cases b)*

obtain *z where abz: ab = Tree_m z by (cases ab)*

from *Tree_m.prems[unfolded bY abz merge-tree.simps]*

have *mz: merge-rt-F_m hash-discrete merge-discrete (hash-tree hash-discrete)*

rv-merge x y = Some z

by (*auto split: option.splits*)

note *mb = mz[unfolded merge-rt-F_m-def]*

show *?case*

```

proof (cases x)
  case (Blinded hx)
    from mb have z = y  $\vee$  ( $\exists q. y = \text{Unblinded } q \wedge z = \text{Unblinded } q$ )
    by (cases y) (auto simp: Blinded split: if-splits)
    then show ?thesis using bY abz Blinded by auto
next
  case (Unblinded px)
    obtain c1 k1 where px: px = (c1, k1) by (cases px)
    show ?thesis
  proof (cases y)
    case (Blinded hy)
      from mb have z = Unblinded (c1, k1)
      by (auto simp: Unblinded px Blinded split: if-splits)
      then show ?thesis using bY abz Unblinded px Blinded by simp
    next
      case (Unblinded py)
        obtain c2 k2 where py: py = (c2, k2) by (cases py)
        from mb obtain k where c12: c1 = c2 and kk: merge-list rv-merge k1 k2
        = Some k
        and zk: z = Unblinded (c1, k)
        by (auto simp:  $\langle x = \text{Unblinded } px \rangle$  px Unblinded py merge-discrete-def
          split: option.splits if-splits)
        have IHc:  $\bigwedge s t u. s \in \text{set } k1 \implies \text{rv-merge } s t = \text{Some } u$ 
           $\implies \text{reg-chains } u = \text{reg-chains } s \cup \text{reg-chains } t$ 
        proof –
          fix s t u assume sset: s  $\in$  set k1 and st: rv-merge s t = Some u
          have m1: (c1, k1)  $\in$  set1-blindablem x by (simp add:  $\langle x = \text{Unblinded } px \rangle$ 
px)
          have sm: k1  $\in$  snds (c1, k1) by (simp add: prod-set-simps)
          show reg-chains u = reg-chains s  $\cup$  reg-chains t
          using Treem.IH[OF m1 sm sset st] .
        qed
        have ( $\bigcup_{s \in \text{set } k. \text{reg-chains } s}$ ) = ( $\bigcup_{s \in \text{set } k1. \text{reg-chains } s}$ )  $\cup$  ( $\bigcup_{t \in \text{set } k2. \text{reg-chains } t}$ )
        using reg-chains-merge-list[OF kk IHc] .
        then show ?thesis
        using bY abz zk c12 by (simp add:  $\langle x = \text{Unblinded } px \rangle$  px Unblinded py)
      qed
    qed
  qed

```

Regression witnesses for the recursive structure. First, deep-nested chains are not dropped: extraction genuinely recurses into subviews. Second, list-merge length alignment is enforced (mismatched lengths are rejected), so it is derived from the hash rather than assumed. Third, a blinded node contributes nothing. The single-consensus invariant (load-bearing, requiring the extraction map) is exhibited with the interpretation below.

lemma *reg-chains-deep-nesting*:

$$\text{reg-chains } (\text{Tree}_m (\text{Unblinded } (0::\text{chain-id}, [\text{Tree}_m (\text{Unblinded } (\text{Suc } 0, []))]))) =$$

$\{0, \text{Suc } 0\}$
by *simp*

lemma *reg-chains-blinded-contributes-nothing*:
 $\text{reg-chains } (\text{Tree}_m (\text{Blinded } h)) = \{\}$
by *simp*

lemma *merge-list-length-aligned*:
 $\text{merge-list rv-merge } [t] [] = \text{None}$
 $\text{merge-list rv-merge } [] [t] = \text{None}$
by *simp-all*

62.2 The recursive transaction-tree instance

A Canton transaction wraps its consensus metadata and its view list in one top-level blindable ($\text{transaction}_m = \text{Transaction}_m (\text{blindable}_m \dots)$). We mirror that exactly: the carrier is a top-level blindable over a payload $\text{reg-state} \times \text{reg-view list}$ — the consensus regulatory state together with the list of recursive view trees. Hash, blinding and merge come from the public product/list/blindable building blocks, so the Merkle interface is inherited (lemma *merkle-reg-tx* below).

Honesty (model-fidelity boundary, read carefully). The recursive view structure is now *faithful*: subviews nest as in Canton, and subview-level blinding is alive (the witnesses below). Two points remain modelling choices, *not* a 1:1 match with Canton’s datatype. First, leaf content is concrete *chain-id/reg-state*, whereas Canton’s leaves are the opaque *view-data/view-metadata*: this is *recursive-faithful*, *content-abstracted*. Second, the consensus *reg-state* is modelled as a *bare, non-independently-blindable* field of the payload, while Canton’s *common-metadata_m* is itself an independently blindable position. This is what structurally forces the single-consensus invariant (a view is reachable only by revealing the whole transaction, hence its consensus), which in turn gives validity with no assumption. The price is a scope limit: the case “a view is revealed while the consensus is blinded” is *out of this model’s scope*. Lifting it would require an independently-blindable consensus and a fresh consistency argument; that is recorded as future work and as the residual fidelity check for the Canton authors, not closed here.

type-synonym *reg-transaction-tree* =
 $(\text{reg-state} \times \text{reg-view list}, \text{reg-state} \times \text{chain-id rose-tree}_h \text{ list}) \text{ blindable}_m$

abbreviation *rtt-chash* :: $(\text{reg-state} \times \text{reg-view list}, \text{reg-state} \times \text{chain-id rose-tree}_h \text{ list}) \text{ hash}$

where $\text{rtt-chash} \equiv \text{hash-prod hash-discrete } (\text{hash-list rv-hash})$

abbreviation *rtt-cbo* :: $(\text{reg-state} \times \text{reg-view list}) \text{ blinding-of}$

where $\text{rtt-cbo} \equiv \text{blinding-of-prod blinding-of-discrete } (\text{blinding-of-list rv-bo})$

abbreviation *rtt-cmerge* :: $(\text{reg-state} \times \text{reg-view list}) \text{ merge}$

where $rtt\text{-}cmerge \equiv merge\text{-}prod\ merge\text{-}discrete\ (merge\text{-}list\ rv\text{-}merge)$
abbreviation $rtt\text{-}hash :: (reg\text{-}transaction\text{-}tree, (reg\text{-}state \times chain\text{-}id\ rose\text{-}tree_h\ list)\ blindable_h) \rightarrow hash$
where $rtt\text{-}hash \equiv hash\text{-}blindable\ rtt\text{-}chash$
abbreviation $rtt\text{-}bo :: reg\text{-}transaction\text{-}tree \rightarrow blinding\text{-}of$
where $rtt\text{-}bo \equiv blinding\text{-}of\text{-}blindable\ rtt\text{-}chash\ rtt\text{-}cbo$
abbreviation $rtt\text{-}merge :: reg\text{-}transaction\text{-}tree \rightarrow merge$
where $rtt\text{-}merge \equiv merge\text{-}blindable\ rtt\text{-}chash\ rtt\text{-}cmerge$

lemma $merkle\text{-}reg\text{-}tx\text{-}content$: $merkle\text{-}interface\ rtt\text{-}chash\ rtt\text{-}cbo\ rtt\text{-}cmerge$
by $(rule\ merkle\text{-}product[OF\ merkle\text{-}discrete\ merkle\text{-}list[OF\ merkle\text{-}reg\text{-}view]])$

lemma $merkle\text{-}reg\text{-}tx$: $merkle\text{-}interface\ rtt\text{-}hash\ rtt\text{-}bo\ rtt\text{-}merge$
by $(rule\ merkle\text{-}blindable[OF\ merkle\text{-}reg\text{-}tx\text{-}content])$

The two list-level corollaries of the recursive lemmas above, lifted over the view list of a transaction.

lemma $reg\text{-}chains\text{-}views\text{-}blinding$:
assumes $list\text{-}all2\ rv\text{-}bo\ va\ vb$
shows $(\bigcup v \in set\ va.\ reg\text{-}chains\ v) \subseteq (\bigcup w \in set\ vb.\ reg\text{-}chains\ w)$
proof $(rule\ UN\text{-}reg\text{-}chains\text{-}mono)$
show $list\text{-}all2\ (\lambda x\ y.\ reg\text{-}chains\ x \subseteq reg\text{-}chains\ y)\ va\ vb$
using $assms$ **by** $(auto\ elim!:\ list\text{-}all2\text{-}mono\ reg\text{-}chains\text{-}blinding)$
qed

lemma $reg\text{-}chains\text{-}views\text{-}merge$:
assumes $merge\text{-}list\ rv\text{-}merge\ va\ vb = Some\ v$
shows $(\bigcup x \in set\ v.\ reg\text{-}chains\ x) = (\bigcup x \in set\ va.\ reg\text{-}chains\ x) \cup (\bigcup y \in set\ vb.\ reg\text{-}chains\ y)$
proof $(rule\ reg\text{-}chains\text{-}merge\text{-}list[OF\ assms])$
fix $s\ t\ u$ **assume** $rv\text{-}merge\ s\ t = Some\ u$
then show $reg\text{-}chains\ u = reg\text{-}chains\ s \cup reg\text{-}chains\ t$ **by** $(rule\ reg\text{-}chains\text{-}merge)$
qed

Extraction: a revealed transaction yields the cross-domain state in which every attested chain (collected recursively from the revealed views) holds the asset in the consensus state; a fully blinded transaction yields the empty view.

fun $rtt\text{-}extract :: reg\text{-}transaction\text{-}tree \Rightarrow global\text{-}state\ option$ **where**
 $rtt\text{-}extract\ (Unblinded\ (r,\ tvs)) = Some\ (auth\text{-}state\ r\ (\bigcup v \in set\ tvs.\ reg\text{-}chains\ v))$
 $| rtt\text{-}extract\ (Blinded\ -) = Some\ (auth\text{-}state\ ACTIVE\ \{\})$

interpretation $reg\text{-}tx\text{-}authenticated$:
 $authenticated\text{-}state\ rtt\text{-}hash\ rtt\text{-}bo\ rtt\text{-}merge\ rtt\text{-}extract$
proof $(rule\ authenticated\text{-}state.intro)$
show $merkle\text{-}interface\ rtt\text{-}hash\ rtt\text{-}bo\ rtt\text{-}merge$ **by** $(rule\ merkle\text{-}reg\text{-}tx)$
next
note $m\text{-}list = merkle\text{-}list[OF\ merkle\text{-}reg\text{-}view]$
show $authenticated\text{-}state\text{-}axioms\ rtt\text{-}hash\ rtt\text{-}bo\ rtt\text{-}merge\ rtt\text{-}extract$

```

proof
  — extract respects merging
  fix  $a\ b\ ab\ sa\ sb$ 
  assume  $Hab: rtt\text{-}hash\ a = rtt\text{-}hash\ b$  and  $Mab: rtt\text{-}merge\ a\ b = Some\ ab$ 
    and  $ea: rtt\text{-}extract\ a = Some\ sa$  and  $eb: rtt\text{-}extract\ b = Some\ sb$ 
  show  $\exists\ sab. rtt\text{-}extract\ ab = Some\ sab \wedge state\text{-}join\ sa\ sb\ sab$ 
  proof (cases a)
    case (Unblinded pa) note  $A = this$ 
    obtain  $ra\ va$  where  $pa: pa = (ra, va)$  by (cases pa)
    show ?thesis
    proof (cases b)
      case (Unblinded pb) note  $B = this$ 
      obtain  $rb\ vb$  where  $pb: pb = (rb, vb)$  by (cases pb)
      have  $hh: ra = rb \wedge map\ rv\text{-}hash\ va = map\ rv\text{-}hash\ vb$ 
        using  $Hab$  by (simp add: A B pa pb)
      then have  $req: ra = rb$  and  $meq: hash\text{-}list\ rv\text{-}hash\ va = hash\text{-}list\ rv\text{-}hash\ vb$ 
        by simp-all
      from  $meq$  obtain  $v$  where  $mlv: merge\text{-}list\ rv\text{-}merge\ va\ vb = Some\ v$ 
        using merkle-interface.merge-respects-hashes[OF mlist] by blast
      have  $ab\text{-}eq: ab = Unblinded\ (ra, v)$ 
        using  $Mab\ req\ mlv$  by (simp add: A B pa pb merge-discrete-def)
      have  $sa\text{-}eq: sa = auth\text{-}state\ ra\ (\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x)$  using  $ea$  by (simp add: A pa)
      have  $sb\text{-}eq: sb = auth\text{-}state\ ra\ (\bigcup_{y \in set}\ vb.\ reg\text{-}chains\ y)$  using  $eb\ req$  by (simp add: B pb)
      have  $uv: (\bigcup_{x \in set}\ v.\ reg\text{-}chains\ x) = (\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x) \cup (\bigcup_{y \in set}\ vb.\ reg\text{-}chains\ y)$ 
        using reg-chains-views-merge[OF mlv] .
      have  $rtt\text{-}extract\ ab = Some\ (auth\text{-}state\ ra\ ((\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x) \cup (\bigcup_{y \in set}\ vb.\ reg\text{-}chains\ y)))$ 
        by (simp add: ab-eq uv)
      moreover have  $state\text{-}join\ sa\ sb\ (auth\text{-}state\ ra\ ((\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x) \cup (\bigcup_{y \in set}\ vb.\ reg\text{-}chains\ y)))$ 
        unfolding  $sa\text{-}eq\ sb\text{-}eq$  by (rule state-join-auth)
      ultimately show ?thesis by blast
    next
      case (Blinded hb) note  $B = this$ 
      have  $ab\text{-}eq: ab = Unblinded\ (ra, va)$  using  $Mab\ Hab$  by (simp add: A B pa)
      have  $sa\text{-}eq: sa = auth\text{-}state\ ra\ (\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x)$  using  $ea$  by (simp add: A pa)
      have  $sb\text{-}eq: sb = auth\text{-}state\ ACTIVE\ \{\}$  using  $eb$  by (simp add: B)
      have  $rtt\text{-}extract\ ab = Some\ (auth\text{-}state\ ra\ (\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x))$ 
        by (simp add: ab-eq)
      moreover have  $state\text{-}join\ sa\ sb\ (auth\text{-}state\ ra\ (\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x))$ 
        unfolding  $sa\text{-}eq\ sb\text{-}eq$ 
        using state-join-auth[of ra  $\bigcup_{x \in set}\ va.\ reg\text{-}chains\ x\ \{\}$ ]
        auth-state-empty-eq[of ACTIVE ra] by simp
      ultimately show ?thesis by blast
  qed

```

```

next
  case (Blinded ha) note A = this
  show ?thesis
  proof (cases b)
    case (Unblinded pb) note B = this
    obtain rb vb where pb: pb = (rb, vb) by (cases pb)
    have ab-eq: ab = Unblinded (rb, vb) using Mab Hab by (simp add: A B pb)
    have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
    have sb-eq: sb = auth-state rb ( $\bigcup_{y \in \text{set } vb. \text{reg-chains } y}$ ) using eb by (simp
add: B pb)
    have rtt-extract ab = Some (auth-state rb ( $\bigcup_{y \in \text{set } vb. \text{reg-chains } y}$ ))
      by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state rb ( $\bigcup_{y \in \text{set } vb. \text{reg-chains } y}$ ))
      unfolding sa-eq sb-eq
      using state-join-auth[of rb {}  $\bigcup_{y \in \text{set } vb. \text{reg-chains } y}$ ]
        auth-state-empty-eq[of ACTIVE rb] by simp
    ultimately show ?thesis by blast
  next
    case (Blinded hb) note B = this
    have ab-eq: ab = Blinded hb using Mab Hab by (simp add: A B)
    have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
    have sb-eq: sb = auth-state ACTIVE {} using eb by (simp add: B)
    have rtt-extract ab = Some (auth-state ACTIVE {}) by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state ACTIVE {})
      unfolding sa-eq sb-eq using state-join-auth[of ACTIVE {} {}] by simp
    ultimately show ?thesis by blast
  qed
qed
next
  — extract under blinding
  fix a b sb
  assume Hab: rtt-hash a = rtt-hash b and BOab: rtt-bo a b
  and eb: rtt-extract b = Some sb
  show  $\exists sa. \text{rtt-extract } a = \text{Some } sa \wedge \text{state-refines } sa \ sb$ 
  proof (cases b)
    case (Unblinded pb) note B = this
    obtain rb vb where pb: pb = (rb, vb) by (cases pb)
    show ?thesis
    proof (cases a)
      case (Unblinded pa) note A = this
      obtain ra va where pa: pa = (ra, va) by (cases pa)
      have bo: rtt-cbo (ra, va) (rb, vb) using BOab by (simp add: A B pa pb)
      then have req: ra = rb and la: list-all2 rv-bo va vb
        by (simp-all add: rel-prod-inject)
      have sr: ( $\bigcup_{x \in \text{set } va. \text{reg-chains } x}$ )  $\subseteq$  ( $\bigcup_{y \in \text{set } vb. \text{reg-chains } y}$ )
        using reg-chains-views-blinding[OF la] .
      have srf: state-refines (auth-state ra ( $\bigcup_{x \in \text{set } va. \text{reg-chains } x}$ ))
        (auth-state ra ( $\bigcup_{y \in \text{set } vb. \text{reg-chains } y}$ ))
        using sr by (rule state-refines-auth)
    
```

```

have rtt-extract a = Some (auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ ))
  by (simp add: A pa)
moreover have state-refines (auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ )) sb
  using srf req eb by (simp add: B pb)
ultimately show ?thesis by blast
next
case (Blinded ha) note A = this
have sr: state-refines (auth-state rb {}) (auth-state rb ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ ))
  using empty-subsetI by (rule state-refines-auth)
have rtt-extract a = Some (auth-state rb {})
  by (simp add: A auth-state-empty-eq[of ACTIVE rb])
moreover have state-refines (auth-state rb {}) sb
  using sr eb by (simp add: B pb)
ultimately show ?thesis by blast
qed
next
case (Blinded hb) note B = this
have a = Blinded hb using BOab by (cases a) (simp-all add: B)
then have rtt-extract a = Some (auth-state ACTIVE {}) by simp
moreover have sb = auth-state ACTIVE {} using eb by (simp add: B)
ultimately show ?thesis using state-refines-refl by auto
qed
next
— extract preserves validity: single consensus per transaction, hence consistent
fix a s
assume rtt-extract a = Some s
then show valid-state s
  by (cases a) (auto simp: auth-state-valid)
qed
qed

```

Multi-level (subview) non-degeneracy witness. A transaction with one view that has two subviews (chains *1* and *2* nested under chain *0*). Revealing everything attests all three chains; blinding *only the chain-2 subview* — a subview-level, not top-level, blinding — drops exactly chain *2*, leaving chains *0* and *1* attested. Selective disclosure is alive below the root.

definition *demo-subview1* :: *reg-view* **where**

demo-subview1 = *Tree_m* (*Unblinded* (*Suc* 0, []))

definition *demo-subview2* :: *reg-view* **where**

demo-subview2 = *Tree_m* (*Unblinded* (*Suc* (*Suc* 0), []))

definition *demo-view* :: *reg-view* **where**

demo-view = *Tree_m* (*Unblinded* (0, [*demo-subview1*, *demo-subview2*]))

definition *demo-view-sv2blinded* :: *reg-view* **where**

demo-view-sv2blinded = *Tree_m* (*Unblinded* (0, [*demo-subview1*, *Tree_m* (*Blinded* (*Garbage* 0))]))

lemma *demo-subview-disclosure*:

reg-chains demo-view = {0, *Suc* 0, *Suc* (*Suc* 0)}

$\text{reg-chains demo-view-sv2blinded} = \{0, \text{Suc } 0\}$
by (auto simp: demo-view-def demo-view-sv2blinded-def demo-subview1-def demo-subview2-def)

lemma demo-multilevel-extract:

$\text{rtt-extract (Unblinded (ACTIVE, [demo-view]))}$
 $= \text{Some (auth-state ACTIVE \{0, Suc 0, Suc (Suc 0)\})}$
 $\text{rtt-extract (Unblinded (ACTIVE, [demo-view-sv2blinded]))}$
 $= \text{Some (auth-state ACTIVE \{0, Suc 0\})}$
by (simp-all add: demo-subview-disclosure)

The single-consensus invariant is load-bearing. If a model allowed a chain to be attested in a regulatory state different from another chain's (which a per-node *reg-state* would permit), validity would collapse: the state below pins chain 0 to *ACTIVE* and chain *Suc 0* to *FROZEN* on the same asset, and is *not* valid. The recursive extraction *rtt-extract* never produces such a state, because every attested chain shares the *one* consensus *reg-state* of its transaction — exactly the structural reason validity holds with no assumption.

definition rogue-inconsistent-state :: global-state **where**

$\text{rogue-inconsistent-state} =$
 $\lambda \text{gs-chains} = (\lambda c \text{ a.}$
 $\quad \text{if } a = 0 \wedge c = 0$
 $\quad \quad \text{then Some } (\lambda \text{as-reg-state} = \text{ACTIVE})$
 $\quad \text{else if } a = 0 \wedge c = \text{Suc } 0$
 $\quad \quad \text{then Some } (\lambda \text{as-reg-state} = \text{FROZEN})$
 $\quad \text{else None}),$
 $\text{gs-locks} = (\lambda \text{. False})$

lemma single-consensus-load-bearing:

$\neg \text{valid-state rogue-inconsistent-state}$
by (auto simp: rogue-inconsistent-state-def valid-state-def consistent-state-def get-reg-state-def get-asset-state-def)

Non-degeneracy of the recursive transaction instance: a single-chain transaction extracts to a genuine one-chain cross-domain state, and a transaction with a non-empty view list genuinely merges at the transaction-tree level (the *extract-respects-merging* path is exercised directly, not only through *reg-chains-merge*).

lemma reg-tx-authenticated-nontrivial:

$\text{rtt-extract (Unblinded (ACTIVE, [Tree}_m \text{ (Unblinded (0, []))]))}$
 $= \text{Some (auth-state ACTIVE \{0\})}$
 $\text{rtt-merge (Unblinded (ACTIVE, [Tree}_m \text{ (Unblinded (0, []))]))}$
 $\quad (\text{Unblinded (ACTIVE, [Tree}_m \text{ (Unblinded (0, []))]))$
 $= \text{Some (Unblinded (ACTIVE, [Tree}_m \text{ (Unblinded (0, []))]))}$
by (simp-all add: merge-discrete-def merge-list-Cons merge-tree.simps merge-rt-F_m-def)

63 Path-level Merkle inclusion for the recursive view tree

The sequence-level result *authenticated-state.sequence-inclusion-integrity* is stated at the level of revealed holdings. Here we sharpen it to the concrete Merkle *inclusion path*. The recursive view *reg-view* is exactly an instance of the generic rose-tree inclusion-proof machinery of *ADS-Constructor.Inclusion-Proof-Construction*: we take the source-content embedding to be the identity on *chain-id* (*rv-embed*) and the content hash to be *id*. An inclusion proof is a zipper in which the target subview is revealed while every node on the path to the root and all off-path siblings are blinded (*blind-path*). Because the discrete content already satisfies the blinding-order locale (*blinding-of-on-discrete*), the two soundness results transfer with no new assumption. This is the same construction the public Canton formalisation applies to its own view tree; it operates on the view rose tree, so the consensus scope limit of the preceding section is untouched.

abbreviation *rv-embed* :: *chain-id* $\Rightarrow$ *chain-id* **where**
rv-embed $\equiv$ *id*

abbreviation *zippers-reg-view* :: *chain-id* zipper $\Rightarrow$ (*chain-id*, *chain-id*) zipper_{*m*} list **where**
zippers-reg-view $\equiv$ *zippers-rose-tree* *rv-embed* *hash-discrete*

abbreviation *blind-reg-path* :: *chain-id* path $\Rightarrow$ (*chain-id*, *chain-id*) path_{*m*} **where**
blind-reg-path $\equiv$ *blind-path* *rv-embed* *hash-discrete*

abbreviation *embed-reg-path* :: *chain-id* path $\Rightarrow$ (*chain-id*, *chain-id*) path_{*m*} **where**
embed-reg-path $\equiv$ *embed-path* *rv-embed*

Inclusion soundness: every enumerated inclusion proof commits to the same authenticating root hash as the fully revealed view tree, so no inclusion proof can fabricate a tree the root does not authenticate.

theorem *reg-view-inclusion-same-hash*:
assumes $z \in \text{set } (\text{zippers-reg-view } (p, t))$
shows $\text{rv-hash } (\text{tree-of-zipper}_m z)$
 $\quad = \text{rv-hash } (\text{tree-of-zipper}_m (\text{embed-reg-path } p, \text{embed-source-tree } \text{rv-embed } t))$
using *assms* **by** (rule *zippers-rose-tree-same-hash*)

Each inclusion proof is a blinding of the canonical inclusion-path tree, which carries the same root hash as the fully revealed tree: a partial, need-to-know view that refines the whole. The single content premise is discharged from the discrete building block, so no assumption is added.

theorem *reg-view-inclusion-blinding-of*:
assumes $z \in \text{set } (\text{zippers-reg-view } (p, t))$
shows $\text{rv-bo } (\text{tree-of-zipper}_m z)$
 $\quad (\text{tree-of-zipper}_m (\text{blind-reg-path } p, \text{embed-source-tree } \text{rv-embed } t))$

by (*rule zippers-rose-tree-blinding-of*[*OF blinding-of-on-discrete assms*])

The attested chains carried by an inclusion proof are contained in those of the fully revealed tree: a path-level reading of need-to-know, from *reg-chains-blinding*.

corollary *reg-view-inclusion-chains-sound*:

assumes $z \in \text{set } (\text{zippers-reg-view } (p, t))$

shows *reg-chains* (*tree-of-zipper_m* z)

$\subseteq \text{reg-chains } (\text{tree-of-zipper}_m (\text{blind-reg-path } p, \text{embed-source-tree } rv\text{-embed } t))$

using *reg-chains-blinding*[*OF reg-view-inclusion-blinding-of*[*OF assms*]] .

Non-degeneracy. Over a genuine two-level view tree the canonical inclusion endpoint reveals all three attested chains, and a selective inclusion proof in the enumeration exposes its target subview (chains 0 and $\text{Suc } 0$) while the unrelated sibling (chain $\text{Suc } (\text{Suc } 0)$) stays blinded. Selective disclosure is alive at the path level.

definition *demo-src* :: *chain-id* *rose-tree* **where**

demo-src = *Tree* (0 , [*Tree* ($\text{Suc } 0$, []), *Tree* ($\text{Suc } (\text{Suc } 0)$, [])])

lemma *reg-view-inclusion-nonvacuous-endpoint*:

reg-chains (*tree-of-zipper_m* (*embed-reg-path* [], *embed-source-tree* *rv-embed* *demo-src*))

= { 0 , $\text{Suc } 0$, $\text{Suc } (\text{Suc } 0)$ }

by (*auto simp: demo-src-def embed-path-def*)

lemma *reg-view-inclusion-nonvacuous-selective*:

$\exists z \in \text{set } (\text{zippers-reg-view } ([], \text{demo-src})). \text{reg-chains } (\text{tree-of-zipper}_m z) = \{0, \text{Suc } 0\}$

by (*force simp: demo-src-def zippers-rose-tree.simps zipper-children.simps blind-path-def embed-path-def*)

Load-bearing regression witness. A forged zipper that fabricates a different chain is neither in the legitimate enumeration nor matched by the same-hash guarantee, so the membership premise of *reg-view-inclusion-same-hash* is load-bearing: dropping it makes the theorem false.

lemma *reg-view-inclusion-forgery-excluded*:

([], *embed-source-tree* *rv-embed* (*Tree* ($\text{Suc } 0$, [])))

$\notin \text{set } (\text{zippers-reg-view } ([], \text{Tree } (0::\text{chain-id}, [])))$

$\wedge \text{rv-hash } (\text{tree-of-zipper}_m ([], \text{embed-source-tree } \text{rv-embed } (\text{Tree } (\text{Suc } 0, []))))$

$\neq \text{rv-hash } (\text{tree-of-zipper}_m (\text{embed-reg-path } [], \text{embed-source-tree } \text{rv-embed } (\text{Tree } (0::\text{chain-id}, []))))$

by (*simp add: zippers-rose-tree.simps zipper-children.simps embed-path-def hash-rt-F_m-alt-def*)

end

64 Acknowledgments

The regulatory state model (five states, seven actions, and the transition exclusions) is motivated by the Regulatory Compliance Protocol (RCP), a framework developed by the Oraclizer research team that systematically classifies requirements from 15 global financial regulatory authorities for tokenized capital markets.